



VELS



INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Band, no 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYASI-CHENNAI
ACCREDITED BY NAAC WITH A++ GRADE

National Conference

on

NextGen Computing and Future Technologies (NCNCFT' 25)

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School of Computing Sciences

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Editors: Dr. R. Parameswari
Dr. S. Perumal

Editorial Board: Dr. A.S Arunachalam
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Dr V. Poornima

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ABOUT VISTAS

Vels Institute of Science, Technology & Advanced Studies (VISTAS), established in 2008, is committed to shaping individuals into well-rounded professionals with both character and competence, dedicated to nation-building and sustainable development. Recognized as a Deemed to be University by the Ministry of Human Resource Development (MHRD), Government of India, and the University Grants Commission (UGC), VISTAS offers a diverse range of undergraduate, postgraduate, and Ph.D. programs. As a multidisciplinary university in Tamilnadu, VISTAS provides academic programs across various fields, including Arts, Science, Engineering, Medicine, Nursing and Allied Health Sciences, Technology, Management Studies, Pharmaceutical Sciences, Physiotherapy, Maritime Studies, Education, Law, Hotel and Catering Management, Aviation, Agriculture, Music, and Fine Arts.

All programs at VISTAS are accredited by the relevant statutory regulatory bodies, including the All India Council for Technical Education (AICTE), National Medical Commission (NMC), Indian Nursing Council (INC), Pharmacy Council of India (PCI), Bar Council of India (BCI), National Council for Teacher Education (NCTE), and the Directorate General of Shipping (DGS). The institution has been awarded an "A++" grade by the National Assessment and Accreditation Council (NAAC) and has been granted 12-B status by the UGC. Additionally, eleven programs have received accreditation from the National Board of Accreditation (NBA). In the NIRF University Rankings 2024, VISTAS secured a position in the 101-150 range, demonstrating its commitment to excellence in education and research. It has also been ranked in the 1201-1500 band in the Times Higher Education (THE) World University Rankings 2025 and has achieved a Diamond rating in the QS I-Gauge University Rating.

ABOUT THE DEPARTMENT

The School of Computing Sciences at our Institution is a hub of Innovation and Technological Advancement, Comprising five specialized departments dedicated to nurturing future-ready professionals.

The Department of Computer Science and Information Technology caters to both undergraduate and postgraduate students with programmes in B.Sc., Computer Science, M.Sc., Computer Science, B.Sc., Information Technology, and M.Sc., Information Technology.

Key Highlights:

- Well Experienced Staff members.
- Internet Facilities for both Staff and Students available.
- Students imparted with ICT based Education.
- High-end Computers with Servers in 2 Laboratories.
- Complete hands on training to perform all experiments in Lab.
- Placement Department gives Soft skill Training every year. 75% of our Computer Science students received offer letter from various Software Companies viz., Infosys, CTS, Mphasis, TCS, Accenture, Ntrust, Savysoft, etc

ABOUT THE CONFERENCE

The National Conference on NextGen Computing and Future Technologies (NCNCFT' 25) brings together leading academicians, researchers, industry professionals, and innovators to deliberate on the emerging paradigms that are shaping the digital future of India and the world. This conference provides a dynamic platform to share research insights, exchange knowledge, and explore technological solutions that address contemporary challenges.

With rapid advancements in computational power, data processing, and intelligent systems, next-generation computing is revolutionizing diverse sectors such as artificial intelligence, robotics, healthcare, energy, and transportation. It paves the way for transformative breakthroughs in medical diagnosis, personalized learning, autonomous systems, and sustainable innovations that redefine how society functions.

NCNCFT'25 organized by the Department of Computer Science and Information Technology emphasizes the importance of research, collaboration, and innovation in building a smarter, resilient, and technology-driven future.

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Dr. Ishari K. Ganesh

Founder – Chancellor, VISTAS.

It is with great pleasure that I present this edited volume, *NextGen Computing and Future Technologies (NCNCFT '25)*—a collection of cutting-edge research contributions from scholars, researchers, and practitioners across diverse disciplines. This book reflects a shared commitment to advancing knowledge and innovation in response to some of the most significant technological challenges and opportunities of our time.

As the world experiences rapid digital transformation and technological dominance, fields such as Artificial Intelligence, Quantum Computing, Advanced Communication Systems, Green Energy, and Biotechnology are fundamentally reshaping industries, societies, and the way we interact as humans. The chapters in this volume not only delve into the theoretical underpinnings of these transformative technologies but also examine the real-world applications, ethical implications, and potential contributions to sustainable development.

I extend my sincere thanks to all the contributors for their insightful outcomes and to the editorial team for their unwavering dedication in bringing this volume to fruition. I am confident that this book will serve as a valuable resource for students, researchers, industry professionals, and policy makers alike—fostering further exploration and innovation in the evolving landscape of emerging technologies.

Wishing all readers an engaging and enriching experience.

Chancellor



Dr. Preethaa Ganesh

Vice President, Vels Group of Institutions

I extend my warm greetings to all participants of the *National Conference on NextGen Computing and Future Technologies* (NCNCFT '25) and to the readers of this accompanying volume of scholarly contributions.

We are at the threshold of a new era where innovations in Artificial Intelligence, Quantum Computing, Advanced Communication, Renewable Energy, and other frontier domains are redefining the way we live, work, and connect with one another. This collection of research reflects not only the creativity and dedication of our academic and professional community but also our shared commitment to building a future driven by knowledge, innovation, and sustainable growth.

I commend the authors, editors, and organizing team for their tireless efforts in bringing together such diverse perspectives under one platform. This book and the discussions it represents will undoubtedly serve as a valuable resource for researchers, industry leaders, and students alike.

May this initiative inspire collaboration, ignite curiosity, and pave the way for transformative ideas that shape a brighter tomorrow.

Vice President



Dr. A. Jothi Murugan

Pro-Chancellor (Planning & Development), VISTAS

I extend my warm greetings to all the distinguished delegates, researchers, academicians, and industry leaders participating in the National Conference on “*NextGen Computing and Future Technologies* (NCNFT’25)”.

In today’s rapidly evolving world technology remains the driving force behind progress and innovation. The advent of next-generation and emerging technologies is not only transforming industries but also redefining how we live, connect, and address complex global challenges NCNFT’25 serves as a vital platform for the exchange of groundbreaking ideas, cutting-edge research, and forward-thinking solutions—laying the foundation for a smarter, more sustainable future.

I am confident that the deliberations, discussions, and collaborations during this event will inspire new ideas, foster meaningful partnerships, and contribute significantly to the advancement of knowledge. I sincerely appreciate the efforts of the organizers, faculty members, and contributors who have worked tirelessly to make this conference a reality.

On behalf of the University, I extend my best wishes for the grand success of NCNFT’25. May this conference ignite curiosity, encourage innovation, and shape the vision of a technologically empowered tomorrow.

Pro- Chancellor



Dr. M. Bhaskaran

Vice-Chancellor (FAC), VISTAS

My warm greetings on the occasion of the *National Conference on “NextGen Computing and Future Technologies (NCNCFT’25)”* and the publication of this volume of scholarly contributions.

In an era defined by rapid technological advancement, universities and research institutions play a pivotal role in fostering innovation and equipping society to navigate the evolving landscape of future challenges and opportunities. Fields such as Artificial Intelligence, Quantum Computing, Advanced Communication, and Green and Sustainable Energy are not only transforming industries but also reshaping the way we live and work.

It is, therefore, essential to cultivate platforms that encourage meaningful dialogue, interdisciplinary collaboration, and the dissemination of cutting-edge research. This conference and the accompanying edited volume stand as a testament to the dedication and intellectual rigor of our academic community.

I am confident that the discussions and ideas emerging from this initiative will inspire further inquiry and contribute to building a more sustainable, inclusive, and technology-driven future.

With best wishes for the continued success.

Vice Chancellor (FAC)



Dr. C. Jothi Venkatewaran

Dean, School of Computing Sciences

It gives me great pleasure to present the Conference Book of the National Conference on *NextGen Computing and Future Technologies* (NCNCFT'25), organized by the School of Computing Sciences. This compilation is a reflection of the creativity, research acumen, and innovative spirit of scholars, researchers, and practitioners from diverse domains of science and technology.

The pace at which computing and allied technologies are evolving has created new opportunities, while also posing fresh challenges. As educators and researchers, it is our responsibility to not only adapt to these changes but also to drive innovation that benefits society at large. This conference book captures the essence of such endeavors by documenting original contributions, critical insights, and futuristic ideas that will serve as a valuable resource for academicians and professionals alike.

I would like to commend the organizing committee, faculty members, reviewers, and students who have worked with dedication to bring this conference and its proceedings to fruition. Their efforts have ensured both academic rigor and quality in the compilation of this volume.

I extend my best wishes to all contributors and participants. May this book inspire continued exploration, collaboration, and breakthroughs in next-generation technologies.

**Dean
School of Computing Sciences**

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Review on MFA Algorithms for Secure Authentication

Mr. K V Vijay Tharun, Student

Mr. K Sasi Raj, Student

Dr. J. Jebamalar Tamilselvi, Associate Professor & Head

Department of CS w/s in Cyber Security,
FSH, SRMIST, Ramapuram,
Chennai.

Abstract

Secure authentication is essential to protecting digital systems against evolving cyber threats. While traditional password-based methods are widely used, they are highly vulnerable to credential theft and brute-force attacks. Multi-Factor Authentication (MFA) addresses these weaknesses by combining independent verification factors such as passwords, one-time codes, and biometrics. Various methodologies and algorithms have been developed to enhance MFA, including biometric recognition systems, minutiae-based fingerprint matching, cryptographic approaches like the HMAC-based One-Time Password (HOTP), and Public Key Infrastructure (PKI). Each method improves security but also introduces challenges such as usability issues, computational cost, or susceptibility to social engineering. This paper compares these approaches based on accuracy, performance, and practicality. The analysis indicates that while all methods contribute to layered security, minutiae-based fingerprint recognition offers a strong balance of accuracy, stability, and usability, making it a practical choice in many MFA deployments.

Keywords: Multi-Factor Authentication, MFA Algorithms, Secure Authentication, Cybersecurity, Identity Verification, Biometric Authentication

Development and Deployment of a Modular Agricultural Robot with Multiple Sensors to Acquire Data, Enabled by a JSON REST API

Ms. K. Vaishnavi

Research Scholar

PG Department of Computer Science
Quaide-E-Millath Government College for Women,
Chennai-02
vaishuangel125.7@gmail.com

Dr. Sumathy Kingslin

Associate Professor,

PG Department of Computer Science,
Quaide-E-Millath Government College for Women,
Chennai-02.
drsumathykingslin@gmail.com

Abstract

This paper discusses the creation and use of a modular farm robot with several environmental sensors to gather data in real time. We set up and fine-tuned each sensor module - temperature, humidity, soil moisture on its own to produce relevant and exact field data. We designed the robot to be modular and expandable allowing us to add unknown sensors later if needed. The robot gathered data and sent it to a back-end server through a JSON REST API where we stored it to analyze and make decisions later. We also thought about how to design the back-end to store structured data, work with analytics programs, and offer more advanced services like dashboards or even a Blockchain marketplace. Our tests show that the robot can gather field data and send it to the back-end, and store the environmental sensor readings in a structured table format. This shows that low-cost scalable and customizable robotic sensing platforms could be useful in modern farming.

Keywords: *IoT, Agricultural Robot Multi-sensor integration, Environmental Sensing, JSON REST API, Data Acquisition.*

A Comparative Analysis of ANN, CNN and RNN with SVM on Diabetic Healthcare Monitoring System

Ms. C. Seetha Lakshmi

Research Scholar

Department of Computer Science and Information Technology
VISTAS, Chennai.

Dr. A. Angel Cerli

Assistant Professor

Department of Computer Science and Information Technology,
VISTAS, Chennai.

Abstract

Monitoring patients entails the analysis of various data types, including glucose time series, retinal images, and clinical records, to deliver timely and precise health evaluations. Artificial Neural Networks are adaptable models that can manage different data types for both classification and prediction purposes. Recurrent Neural Networks, particularly LSTM variants, are proficient in capturing temporal dependencies in glucose level variations and physiological signals, which assists in the early identification of hypo- and hyperglycemic incidents. Convolutional Neural Networks are mainly skillful at extracting realistic features from retinal images to identify diabetic retinopathy and related problems. The combination of these neural architectures with Support Vector Machines significantly improves classification performance by merging deep feature extraction with robust margin-based decision boundaries, which is especially beneficial in small or imbalanced datasets. This analysis highlights accuracy, interpretability and efficiency of the model and signifying a detailed plan for monitoring patients, encouraging improved diagnosis, personalized treatment, and improved health impacts presenting real-world perceptions for selecting the right models in healthcare.

Keywords: Healthcare, physiological signals, diabetic retinopathy and hyperglycemic.

Analysis of Task Scheduling Techniques in Cloud Computing Environment

Ms. V. Mahalakshmi

Research Scholar

Department of Computer Science and Information Technology

VISTAS, Chennai

maha23.mca@gmail.com

Dr. V. Poornima

Associate Professor,

Department of Computer Science and Information Technology,

VISTAS, Chennai.

poornimasudhaagar@gmail.com

Abstract

Cloud computing provides on-demand services and resources, such as storage, applications, and processing power, delivered over the internet. For effective utilization, these resources require efficient management and scheduling mechanisms. Task scheduling plays a critical role in minimizing idle time, balancing workloads, and maximizing system throughput. Proper scheduling ensures accuracy, fairness, and efficiency in task execution within cloud environments. The ideas and uses of several task scheduling algorithms that have been suggested in cloud computing are reviewed in this study. Furthermore, it discusses evaluation parameters such as completion time, throughput, and cost, which are commonly used by researchers to assess the effectiveness of scheduling techniques.

Keywords: Cloud Computing, Task Scheduling, Cloud Services

Exploring People Awareness and Perception of Augmented Reality Virtual Reality

Mr. S. Gunasekaran

Assistant Professor,
Department of Mathematics
Sona College of Arts and Science, Salem.

Mrs. Mareena Abraham

Assistant Professor
Department of Commerce
Sona College of Arts and Science, Salem

Dr. M. Srividya

Assistant Professor,
Department of Mathematics,
Sona College of Arts and Science, Salem.

Abstract

This study explores people awareness and perceptions of Augmented Reality (AR) in virtual environments. It gathers demographic information, assesses participants' familiarity with AR, and examines their experiences using AR in virtual reality settings. Additionally, the paper investigates their attitudes toward AR applications, with a focus on how AR influences user engagement and purchasing decisions. By analyzing these insights, the study aims to identify current and future expectations for AR technologies. The study underscores the following principles of AR implementation: A continuous effort to reduce variation in process outputs is crucial for success. Manufacturing and processes can be effectively measured, analyzed, improved, and controlled through AR. Achieving sustained quality improvement with AR requires commitment from all levels of the organization, especially from top management. This paper also introduces a new procedure for evaluating perceptions of AR in virtual reality, offering a streamlined approach to selecting AR application strategies.

Keywords: People Satisfaction, Awareness of Augmented Reality, AR Applications, Augmented Reality (AR), Virtual Reality (VR) Awareness, User Perception, AR/VR Applications

AI-Powered Digital Twin and Blockchain-Enabled Smart Farming: A Next-Generation Framework for Transparent, Sustainable, and Resilient Agriculture

Ms. P. Gayathri,
Ph.D Scholar (Full Time)
P.G and Research
Department of Computer Science,
Government Arts College, Karur-05.
msgayathriphd@gmail.com

Dr. C. Jayanthi,
Associate Professor,
P.G and Research,
Department of Computer Science,
Government Arts College, Karur-05.
jayanthibabu117@gmail.com

Abstract

The convergence of Artificial Intelligence (AI), Digital Twin (DT), Blockchain, and Internet of Things (IoT) presents a paradigm shift in precision agriculture. This study proposes a hybrid AI-driven digital twin framework that creates a real-time virtual farm replica by integrating multi-modal IoT sensor data, UAV imagery, and satellite information for predictive simulations and adaptive decision-making. The framework incorporates block chain-enabled smart contracts for secure, transparent, and traceable farm operations, ensuring trust across the supply chain. Key innovations include multi-modal deep learning for crop health diagnostics, achieving 96% accuracy in early disease detection, graph neural networks (GNNs) for soil–crop–pest interaction modeling, reducing pesticide use by 35%, and block chain-driven resource automation, enabling water optimization efficiency of 33%. Comparative evaluations reveal a 28% RMSE reduction in yield forecasting compared to baseline models, alongside enhanced scalability and privacy. This research establishes a resilient, climate-adaptive, and transparent smart farming ecosystem, aligning with global sustainability and food security objectives.

Keywords: AI in Agriculture, Digital Twin, Blockchain, Smart Farming, Deep Learning, IoT agriculture, Graph Neural Networks, Supply Chain Transparency, Sustainable Farming.

Signbot: A Real-Time Virtual Assistant for Sign Language Interpretation

Ms. L. Benzera

MCA (Student),

SRM Easwari Engineering College, Chennai.

Ms. Sai Sivatmika

MCA (Student),

SRM Easwari Engineering College, Chennai.

Ms. M. Priyanayaki

MCA (Student),

SRM Easwari Engineering College, Chennai.

Abstract

Hearing loss continues to be one of the largest barriers to inclusive communication in the modern digital world. According to the World Health Organization (2024), over 430 million people worldwide experience disabling hearing loss, with more than 70 million being completely deaf. While sign language remains the most effective mode of communication for these communities, access to interpreters is often expensive, unreliable, or unavailable in real time. Current assistive technologies are either limited to gesture recognition or offer only partial translation for written text, failing to integrate naturally into daily digital environments. This paper introduces SignBot, a real-time virtual assistant that provides seamless interpretation of spoken words, text, and videos into sign language using customizable animated avatars. Unlike existing solutions, SignBot supports multiple sign languages, integrates with video platforms such as YouTube, and allows users to choose avatars ranging from robotic to male or female figures. Built using speech recognition, natural language processing, and real-time 3D animation, SignBot delivers instant, clear, and human-like sign translations on desktop and mobile platforms. By enabling education, workplace communication, digital content access, and social inclusion, SignBot represents a scalable and human-centered solution for bridging the communication gap between hearing and hearing-challenged individuals.

Keywords: Sign Language, Accessibility, Virtual Assistant, Real-Time Translation, Avatars.

Focus Flow: An AI-Driven Behavioral Psychology Framework for Enhancing Student Productivity

Ms. Hema Harrini.T

Student,

Kumaraguru Collage of Liberal Arts And Science, Coimbatore

hemaaharrini@gmail.com

Dr.Mary Louis.L

Professor,

Kumaraguru Collage of Liberal Arts And Science, Coimbatore

martlouis@kclas.ac.in

Abstract

Student productivity is often reduced due to distractions, irregular study and sleep schedules and the absence of a personalized learning strategy. This paper introduces Focus Flow which is a next-generation framework combining behavioral psychology and Artificial Intelligence to optimize student productivity. The main objective of this work is to analyze individual learning patterns and provide improved strategies to enhance productivity. The methodology includes structured data collection through Google Forms; preprocessing in Jupyter and the application of machine learning models such as Random Forest in Python. Key behavioral factors such as study duration focus and distraction levels are analyzed to identify unique patterns. Experimental results show that the system achieved a prediction accuracy of 70% in identifying focus trends and recommended optimized study schedules. Students who followed these recommendations reported an average improvement of 18 to 22% in focused and effective study time compared to their previous performances. This system differs from existing tools as it offers data-driven, personalized strategies, addressing a common academic challenge with an interdisciplinary solution. It aims to build an intelligent academic support system that could be widely used in the field of education which ultimately changes study practices in the digital age.

Keywords: Artificial Intelligence, Behavioral Psychology, Student Productivity, Focus Optimization, Next-Generation Computing.

The Future of Artificial General Intelligence in Redefining Human–Machine Interaction

Ms. R. B. Deebika ,
Student

Bsc.AI &DS,
Saveetha college of liberal arts and Science.

Ms. V Sanjana
Student

Bsc.AI &DS,
Saveetha college of liberal arts and Science.

Abstract

Artificial General Intelligence (AGI) signifies a paradigm shift in Artificial Intelligence research, aiming to design systems capable of human-like reasoning, contextual awareness, and adaptability across diverse tasks. Unlike narrow AI, which thrives only within predefined boundaries, AGI envisions machines that can self-learn, reinterpret prior knowledge, and interact seamlessly with human users in dynamic environments. This paper investigates the transformative potential of AGI in redefining human–machine interaction by exploring current AI methodologies, emerging approaches, and practical case studies. While large language models, autonomous agents, and multi-modal systems illuminate progress toward generalization, achieving true AGI requires overcoming challenges of ethical alignment, interpretability, and safety. The findings underscore that AGI is not simply a technological goal, but a societal milestone demanding responsible innovation, human-centered design, and interdisciplinary collaboration.

Keywords: Artificial General Intelligence, AGI, Human–Machine Interaction, Intelligent Systems, AI Ethics, Machine Learning

Cloud-Enabled Framework for Efficient Medical Diagnosis: A Case Study on Hydrocephalus Detectio

Ms. R.B. MariaSofia

Research Scholar

Department of Computer Science and Information Technology
VISTAS, Chennai.

mariasofia.britto@gmail.com

Dr. R. Parameswari

Professor & Head

Department of Computer Science and Information Technology,
VISTAS, Chennai.

parameswari.scs@vistas.ac.in

Abstract

Medical diagnostics have revolutionized in great part to even more efficient data processing, storage, and analysis made possible by recent developments in cloud computing. Early and accurate diagnosis of hydrocephalus, a neurological condition defined by aberrant cerebrospinal fluid accumulation in the brain, and the limited access to diagnostic tools in geographically isolated areas continue to be clinical challenges early and accurate diagnosis of hydrocephalus presents. This work can show a cloud-based framework for hydrocephalus detection and classification using medical image analysis. The scalability and computational efficiency of the cloud support the framework. The proposed architecture can effectively classify medical images using cloud-based resources and machine learning methods. Preprocessing a dataset comprising 5,000 anonymised medical images—which included MRI and CT scans—used noise reduction and normalization techniques. This was done before features were added. By means of a hybrid model comprising Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks for both spatial and sequential analysis, the classification task acquired an overall accuracy of 96.8%. For the system, the measurements turned out to be 97.3% precision, 96.2% recall, and 96.7% F1-score respectively. The cloud architecture of the framework guarantees scalability and real-time diagnostics, so enabling healthcare provider's remote access and efficient data processing of enormous volumes. Comparisons with modern methods, such as standalone CNNs and conventional classifiers revealed that the computational overhead was dropped by 25% and the diagnosis accuracy was raised by 12%.

Keywords: *Cloud Computing, Hydrocephalus Detection, Medical Image Classification, CNN-LSTM Hybrid Model, Diagnostic Accuracy*

A Comprehensive Survey of Light GBM, KNN, and Voting Classifiers in Coronary Artery Disease Detection

Ms. Jeevika

Student, II MCA

School of Computer Applications

Karpagam College of Engineering, Coimbatore.

Dr. P. Shanthi

Professor,

School of Computer Applications,

Karpagam College of Engineering, Coimbatore.

Abstract

Heart disease remains one of the primary causes of mortality worldwide, making early detection a vital component in preventing life-threatening complications. This study presents a machine learning-based classification model to predict the presence of heart disease using publicly available patient datasets. The proposed method employs two widely used supervised learning algorithms—K-Nearest Neighbors (KNN) and Random Forest (RF)—to build a dual-model prediction framework. Standardization is applied during preprocessing, and the classifiers are assessed using performance measures like accuracy, precision, recall, and F1-score. To improve robustness, a soft voting ensemble approach combining both classifiers is also implemented. The experimental findings reveal that the ensemble classification model achieves superior predictive performance compared to the individual base learners. This work aims to support clinical decision-making by offering an interpretable, efficient, and reliable tool for the early prediction of heart disease. The entire framework is designed to be scalable and suitable for deployment using cloud-based platforms such as Azure Machine Learning.

Keywords: Heart Disease Prediction, Coronary Artery Disease, Machine Learning, K-Nearest Neighbors, Random Forest, Ensemble Learning.

Crowd-Sourced Disaster Response Web Application

Ms. V. Leelavathi

Department of Computer Science and Engineering
VISTAS, Chennai

Ms. S. Pavithra

Department of Computer Science and Engineering
VISTAS, Chennai

Dr. A. Banushri

Department of Computer and Engineering
VISTAS, Chennai

Abstract

The increasing frequency of disasters requires innovative approaches to improve disaster preparedness, response, and recovery. Crowd sourced disaster management leverages public participation, digital platforms, and real-time data collection to enhance situational awareness and response. This paper presents the crowd project, a system designed to integrate crowd sourced inputs with institutional frameworks. The paper outlines the problem statement, motivation, objectives, existing systems, gaps in research, and a proposed architecture. It also discusses implementation details, technologies used, system outputs, and performance metrics.

Keywords: *Data Synchronization, Community-driven Response, Resilient Architecture, Offline-first Architecture, Data Security, Web Application Architecture, Implementation Workflow.*

Multi Variant Deep Transfer Learning Analytics Model for Improved Health Care Solution Using Big Data

Ms. P. Malathi

Research Scholar

Department of Computer Science and Information Technology

VISTAS, Chennai.

malathipandiyan13290@gmail.com

Dr. R. Parameswari

Professor & Head

Department of Computer Science and Information Technology,

VISTAS, Chennai.

parameswari.scs@vistas.ac.in

Abstract

The problem of healthcare analytics with big data and deep transfer learning has been well studied. There exist number of approaches which consider limited factors and homogeneous data maintained by various organizations. Still they suffer to achieve higher performance in predicting the disease to support healthcare solutions. With the motivation to improve the performance, this article presents a novel Multi Variant Deep Transfer Analytics Model (DVTAM), which normalize the big data set using Deep Variant Feature Normalizer. The normalization is performed by at feature level and adjusts the feature values by computing Feature Covariance Value (FCV). Further, the method transforms the data using Multi Variant Transformer (MVT) using transfer learning. Transformed data has been grouped and trained. At the test phase, the method computes Multi Factor Disease Absorption Weight (MFDW) towards various disease classes. Based on the value of MFDW, the method identifies the class of disease as result. The proposed model produces higher accuracy on disease prediction with less false ratio.

Keywords: *Multi-Variant Learning, Deep Transfer Learning, Healthcare Analytics, Big Data, Predictive Healthcare, Machine Learning*

Secure Data Storage and Access in Cloud Environments: Current Approaches and Future Directions

Ms.S.Deepika

Assistant Professor,

Department of Master of Computer Application,

K.L.N. College of Engineering, Pottapalayam.

deepikarakshi123@gmail.com

Abstract:

Cloud computing is a vital tool for managing and storing massive amounts of data. However, it is still very challenging to guarantee secure data storage and controlled access in cloud environments due to the growing privacy concerns and cyber threats. This study looks at modern approaches to safe data storage and access control, such as encryption techniques, role-based and attribute-based access control models, and auditing systems. The objective is to identify the benefits and drawbacks of existing solutions with regard to security, performance, and scalability. Comparing methods based on criteria including user identity, data integrity, and secrecy is done using recent academic and commercial research. The paper also discusses emerging trends including blockchain, AI-powered access control, and zero-trust security measures. It ends by listing possible research directions for developing more integrated, scalable, and efficient cloud security frameworks.

Keywords: Auditing, Data Storage, Access Control, Blockchain, Zero Trust, Encryption, Cloud Security, Institutional Cloud.

Coughsense: Deep Learning based Audio-Driven Tuberculosis Detection using Spectrogram

Mr. D. Manikandan, Mr. S. Karthick, Mr. R. Santhosh, Ms. S. Santha kumari

Department of Computer Science and Engineering, VISTAS.

ssanthakumari2004@gmail.com

Abstract

Tuberculosis (TB) continues to challenge global health, particularly in areas where conventional diagnostic methods are costly, invasive, and slow. Leveraging the acoustic properties of patient coughs, audio-based diagnostics offer a non-invasive and scalable alternative. In this approach, raw cough recordings undergo a multi-stage preprocessing pipeline, including noise filtering, amplitude normalization, clipping, and padding, to produce standardized and clean audio signals. These signals are then transformed into spectrograms through Fast Fourier Transform, frequency sampling, windowing, Mel-scale conversion, and logarithmic scaling, effectively converting temporal sound patterns into visual representations suitable for image-based analysis. Feature extraction employs Custom Histogram of Oriented Gradients (CHOG) to capture localized spectral characteristics, and Principal Component Analysis (PCA) reduces dimensionality while retaining discriminative information. Classification is performed using YOLO V10, which detects and differentiates TB-specific acoustic patterns from non-TB signals within the spectrograms. This integrated pipeline demonstrates a precise, low-cost, and non-invasive method for TB detection, offering rapid, real-time analysis suitable for deployment in resource-limited healthcare environments. By combining audio signal processing with advanced machine learning techniques, the methodology provides a robust framework for accessible and efficient tuberculosis screening.

Keywords: Tuberculosis Detection, Cough Analysis, Spectrograms, Chog, PCA, YOLO V10, Audio Classification, Non-Invasive Diagnostics, Healthcare AI.

SmartEcoGrow: A Sustainable Ecosystem Approach to Organic Fertilizer Recommendation

Ms. M. Sreebhavana

Student

Kumaraguru College of Liberal Arts and Science, Coimbatore.

sribhavs8220@gmail.com

Dr.M.Emimanimcy,

Assistant Professor,

Kumaraguru College of Liberal Arts and Science, Coimbatore.

emimanimcy.m.mat@gmail.com

Abstract

In today's world, maintaining crop productivity while preserving ecological integrity is a twin challenge for modern agriculture. Without prior knowledge, careless application of chemical fertilizers has led to long-term environmental imbalance, biodiversity loss, and soil degradation. In the theme of SMART ECOSYSTEM, the knowledge-driven framework addressed in this paper, A Sustainable Ecosystem Approach to Organic Fertilizer Recommendation reframes fertilizer preference as an essential part of a smart ecosystem rather than an isolated agronomic choice. The primary goal is to build a system that helps farmers adapt to ecologically and regionally appropriate organic alternatives, which will reduce their dependency on chemicals and use of inorganic fertilizer and promote soil regeneration. By mapping farming conditions like crop requirements, climate conditions, and cultivation patterns to recommend sustainable organic fertilizer for crops using predetermined decision rules, the methodology combines agricultural knowledge with rule-based decision-making. The system analyzes these conditions and produces context-aware recommendations by using forward chaining. This design provides suitable outputs while remaining adaptable to different regions and datasets, making it scalable for diverse farming contexts. Key findings show that incorporating sustainability into the recommendation process promotes ecological restoration as well as increased yield. The method promotes soil health, resilient food systems, and alignment with global goals by framing fertilizer choice in a smart ecosystem. The system identifies modern agriculture with environmental stewardship by providing farmers with sustainable, practical alternatives through using advanced decision-support technology.

Keywords: Smart Ecosystem, Organic Fertilizer Recommendation, Ecological Restoration.

Smart College Ecosystem

Ms. D. Indhuja

Department of Computer Science and Engineering, VISTAS.

Ms. B. Shakthi

Department of Computer Science and Engineering, VISTAS.

Dr. A. Banushri

Department of Computer Science and Engineering, VISTAS.

Abstract

Modern college students face a range of everyday challenges, including time constraints during lunch breaks, difficulty accessing administrative services, a lack of centralized academic updates, and inefficient communication systems. To address these pressing concerns, this project proposes the development of a Smart Campus Utility Mobile Application, designed to digitize and streamline essential student services into a single, easy-to-use platform. The application will offer features such as classroom food delivery and pre-ordering, a real-time campus map for navigation, a maintenance request module, peer-to-peer services like tutoring and second-hand book exchanges, digital notice boards for events and lost & found, and a direct complaint mechanism that connects students to campus authorities. By providing both convenience and transparency, the app aims to significantly reduce the time and effort students spend on routine tasks, enabling them to focus more on academics and overall well-being. Developed using modern mobile technologies (React.js, Node.js, Tailwind CSS), the project will follow an agile development model with continuous feedback from real users. The expected outcome is a scalable, modular, and secure mobile application that enhances campus productivity and improves student satisfaction. Budget requirements will primarily focus on development tools, testing devices, cloud services, and promotional materials. Long-term sustainability will be supported through partnerships with campus facilities and optional premium services.

Keywords: *Smart Campus, Mobile Application, Digital Services, Student Utility, Campus Management*

Towards Secure and Transparent Elections: A Survey on AI-Powered Voting Systems with Aadhaar Integration and Emerging Enhancements

Lorina Rogin S, Mirthula Sri A, Divya Raj

PG Student, Department of Master of Computer Applications
Easwari Engineering College, Ramapuram

Dr. G. Gayathri

, Assistant Professor, Department of Master of Computer Applications,
Easwari Engineering College, Ramapuram.

Abstract

Free and fair elections are the backbone of democracy, yet traditional voting systems continue to face challenges such as duplicate voting, booth-level fraud, and the rapid spread of misinformation. Recent advances in Artificial Intelligence (AI) have introduced innovative techniques to strengthen electoral integrity by improving identity verification, detecting irregularities, and safeguarding public trust. This survey paper reviews current AI-driven approaches in voting systems, focusing on biometric authentication, blockchain-based vote recording, and machine learning models for fraud detection. The study highlights their effectiveness in preventing impersonation, ensuring transparency, and improving scalability, while also identifying limitations related to privacy, cost, and inclusivity. Building on these insights, the paper proposes three future enhancements: first, Aadhaar-linked fingerprint authentication combined with AI verification to ensure one-person-one-vote and prevent duplicate entries; second, smart anomaly detection in voting patterns using models such as LSTMs to flag suspicious turnout spikes or booth-level irregularities in real time; and third, AI-powered misinformation and deep fake detection, employing NLP models for fake news in Indian languages and computer vision techniques for manipulated media, to counter digital propaganda during elections. Together, these approaches present a pathway toward more secure, transparent, and inclusive elections. The survey concludes by emphasizing the importance of privacy-preserving AI models, strong cybersecurity frameworks, and ethical adoption to build trustworthy next-generation voting systems.

Keywords: Artificial Intelligence, Voting Systems, Aadhaar Authentication, Biometric Verification, Fraud Detection, Anomaly Detection, Misinformation Control.

Beyond the Screen: An HCI - Driven Framework to Study Digital Behavior and Encourage Responsible Technology Use

Ms. S. G. Reganya, Student
reganya.24bds@kclas.ac.in

Dr. M. Sindhana Devi, Assistant Professor
sindhanadevi.m.dsc@kclas.ac.in

Department of Data Science,
Kumaraguru College of Liberal Arts and Science, Coimbatore.

Abstract

Nowadays, technology has become deeply woven into everyday life, shaping how individuals connect, learn, and entertain themselves. However, the excessive and unmindful use of digital platforms has raised concerns such as media addiction, reduced productivity, and unhealthy online behaviors. Within the theme of Human–Computer Interaction (HCI), this paper presents Beyond the Screen - an HCI Driven Framework to Study Digital Behavior and Encourage Responsible Technology Use. The core objective of this framework is to analyze users’ digital engagement patterns, identify early signs of technology overuse, and deliver awareness-driven interventions that promote mindful and balanced media habits. The methodology integrates behavior tracking, interaction design principles, and intelligent data analysis to evaluate parameters such as screen time, platform preference, and usage frequency, the framework generates personalized insights along with awareness messages to guide users toward healthier digital routines. The system is designed to remain scalable across different age groups and platforms, offering a versatile tool to promote digital wellness. The findings show that this framework can spot early signs of digital overuse, send personalized awareness messages, and help users manage their screen time more effectively. By applying HCI principles, the awareness prompts became more engaging and persuasive, encouraging users to reflect on their habits. The system also proved flexible across different age groups, making it suitable for a wide range of users. Most importantly, it helped shift people’s view of technology from something that fuels compulsive use to something they can consciously control. In this way, the project demonstrates how HCI can guide people toward healthier digital habits and ensure technology supports growth instead of dependency.

Keywords :*Human–Computer Interaction, Digital Behavior, Media Addiction, Responsible Technology Use, Awareness Framework.*

Multi-Sensor Deep Learning for Real-Time Landslide Detection and Early Warning using Satellite Imagery and DEM Data

Ms. M. Sowmiya, Ms. P Dheiveeka Nachiyar, Ms. S. Lakshmi Pratha

Students, Department of Master of Computer Applications,
DDGD Vaishnav College, Chennai, Tamil Nadu, India

Abstract

One of the most damaging natural disasters, landslides pose serious risks to infrastructure, human life, and the environment, particularly in mountainous and geologically unstable regions. Traditional ground monitoring methods are costly, have a limited scope, and are insufficient for comprehensive monitoring. Scalable, effective disaster management techniques may now be implemented because of developments in artificial intelligence and remote sensing. In this work, a multi-sensor deep learning system for instantaneous landslide diagnosis and early warning is presented. It integrates Synthetic Aperture Radar (SAR) data (Sentinel-1), optical satellite images (Sentinel-2, Landsat-8/9), and digital elevation models (DEM). The system uses transformer-based temporal modeling and convolutional neural network to recognize geographical parameters (landscape, vegetation, soil visibility) and as well as temporal fluctuations data (precipitation patterns, earth movement). Multi-spectral and radar signals are combined with DEM characteristics including elevation, slope, and aspect to enhance landslide susceptibility mapping. The model is trained and validated using time-series satellite data and global landslide inventories to address class imbalance and enhance detection accuracy. Measures of evaluation such as recall, F1-score, and intersection over union (IoU) are employed. According to experimental results, multi-sensor fusion significantly increases the detection precision when compared to other single-sensor approaches, especially in cloud-based scenarios when SAR data provides crucial resilience. Nearly real-time landslide likelihood maps are generated by the proposed method, offering useful data for post-disaster assessment and early warning systems. This research offers a strong, scalable, and data-informed strategy for landslide risk management, connecting remote sensing, deep learning, and disaster preparedness. Incorporating it into national disaster management strategies which could lower fatalities, protect infrastructure, and enhance climate-resilient sustainable development.

Keywords: Landslide detection, Multi-sensor fusion, Deep learning, Satellite imagery, DEM, Disaster management, Early warning system, Remote sensing.

Current Learning Systems for Higher Education in India: A Review

Mr. R. Rajapriyan, Research Scholar
Department of Nautical Science
VISTAS, Chennai

Dr. Capt. N. Kumar, Professor & Director
School of Maritime Studies
VISTAS, Chennai

Dr. T. Kamalakannan, Professor
School of Computing Sciences
VISTAS, Chennai

Abstract

The landscape of Indian higher education is now undergoing a significant transformation, rapid driven adaption of technology-enabled learning system with the urgent to expand access, quality and equity. The traditional classroom-based instruction is very much increased with supplemented and also in many cases redefined by digital platforms such as Learning Management System (LMS), MOOCs and government-led initiatives like SWAYAM and the National Digital Library of India. The current adopted systems have opened avenues for flexible and blended learning, enabling students across diverse geographic location to access the quality resources. Moreover, the Artificial Intelligence (AI), adaptive learning tools and the data driven analytics is beginning to personalize the learning pathways, while virtual and augmented reality offer immersive opportunities for skilled development. Despite with these advancements, The Indian higher education continues to face many challenging aspects including the digital divide, infrastructure gaps, readiness of faculty and concerns over inclusivity. This review paper critically examines the current learning system shaping the Indian higher education compares their pedagogical and technological impact and highlights both opportunities and limitations. It further explores how emerging models such as hybrid classrooms, gamified learning and the competency-based frameworks which can be strategically leveraged to meet the goals of the National Education Policy (NEP 2020). This paper concludes the insights the further directions for building equitable, resilient and technology-driven ecosystems that can enhance the global competitiveness of higher education system in India.

Keywords: *E-learning, Higher Education, India, MOOCs, Blended Learning, NEP 2020, Educational Technology.*

Eco Score Sustainability Index for Cities

Ms. R. Sneha

Student, BSc .Data Science

Kumaraguru College of Arts and Science, Coimbatore

Dr. J. Soonu Aravindhan

Professor

Kumaraguru College of Arts and Science, Coimbatore

Abstract

Urban waste management is among the major challenges for sustainability, with poor practices like open dumping and unaccounted waste playing significantly in environmental degradation. To analyse and compare city-level performance, this project presents a data-driven Eco Score model for assessing how efficiently the cities manage their wastes. The Eco Score is obtained by calculating weighted contributions through major treatment approaches—recycling, composting, incineration, sanitary landfilling, open dumping, and unaccounted wastes. Sustainable practices are rewarded with larger weights, whereas unsustainable ones are penalized, obtaining the composite score normalized on the scale of 0–100 for comparison across cities on an even scale. To test the validity of the model, hypothesis testing through linear regression is performed. Results indicate that recycling and composting significantly increase the Eco score, while open dumping and unaccounted waste strongly reduce it. This confirms that sustainable treatment methods are statistically important drivers of urban environmental performance. Moreover, a number of the results are visualized through the following techniques: top-performing city bar graphs, recycling-Eco Score scatter plots, Eco Score-recycling correlation heatmaps, and distribution of the waste treatment by stacked bars. These reveal additional details on the strengths as well as weaknesses on the current waste management practices. The demo showcases how the Eco Score will act as an intelligence-based assessment tool, allowing policymakers and urban planners to monitor, compare.

Keywords: Eco-Score, Sustainability Index, Urban Sustainability, Green Cities, Environmental Impact

Hybrid Image Processing and Data Mining Framework for Accurate Lung Cancer Detection Using FBAC-NET And BI-LRCN

Ms. K. Geetha

Research Scholar (Part time Internal)
Department of Computer and Information Science
Annamalai University
geetha2kumar@gmail.com

Dr.Karthikeyan Elangovan

Research Supervisor,
Assistant Professor/Programmer,
kanchikarthi2010@gmail.com

Abstract

Early lung cancer detection is significant for enhancing patient survival and aiding early clinical decision-making. This paper introduces a hybrid image processing and data mining model combining Feature-Based Attention Convolutional Network (FbAC-NET) and a Bidirectional Long Recurrent Convolutional Network (Bi-LRCN) to detect lung cancer accurately. FbAC-NET optimizes feature extraction by emphasizing discriminative areas in lung CT scans, whereas Bi-LRCN models spatial and temporal dependencies for stable classification of malignant and benign nodules. Experimental results on standard lung cancer datasets show that the hybrid model attains overall accuracy of 95.4%, sensitivity of 94.2%, and specificity of 95.0%, surpassing traditional CNNs as well as individual deep models. The findings emphasize the power of integrating high-level image processing and data mining methods with deep learning models to offer a consistent diagnostic tool for early detection and clinical decision support for lung cancer management.

Keywords: *Lung Cancer Detection, Hybrid Image Processing, Data Mining, FbAC-NET, Bi-LRCN, Deep Learning, Medical Imaging*

A Survey-Based Comparative Study of WEKA Classifiers for Coconut Tree Health Prediction using Climate and Soil Parameters

Ms. S.T Pavithra devi

Research Scholar

Department of Computer Science,

A.V.V.M Sri Pushpam College, Poondi, Thanjavur

pspavithrasugumar@gmail.com

Dr.V. Maniraj

Associate Professor & Research Supervisor

Department of Computer Science,

A.V.V.M Sri Pushpam College, Poondi, Thanjavur

maniraj_vee@yahoo.co.in

Abstract

Coconut trees are a vital source of livelihood, nutrition, and economic support for millions of farmers worldwide. However, their productivity is often threatened by diseases and unfavourable environmental conditions that are difficult to detect early through traditional methods. This study explores the potential of WEKA, a widely used machine learning platform, to predict coconut tree health by analyzing climate and soil parameters. A comparative evaluation of classifiers Support Vector Machines (SVM), Random Forests (RF), J48 Decision Tree, and Naïve Bayes (NB) was conducted to assess their performance in disease classification. Models were tested using cross-validation and split datasets, and performance was measured through accuracy, precision, recall and F1-score. Results indicated that SVM and RF achieved the highest accuracy; while J48 offered better interpretability and Naïve Bayes provided faster predictions with minimal computational resources. The findings highlight how farmer-friendly decision-support systems built on machine learning can help in smarter, more sustainable coconut farming.

Keywords: Coconut Tree Health, Machine Learning, WEKA, Classifier Comparison, Climate and Soil Parameters

Speech-to-Speech Translation using Deep Learning

Ms.R. Preethi

preethi3472@gmail.com

Ms. J. Sajetha

sajethajeeva@gmail.com

Ms. G. Yogalakshmi.

yoga.lakshmi2021@gmail.com

Dr. A. Banushri

banushri.se@vistas.ac.in

Department of Computer Science and Engineering, VISTAS.

Abstract

Speech-to-speech translation (S2ST) has emerged as a breakthrough technology that eliminates language barriers and enables seamless multilingual communication in areas such as diplomacy, healthcare, tourism, and global trade. This paper reviews the development of S2ST systems, contrasting traditional cascade architectures—which sequentially employ automatic speech recognition (ASR), machine translation (MT), and text-to-speech (TTS)—with more recent end-to-end direct speech translation (DST) models that bypass text-based intermediaries. While cascade systems are modular and benefit from optimized subcomponents, they often face problems such as error accumulation, higher latency, and the inability to preserve prosody. In contrast, direct S2ST approaches, including Translatotron 2 and UnitY, achieve more natural results by retaining speaker identity, reducing translation delays, and maintaining vocal expressiveness. Nonetheless, these advanced models remain challenged by limited parallel speech data, heavy computational requirements, and reduced generalization in low-resource language settings. This study critically examines the strengths and weaknesses of these approaches and explores future directions for building robust, real-time multilingual communication systems.

KEYWORDS - *Speech-to-Speech Translation (S2ST), Deep Learning, Automatic Speech Recognition (ASR), Neural Machine Translation (NMT), Text-to-Speech (TTS), Direct Speech Translation (DST), Natural Language Processing (NLP), Transformer Models.*

Enhancing Awareness of Cyber Security in E-Learning Techniques

Dr M Anita Priscilla Mary, Assistant Professor,
Department of Computer Science, Anna Adarsh College For Women (Autonomous)
anitapriscillamarym@annaadarsh.edu.in

K. Bhavya Pravaika, Student,
Department of Computer Science, Anna Adarsh College For Women (Autonomous)
kbhavyapravalika@gmail.com

S. Rakshitha, Student,
Department of Computer Science, Anna Adarsh College For Women (Autonomous)
srrakshitha0707@gmail.com

Abstract

E-learning is now a key part of modern education. It provides flexibility, accessibility and a global reach. However, the increasing reliance on digital platforms also makes learners and institutions vulnerable to cyber security threats like phishing, malware, identity theft, and unauthorized access. These risks put data security at risk, lower trust in virtual learning systems, and show the urgent need for cyber security awareness in educational settings. This research examines methods to enhance digital safety practices and emphasizes the necessity of promoting cyber security awareness within e-learning platforms. Insecure passwords, perilous browsing habits, and susceptibility to social engineering contribute to substantial risks. Therefore, educating individuals and raising awareness is equally crucial as establishing technological defenses. Approaches like scenario-based learning, interactive simulations, gamified training and real-time security guidance can effectively engage participants. The integration of emerging technologies also supports this goal. Artificial intelligence can personalize cyber security learning by identifying user weaknesses, while block chain and cloud-based monitoring systems can provide secure data handling and instant threat detection. Collaboration between educators, institutions and cyber security experts is essential to embed digital safety into curricula and create a culture of security consciousness. Enhancing cyber security awareness in online education enables schools to safeguard their critical digital resources, maintain the confidentiality of student information and foster a secure and reliable learning atmosphere. This approach is not limited to the deployment of advanced technologies like firewalls, encryption or secure authentication systems but extends to the integration of instructional techniques and behavioral changes.

Keywords: *Cyber security, Awareness, E-learning, Digital Literacy, Online safety, Malware prevention.*

From Passive Predictions to Proactive Autonomy: A Literature Review on Agentic Artificial Intelligence and Its Transformative Role in Healthcare Systems

Ms. Siji K B

Research Scholar

Department of Computer Science and Information Technology

School of Computing Sciences

VISTAS.

Dr K Sheela

Assistant Professor

Department of Computer Science and Information Technology

School of Computing Sciences

VISTAS

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Abstract

Agentic Artificial Intelligence (AI) represents a paradigm shift in computational intelligence, moving beyond static decision-support models toward autonomous, goal-directed systems that can plan, act, reflect, and adapt in dynamic environments. This literature review synthesizes theoretical foundations, applications, challenges, and future trajectories of agentic AI, with an emphasis on healthcare as a high-stakes, ethically complex domain. By examining key agent architectures—such as ReAct, Reflexion, and multi-agent simulations—the review situates healthcare within a broader landscape that includes finance, manufacturing, and education, while highlighting the unique clinical, regulatory, and ethical stakes of medical contexts. Applications in diagnostics (e.g., FDA-approved IDx-DR), therapeutics (closed-loop insulin delivery), clinical decision support, and workflow optimization demonstrate the transformative potential of autonomous agents to improve patient care, personalization, and operational efficiency. However, pressing challenges persist in accountability, transparency, bias, safety, privacy, and regulatory governance. The review concludes that the promise of agentic AI in healthcare will only be realized through explainable, ethically aligned, and human-centered systems supported by robust evaluation frameworks and multidisciplinary oversight.

Keywords: *Agentic Artificial Intelligence; Autonomous agents; Healthcare applications; Clinical decision support; Explainable AI; Ethics and governance.*

AI for Workforce Reskilling: Building Human Capital for a Sustainable Future

Dr.R.Vijayashree

Assistant Professor,

Dept of CT& DS

Sri Krishna Arts and Science college

vijayashree.nehru@gmail.com

Mr. Nagulan

Student,

Dept of CT& DS

Sri Krishna Arts and Science college.

nagulj4@gmail.com

Abstract

The rapid growth of Artificial Intelligence (AI) is reshaping industries and redefining the global workforce. While AI automates repetitive tasks and improves efficiency, it also disrupts traditional jobs, creating an urgent need for large-scale reskilling and up skilling. This paper investigates how AI can be leveraged as a tool for workforce reskilling, enabling workers to transition into emerging roles within the digital and green economies. The study explores AI's role in personalized learning, skill assessment, and lifelong education, while addressing challenges such as the digital divide, data privacy, and governance. Using a qualitative research design supported by secondary data analysis, the paper synthesizes existing reports, case studies, and global initiatives. Findings indicate that AI-powered platforms enhance scalability, inclusivity, and adaptability of training but face significant cost, ethical, and adoption challenges. The paper concludes with practical recommendations for policymakers, educators, and industries to ensure AI-driven reskilling supports sustainable economic growth and inclusive human development.

Keywords: Artificial Intelligence, Workforce Reskilling, Human Capital Development, Future of Work, Sustainable Employment, Digital Transformation, Lifelong Learning.

A Survey on Intelligent E-Commerce Platforms: From Lexical Search to Semantic Retrieval

Ms. Jeevitha. K, Ms. M. Shalini, Ms. R. Priyadharshini, Ms. K. Nithiya Gokula Vasaki, Ms. R. Vinodhini

Student, Department of Artificial Intelligence and Data Science, Sri Manakula Vinayagar Engineering College.

Abstract

The recent growth of e-commerce has brought a new focus on product search, personalized interaction, and user engagement, which all depend on effective and efficient product search. More traditional methods of user search, generally based on inverted index and lexical matching algorithms, often have shortcomings as users move forward in the e-commerce process. These limitations can include lexical mismatches, constraints and restrictions of semantics, user experience, preference modeling, and constraints to personalized configurations. However, with the growth and advancement of deep learning (DL) and natural language processing (NLP), the landscape for search has changed to include semantic search, neural embedding's for product representations, structured product representations, and evolving personalized recommendation models. This survey reviewed many of the state-of-the-art work, including Walmart's hybrid retrieval system called Deep Personalized Semantic Retrieval (DPSR), and Structured Matching Modules (SMM). These are examples of the promise discoveries from hybrid neural-lexical retrieval and structured semantic modeling to improve accuracy and scalability. In addition, other new developments in transformer-based architecture such as GPT, BART, T5 and, Pegasus have ability to accomplish new advancements in semantic retrieval, contextual query understanding, and summarization of product content. Moreover, models like PAAG extend the idea of conversational commerce to go beyond intelligent FAQ chatbots. And, new approaches to address emotional preference modeling and sentiment aware recommendations can broaden personalization to implicit user experience modeling.

Keyword: E-Commerce Platforms, Deep Learning, Lexical Search Semantic Retrieval, Chatbots

Sixth-Sense Gestures in Touchless Healthcare

Ms. R. Allirani

Assistant Professor,
alliranir@annaadarsh.edu.in

Ms. P. Jaishree

Student,
jaishree2006happy@gmail.com

Ms. S.Priyatarsini

Student,

Department of Computer Science,
Anna Adarsh College for Women
(Autonomous),
priyatarsinisenithil@gamil.com

Abstract

Modern healthcare requires interaction systems that are efficient, hygienic, and intuitive. Conventional input methods such as keyboards, mice, and touchscreens are unsuitable in sterile environments, as they raise contamination risks and disrupt workflows. This paper presents Sixth-Sense Gestures in Touchless Healthcare, a framework enabling seamless, contact-free interaction between medical professionals and digital systems. The system integrates 2D gestures (swipes, pointing, selections) and 3D gestures (mid-air motions, spatial navigation) into an intelligent multi-modal platform. Through computer vision, depth sensing, and machine learning, it achieves accurate, real-time recognition of hand movements, supporting natural human-machine interaction. Key applications include gesture-controlled surgical navigation, where clinicians can manipulate medical images without breaking sterility; hands-free access to electronic patient records; gesture-based diagnostic imaging analysis; and rehabilitation monitoring through motion tracking. These applications demonstrate how touchless interfaces can enhance precision, reduce workflow interruptions, and maintain strict hygiene standards in critical care settings. The framework suits diverse healthcare settings, from operating rooms to telemedicine, by prioritizing efficiency, accessibility, adaptability, and sterility. Findings suggest sixth-sense gesture technology could revolutionize medical interaction, offering a safer, smarter, more intelligent, innovative, and patient-centered approach to digital healthcare. The study explores integrating sixth-sense gestures with augmented reality (AR), virtual reality (VR), and AI-driven decision support to extend applications into remote diagnostics, training, and telesurgery. Future work will refine recognition, improve scalability, and validate real-world usability. Bridging human intuition and digital intelligence, sixth-sense gestures represent a transformative step toward next-generation touchless healthcare.

Keywords: *Sixth-Sense Gestures, Touchless Healthcare, Multi-Modal Interaction, Human-Computer Interaction, Computer Vision, Machine Learning, Telemedicine, Surgical Navigation.*

Analyzing Learning Behaviors with GRU Networks and HMM Models: A Deep Learning based Approach to Personalized Feedback in Education

Dr.G. Kanimozhi

Assistant Professor

SRM Institute of Science and Technology, Chennai

L. Bhuvaneswari

Assistant Professor,

Guru Nanak College, Chennai.

Abstract

Understanding student learning behaviors is essential for delivering effective personalized education. Personalized learning requires accurate identification of student behaviors and timely feedback to address individual needs. This paper presents a hybrid framework that integrates Gated Recurrent Unit (GRU) networks with Hidden Markov Models (HMM) to analyze sequential learning data and provide adaptive feedback. GRU networks are employed to capture temporal dependencies in student interactions, while HMM models the hidden learning states and their transitions. The proposed GRU-HMM approach enables accurate prediction of student performance and uncovers latent cognitive patterns that influence learning outcomes. By combining the strengths of deep learning and probabilistic modeling, the proposed approach enables dynamic feedback generation, supports adaptive learning pathways, and enhances student engagement. Experiments conducted on educational datasets demonstrate improved prediction accuracy and interpretability compared to traditional sequence modeling techniques. Moreover, the system generates personalized feedback that supports instructors in tailoring interventions and assists learners in addressing specific challenges. This paper underscores the potential of combining deep learning and probabilistic modeling to advance adaptive, student-centered education.

Keywords: *Learning behaviors, GRU networks, Hidden Markov Model, personalized feedback, educational deep learning, adaptive learning.*

Combating Deepfakes : A Hybrid Framework for Face Authenticity Detection and User Attention Monitoring

Barath Kumar E, Dharshish K G, Jai Surya S
UG Students

Ms. S J Kavitha
Assistant Professor
Department of Computer Science and Engineering,
Vels Institute of Science Technology and Advanced Studies, Chennai

Abstract

The increasing realism of AI-generated human faces presents a serious challenge to identity verification and biometric security. As synthetic images become harder to tell apart from real ones, traditional face recognition methods are vulnerable to spoofing, impersonation, and deepfake attacks. To address this, we developed a web-based system combining real-time head pose estimation, attention tracking, and face authenticity detection. Using MediaPipe to precisely locate facial landmarks, the system estimates head position and gaze direction to monitor user attention in live video. A custom-trained classifier then analyzes facial features to decide if the face is genuine or AI-created. This combined approach strengthens security by using both behavioral and authenticity checks to guard against synthetic media threats. The system works with real-time webcam feeds and offline image uploads, making it suitable for applications like remote proctoring, secure logins, financial onboarding, and surveillance. Built to run efficiently in a web browser, it requires no extra hardware or software, making deployment simple and accessible. By tracking attention and verifying authenticity, the system helps build trust in situations where confirming a user identity and presence is crucial. As deepfake technology advances, this solution provides a practical and scalable way to protect biometric systems and maintain digital trust.

Keywords - Deepfake Detection, Face Authenticity, Attention Tracking, Mediapipe, Biometric Security, Real-Time Verification, Remote Proctoring

Big Data Analytics for Disaster Prediction and Climate Adaptation

Ms. N. Amrath Fathima Shifa,
Master of Computer Applications (II Year)
DDGD Vaishnav College, Chennai, Tamilnadu
Shifanazeer04@gmail.com

Ms. A. A. Hansikapriya
Master of Computer Applications (II Year)
DDGD Vaishnav College, Chennai, Tamilnadu
hnsk.aa.05@gmail.com

Abstract

The frequency and severity of natural disasters have increased due to the increasing effects of changes in the climate, posing major risks to ecosystems, infrastructure, and human lives. The volume and complexity of environmental data produced by satellites, sensors, and social media platforms are frequently too much for traditional forecasting methods to handle. By combining and evaluating various information in real time, big data analytics offers a sophisticated way to get beyond these restrictions. Large-scale data streams can be converted into early warning signals for natural disasters including hurricanes, wildfires, droughts, and floods using machine learning, geospatial modeling, and predictive algorithms. Beyond making predictions right away, big data is essential for climate adaptation because it helps develop robust infrastructure, allocate resources, and improve community readiness. This study demonstrates how Big Data Analytics may increase the precision of disaster predictions and the effectiveness of adaption plans. The work helps create more resilient and sustainable communities that can lessen their vulnerability to natural calamities and mitigate the effects of climate change by tying large-scale data processing to practical decision-making.

Keywords: *Big Data Analytics, Disaster Prediction, Climate Change, Climate Adaptation, Machine Learning, Geospatial Modeling, Predictive Algorithms, Early Warning Systems, Resilient Infrastructure, Community Preparedness.*

Reviving Indian Agriculture : Sustainability as the Key to Food Security

.Gunasekaran

Assistant Professor in Statistics,
Sona college of Arts and Science, Salem

P.Anandhi

Assistant Professor in Statistics,
Sona college of Arts and Science, Salem

R.Imran

Assistant Professor in Mathematics,
Sona college of Arts and Science, Salem

Abstract

This study explores the critical role of sustainable agriculture in ensuring global food security amid growing environmental, economic, and social challenges. Sustainable agriculture practices aim to maintain productivity while conserving natural resources and promoting ecological balance. By integrating principles such as soil conservation, water management, crop diversification, and reduced chemical inputs, these practices enhance the resilience of food systems to climate change and resource depletion. Food security, defined by the availability, access, utilization, and stability of nutritious food, is directly influenced by the sustainability of agricultural systems. The study examines the interdependence between sustainable farming methods and food security outcomes, identifies key challenges including climate variability, soil degradation, and economic constraints, and highlights strategic interventions to promote sustainable agriculture. Findings suggest that adopting sustainable agricultural practices is essential for achieving long-term food security, supporting rural livelihoods, and preserving environmental health. The study concludes with recommendations for policy frameworks, technological innovations, and community-based approaches to foster sustainable agricultural development globally.

Keywords: Food Security, Sustainable Agriculture, Indian Agriculture

Agri Assist: AI-Driven Farm Solutions with Farmer Marketplace System

Nikitha Chidambaram

BSc Data Science

Kumaraguru College of Liberal Arts and Science

chidambaramnikitha@gmail.com

Dr. M. Emimanimcy

Assistant Professor,

Kumaraguru College of Liberal Arts and Science

Emimanimcy.m.mat@gmail.com

Abstract

Agriculture plays an important role in the overall growth of our Indian economy, yet farmers face hardships like the unpredictable prices, high transportation costs, and being tricked by the vendors, which disturb their cost of living, as it does not bring out decent profits. To address these challenges, AGRI-ASSIST: AI-DRIVEN FARM SOLUTIONS WITH FARMER MARKETPLACE SYSTEM is innovated as an intelligent decision-support tool that utilizes artificial intelligence, machine learning, big data analysis, and image processing. Using various dashboards and tools like Excel and Python for data preprocessing and Google Colab to perform Exploratory Data Analysis (EDA), this project also uses the Random Forest Model to predict the price trends and to identify nearby markets that are profitable, and the Power BI dashboard is being used to provide better data visualization to make better data-driven decisions. Alongside, this project also integrates a farmer marketplace system that enables digital connectivity among farmers and stakeholders, which also contributes to a smart ecosystem. Overall, this project demonstrates how emerging technologies can be integrated with agriculture to create more efficient, sustainable, and farmer-centric solutions that boost productivity and decision-making.

Keywords: *Market optimization, Random Forest classifier, Marketplace system, Smart Ecosystem, Digital connectivity*

AI Based Detection of Voice Deepfakes using Spectrogram Analysis and Deep Learning Models

Mrs.A. Jaya Priya, Ms. S. Nithya Sri, Mr. B. Pandeewaran

Assistant Professor, Department of AI&DS Meenakshi Sundararajan Engineering College, Chennai, India.

Mr.P. Kumaraguru

Student, Department of AI & DS, Meenakshi Sundararajan Engineering College, Chennai

Abstract

The rise of generative artificial intelligence has led to the creation of highly convincing synthetic voices, widely referred to as audio deep fakes. While deep fake research in image and video domains has advanced rapidly, the study of voice deep fakes is still emerging despite their growing use in fraud, impersonation, and misinformation campaigns. This paper investigates the potential of AI to detect fake voices created by AI itself through the development of a deep learning-based detection framework. The framework employs spectrogram analysis combined with hybrid deep learning models, including Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal sequence learning. An ensemble classifier is further integrated to enhance generalization across diverse types of synthetic voices. Experiments conducted using benchmark datasets such as ASV spoof demonstrate the effectiveness of this approach, achieving promising accuracy in distinguishing real from fake speech. The expected outcome is a scalable and real-time detection system that can be applied in high-risk sectors such as banking, telecommunications, and social media verification. This work contributes to the broader field of AI and cyber security by addressing one of the most urgent challenges in next-generation computing.

Keywords— *Voice Deepfakes, Spectrogram Analysis, CNN, LSTM, Cyber Security, Audio Forensics*

Smart Urban Planting: A Fusion of IoT Embedded Systems and AI-Based Environmental Monitoring

Suwastika V

Student

Department of Data Science
Kumaraguru College of Technology
suwastika2007@gmail.com

Dr. R. K. Kavitha

Assistant Professor

Department of Computer Applications
Kumaraguru College of Technology
kavitha.rk.mca@kct.ac.in

Abstract

The swift expansion of urbanization has significant privileged access to the greenery spaces, which creates challenges for plant cultivation in smaller spaces. This study presents an AI- and IoT-integrated system combining environmental sensing, computer vision, and predictive analysis to improve the success rate of plant survival in urban households. The primary focus areas are to formulate IoT-embedded planters that can self-regulate irrigation and nutrient delivery, to use a mobile-based vision for speculating the spatial features like sunlight, area, exposure, shading and then infuse the weather APIs to determine the accurate weather conditions. The system architecture constitutes of the IOT nodes like (moisture, pH, temperature) in smart planters along with a mobile app that applies deep learning models for image-based environmental analysis. Weather APIs provide climate insights, while machine learning algorithms combine sensor and external data to create responsive care strategies. A prototype was tested using controlled datasets and compared with standard horticultural practices. Experimental results show that the proposed system reduces manual work by 65%, improves water-use efficiency by 40%, and achieves 92% accuracy in environmental detection with mobile vision models. Furthermore, it assists the urban dweller to plant efficiently, saves resources, and encourages people to enrich the green landscape, providing mutual benefit between the ecosystem and communities. By integrating AI, IOT, and real-time analysis, this research fosters the concept self-sustaining garden plants and supports global sustainability goals while redefining plant care as an intelligent part of smart living environments.

Keywords: *Internet of Things (IoT), Artificial Intelligence (AI), Computer Vision, Weather API Integration, Autonomous Plant Care.*

Deep Training-Based Automatic Facial Expression Recognition System

Mrs. V. Priyanandhini, Mr. M. Ramesh

Assistant Professor,
Department of Computer Science with Artificial Intelligence,
SRM Arts and Science College,
Kattankulathur.

Dr. R. Rathidevi

Assistant Professor,
Department of Computer Applications and Technology,
SRM Arts and Science College,
Kattankulathur.

Abstract

The more we move towards the digital age, the more important human-computer interaction becomes. One of the most important elements of nonverbal communication and a prerequisite for human-computer interaction is facial expressions. Facial expressions are crucial in fostering social interactions as they convey emotions and intentions. Recognition of facial expressions is gaining popularity in domains such as cognitive computing, security, and health due to its numerous applications, such as online conversations, medical monitoring, security surveillance, and detecting driver fatigue. Machines struggle with reliable facial recognition, whereas humans excel at interpreting emotions. Complex algorithms, such as face recognition, feature extraction, and emotion recognition, are used in facial expression identification to identify seven common expressions. Convolutional Neural Networks (CNNs) perform effectively in controlled environments; however, they encounter difficulties in real-time scenarios due to occlusion, variations in posture, lighting inconsistencies, and fluctuations in image quality. The goal of this study is to make it easier to tell what emotions are on a person's face using Convolutional neural networks with deep con (DCNNs). Using MATLAB, an automatically generated Facial Expression Recognition (AFER) system is made that can reliably tell seven different facial expressions apart. You can make hyperparameters work better on benchmark datasets like CK+ and JAFFE by testing using different frameworks and architectures. The paper looks at the pros and cons of using computational neural network models in this field.

Keywords: *databases, feature extraction, machine learning, convolutional neural networks, and recognising faces.*

Gastric Cancer Detection using Intelligent Decision-Making Method for GC Screening

Dr. R. Rathidevi

Assistant Professor,
Department of Computer Applications and Technology,
SRM Arts and Science College,
Kattankulathur

Mrs.V. Priyanandhini

Assistant Professor
Department of Computer Science with AI
SRM Arts and Science College
Kattankulathur

Mr. M. Ramesh

Assistant Professor
Department of Computer Science with AI Arts and Science College
SRM Arts and Science College,
Kattankulathur

Abstract

Gastroscopy is a commonly used technique for early screening and detection of gastric cancer (GC) as well as for identifying gastric abnormalities. However, the gastroscopy specialist's medical expertise determines how well traditional GC screening techniques work. Inexperience and ignorance can result in incorrect diagnoses and mistreatment, particularly in small institutions. Studies on data-driven computer-aided diagnosis methods have significantly increased recently. In this paper, we provide a new intelligent GC screening approach (ID-GCS), a data-driven decision-making system based on multimodal semantic fusion. In order to increase the interpretability of gastroscopy findings, ID-GCS uses a hybrid attention method to extract textual semantics from multimodal gastroscopy reports and integrates the semantics of textual and image gastroscopy reports through semantic fusion. The suggested cancer cell detector's Faster R-CNN architecture. The RPN uses sliding windows (SW) to connect the final convolution feature map of the Convolution Network, as indicated by the yellow arrows. The proposal regions' bounding boxes (Bbox) and classification scores are produced by the two RPN output layers. Red arrows indicate the Fast R-CNN Network's distinctive feature.

Keywords: *Gastric Cancer, Intelligent Decision-Making Method for Gastric Cancer, Semantic Fusion.*

Towards Trustworthy AI in Agriculture: An XAI-Driven Ensemble Model for Sugarcane Leaf Disease Detection

Ms. K.P. Sreelakshmi, Research Scholar

Dr. R. Parameswari, Professor & Head

Department of Computer Science and Information Technology
Vels Institute of Science Technology and Advanced Studies

Abstract

Sugarcane farming, like much of agriculture, is highly vulnerable to diseases. If these diseases are detected only at later stages, they can cause major yield losses. Among the most serious threats are Bacterial Blight and Red Rot, which spread quickly and damage crops heavily. In recent years, deep learning has improved disease detection, but many of these models work like “black boxes.” Because their decisions are not clear, farmers and agronomists often hesitate to trust them. In this paper, we introduce an Explainable Ensemble Framework for Sugarcane Leaf Disease Prediction. The core of the framework is a Tversky Index-based Projected Graph Neural Network (TIPGNN). Before this, an Adaptive Dual-Layer (ADL) method prepares the data in two stages, helping the model capture fine details of leaf lesions that older methods often miss. To make feature extraction stronger, we use a Gaussian Distributed Convolutional Neural Network (GDCNN), which is good at spotting textures and color variations that standard CNNs usually miss. The real difference in our framework is the explainability Layer. It brings together Grad-CAM, which visually highlights the diseased regions of a leaf; SHAP values, which explain the importance of different features; and a Graph Attention Network (GAT), which handles graph-structured data more effectively. With these combined, the system becomes not just accurate but also transparent. We evaluated the framework using a sugarcane disease dataset from China, and the outcomes were very encouraging. It performed better than traditional CNN and GNN models in terms of accuracy, precision, and recall, while also giving clear, easy-to-understand explanations. This makes it possible for experts to verify predictions in real time, catch mistakes, and gain more confidence in the system’s decisions.. The proposed framework thus fills the chasm between high performance machinery and credibility, creating a working model suitable to new applications of intelligent agriculture.

Keywords: *Agriculture, Sugarcane leaf disease prediction, Explainable Artificial Intelligence (XAI), Graph Neural Network (GNN), Convolutional Neural Network (CNN), Ensemble Deep Learning, Grad-CAM, SHAP, Graph Attention.*

A Survey of Big Data Analytics in Healthcare

Mr. M. Ramesh

Assistant Professor

Department of Computer Science with AI
SRM Arts and Science College
Kattankulathur,
Chennai.

Dr. R. Rathidevi

Assistant Professor

Department of Computer Applications and Technology
SRM Arts and Science College
Kattankulathur.
Chennai.

Mrs. V. Priyanandhini

Assistant Professor

Department of Computer Science with AI
SRM Arts and Science College
Kattankulathur
Chennai.

Abstract

This article describes how we may extract additional value from data created by healthcare. These agencies generate vast amounts of diverse data. However, without suitable data analytics methodologies, this data become meaningless. Big Data Analytics utilizing Hadoop is effective at performing relevant real-time analysis on massive amounts of data and predicting emergency scenarios before they occur.

Keywords: *Big Data, Hadoop, Healthcare, MapReduce*

Onboard Edge Big Data Analytics for Autonomous Deep-Space Exploration

Ms. E.G. Krishnaveni

Master of Computer Applications
krishna456veni21@gmail.com

Ms. S. Harshini

Master of Computer Applications
harshinishan12@gmail.com

Ms. P. Kavitha

Master of Computer Applications
Dwaraka Doss Goverdhan
Doss Vaishnav College
Chennai, Tamil Nadu, India
kavithasachin26@gmail.com

Abstract

Modern space exploration is generating enormous volumes of data – sometimes kept in petabytes – from satellites, rovers, and telescopes. Managing this, big data analytics has developed into an essential basis helping scientists convert unprocessed spatial data into meaningful knowledge. Researchers have already made tremendous strides with methods like machine learning models and dispersed systems such Hadoop and Apache Spark to create detailed planetary mapping. Predictive maintaining spacecraft, climate research, and asteroid discovery. Still, one major problem remains: still much of the data processing takes place on Earth. This dependence results in high latency and bandwidth issues especially in deep-space missions when signal delays might vary from minutes to hours. Among the fields affected by such delays are real-time activities including rover navigation, early anomaly detection, and astronaut health monitoring. To cross this gap, our study suggests including Edge Big Data Analytics directly in rovers and spacecraft. Onboard AI accelerators and dispersed mini-clusters let data be processed locally in real time. This structure makes everything possible, among which information has to be sent back to Earth. Rather than sending everything, only the most essential insights are disclosed, hence conserving bandwidth and speed up emergency reactions. This approach prepares for future deep-space studies where human control would be limited as well as improves mission safety and autonomy.

Keywords: *Space Data Analytics, Edge Computing in Space, Machine Learning Applications, Real-Time Processing, Autonomous Space Missions, Deep-Space Communication, AI-Powered Rovers.*

Review on Big Data Analytics

Ms. S. Aakhash

Student,

Department of Computer Application and Technology
SRM Arts and Science College,
Chennai

Dr. R. Rathidevi

Assistant Professor,

Department of Computer Applications and Technology,
SRM Arts and Science College, Chennai

Abstract

In the age of information, decision-makers now have access to huge quantities of data. Datasets that are challenging to manage with conventional tools and methods due to their size, variety, and velocity are referred to as big data. In order to manage and extract value and knowledge from big datasets, solutions must be researched and offered due to their rapid growth. Additionally, decision-makers must be able to extract insightful information from such diverse and quickly evolving data, which includes any kind from everyday transactions to consumer interactions and social network data. Big data analytics, or the application of advanced analytical techniques to large data, can offer this kind of benefit. This study aims to The purpose of this paper is to examine some of the various analytics techniques and resources that may be used with big data, along with the potential that come with using big data analytics in different areas of decision-making.

Keywords: Big Data, Datasets, Social Network, Data Analytics

Animated AI Teacher: An Intelligent Avatar-Based Pedagogical Framework for Remote and Hybrid Learning

Mrs T. Dhivya

Ms.V.S. Veebika

Ms.B. Tanvi

Assistant Professor

Department of AI&DS Meenakshi Sundararajan Engineering College,
Chennai.

Mr.Y. Saidharan

Student,

Department of AI & DS, Meenakshi Sundararajan Engineering College,
Chennai

Abstract

The COVID-19 pandemic exposed the limitations of conventional e-learning platforms such as Zoom, Google Meet, and mainstream EdTech applications, which primarily replicate traditional classrooms online without addressing engagement, personalization, and accountability. This paper proposes the design of an Animated AI Teacher, a next-generation intelligent system where an AI-driven avatar functions as a fully autonomous digital teacher capable of delivering lessons, interacting with students, assigning and tracking homework, and adapting to individual learning needs. The system integrates Artificial Intelligence (AI), Natural Language Processing (NLP), speech synthesis, computer vision, and avatar animation to create an interactive and human-like teaching experience. It provides AI-driven instructional delivery using a 2D/3D animated avatar on a digital board, real time conversational interactivity, automated homework assignment and progress tracking, adaptive pedagogy powered by knowledge-tracing models, and gamification features such as badges and leaderboards. Unlike existing EdTech solutions that provide static video lessons or text-based tutoring, the proposed framework offers a holistic, scalable, and cost-effective platform. The expected outcome is a transformative remote learning ecosystem that addresses teacher shortages, enhances engagement, and ensures accountability in learning outcomes. This innovation aligns with the Sustainable Development Goals (SDGs) and India's National Education Policy (NEP 2020), offering a pathway towards accessible, inclusive, and experiential education in the post-pandemic era.

Keywords—Artificial Intelligence in Education, Animated Teacher Avatar, Remote Learning, Intelligent Tutoring Systems, Gamification, Adaptive Pedagogy

Unveiling Dataset Bias: A Logistic Regression Framework with Fairness Metrics and Statistical Validation

Ms. S. Bhoomika

UG Scholar

Department of Data Science

Kumaraguru College of Liberal Arts and Science, Coimbatore.

bhoomika24bds@kclas.ac.in

Dr. R.K.Kavitha

Assistant Professor,

Department of Computer Science

Kumaraguru College of Liberal Arts and Science, Coimbatore.

kavitha.rk.mca@kct.ac.in

Abstract

Machine learning and training models are being deployed in various sectors such as banking, human resource management, medicine etc. This adoption has significantly reduced the time and human error with high accuracy. The success of a machine learning model lies in the dataset and how well the model understands and recognises intricate patterns and identify dependencies without overfitting to noise. If the training data does not represent underlying population or if any systematic correlation exist between sensitive attributes and target variables, the dataset is said to be biased. A model trained on biased dataset produces outcomes which are discriminatory in nature and disproportionately affects the smaller group in the dataset. This research paper addresses the issue of dataset induced bias by developing a framework, which is used for fairness analysis with solid statistical and fairness metrics evidence. The Logistic Regression model is employed to test the bias due to its interpretability and mathematical traceability allowing clear inspection of boundaries. Using both the predictive performance metrics - Accuracy, Precision, Recall, and F1-score and fairness metrics such as Demographic Parity Difference and Equalized Odds Difference. With statistical integration of concepts like ANOVA, Chi-square and hypothesis testing variance across subgroups are analysed, providing evidence of skewness and imbalance in the dataset. An interactive web application is implemented using Streamlit which allows users to upload datasets, select target variables and visualize fairness metrics alongside predictive performance. The platform performs preprocessing through scaling and encoding, and highlights trade-offs introduced by varying Prediction Probability Threshold. This work contributes to the fields of responsible AI and data science by combining ML evaluation with statistical inference, offering a practical and transparent tool for auditing bias in real-world datasets.

Keywords: *Bias, Machine learning, Logistic Regression, fairness analysis, fairness metrics, inferential statistics, data science, dataset.*

Reimagining Virtual Recruitment through Interactive Augmented Reality

Mr. Pradeepa Suryakumar

Assistant Professor

Department of Computer Applications and Technology

SRM Arts and Science College

Kattankulathur, Chengalpattu India

prathee272@gmail.com

Ms. M. Jasmine

Assistant Professor

Department of Computer Science

with Artificial Intelligence

Anna Adarsh College for Women

Chennai, India.

mjasminejas23@gmail.com

Abstract

The increasing use of AI in online interviews process had opened gates to innovative foul plays which had eventually led Google to ban virtual/online interviews in recent times. The need for trustworthy and transparent hiring processes became paramount to maintain recruitment integrity and candidate confidence. By reintroducing in-person, it increases the time frame of the interview process and will create more hustle for the candidates. This paper aims to examine the systematic and technical background of the AI foul plays in an interview process and explore the alternate recruitment process incorporating interactive Augmented Reality (AR) ensuring transparency and equity. To address the AI-assisted cheating and inequality issues in the virtual recruitment process and incorporate interactive Augmented Reality based tool in recruitment process. Conduct risk assessments, reviewed feedback, security vulnerabilities and challenges in in-person interview models. Examine and evaluate the potential and challenges of available integrated AR technologies. Qualitative and quantitative analyses performed to measure the effectiveness and fairness of the in-person and AR based interview formats and process.

Keywords: Augmented Reality (AR); virtual interview; recruitment process; AI foul play

A Comprehensive Exploration of Agro-Based Pest Identification Using Image Processing and Deep Learning

Mr. S. Saravanan,

Student

Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies, Chennai
Mail ID: sarisrigym1976@gmail.com

Abstract

Effective pest control is crucial for ensuring healthy crop production and sustainable agriculture. This research presents an ontology-based knowledge mining model for pest control in plants, integrating image processing techniques and deep learning. The model leverages image-based pest identification using Convolutional Neural Networks (CNNs), enabling the accurate detection and classification of pest species from plant images. The identified pests are mapped to a domain-specific ontology, which encodes expert agricultural knowledge about pest behaviour, plant-pest relationships, and appropriate control measures. This research is an effective exploration of pest identification using different methods such as support vector machine, random forest, K-Nearest Neighbours, Gradient Boosting Machines, ResNet, VGGNet, YOLO, and faster R-CNN. This paper highlights the various methods used for data pre-processing, image acquisition, and feature extraction. Additionally, this paper explores ontology-based knowledge representation for a pest detection model using classes, objects, and relationships between them in the OWL language. Finally, this study also discusses about decision support system for pest management, treatments recommended, and new knowledge inference on pest management in plants. The ontology facilitates semantic reasoning, enabling intelligent decision-making and the recommendation of targeted pest management strategies. By combining visual intelligence with structured domain knowledge, the proposed system enhances the precision, scalability, and interpretability of automating pest control. Experimental results demonstrate high accuracy in pest detection and effective integration with ontology-based recommendations, offering a powerful tool for precision agriculture and smart farming applications.

Keywords: Agriculture, Ontology, Deep Learning, Image Processing, Neural Networks

Integrating Pierce Criterion for Outlier Detection in Extreme Learning Machine Classification

Ms. S.V. Arulvani
Ph.D Research Scholar
arulphdcs@gmail.com

Dr. C. Jayanthi
Associate Professor
PG & Research
Department of Computer Science,
Government Arts College
Affiliated to Bharathidasan University,
Karur-05
jayanthibabu117@gmail.com

Abstract

Outlier detection plays a crucial role in the performance and reliability of machine learning models, especially in high-dimensional datasets. In this study, we propose an innovative approach for outlier detection in Extreme Learning Machine (ELM) classification using the Peirce criterion. The Peirce criterion, traditionally used in statistical analysis, is applied here to effectively identify and exclude outliers, thereby enhancing the classification performance of ELM models. The ELM, known for its rapid training speed and effectiveness in handling large-scale problems, is combined with the Peirce criterion to filter noisy or anomalous data points that could negatively impact the model's accuracy and generalization ability. We demonstrate the effectiveness of this hybrid approach through extensive experimentation on benchmark datasets, showing significant improvements in classification performance, robustness, and stability. This work presents a novel synergy between classical statistical methods and modern machine learning techniques, offering a promising direction for further advancements in reliable data-driven decision-making systems.

Keywords: *Outlier Detection, Peirce Criterion, Extreme Learning Machine (ELM), Classification, Noise Filtering, Data Quality, Machine Learning, Model Performance, Robustness, Anomalous Data, Statistical Methods, High-dimensional Datasets.*

Pawcare: A MERN Stack-Based Dog Health & Wellness Tracker

Madhumitha S
Jeevitha R
Bandi Yamini
UG student

Ms. Parvathavarthini K
Assistant Professor,
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING,
VISTAS.

Abstract

In recent years, pet ownership has significantly increased in Tamil Nadu and across India, creating a strong demand for structured and reliable healthcare solutions for dogs. Many pet owners face real-time, challenges such as forgetting vaccination schedules, mismanaging diet, delayed recognition of health symptoms, lack of breed-specific guidance, and difficulty in storing veterinary records. Most existing applications available in the market are either general in nature or focus on only a few aspects such as vaccination or grooming, leaving a large gap in personalized and breed-based wellness management. This project introduces a comprehensive Dog Health & Wellness Tracker application developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) to provide a robust, scalable, and user- friendly full-stack solution. The application provides five core features: (i) Vaccination Tracking, with automated reminders to prevent missed doses, (ii) Growth and Diet Monitoring, where nutrition and feeding suggestions are personalized based on breed requirements, (iii) Disease Prediction, implemented using a rule-based system that maps symptoms and health records to common canine illnesses, (iv) Grooming and Hygiene Reminders, to promote preventive healthcare, and (v) Breed- Specific Recommendations, where each selected breed receives wellness guidelines suited to its nature, lifestyle, and risks. Unlike existing systems, the proposed tracker also emphasizes native Tamil Nadu breeds along with popular foreign breeds, making the solution both globally relevant and regionally valuable. A personalized health and wellness model is created for five popular breeds: Labrador retriever, German shepherd, Pomeranian, Rajapalayam, and Chippiparai. The system empowers pet owners with organized health records, automated alerts, predictive disease insights, and breed-wise guidance, ultimately bridging the gap between veterinary consultation and everyday pet management.

Keywords: *MERN Stack, Dog Health Monitoring, Pet Wellness, Breed-Specific Care, Vaccination Tracking, Disease Prediction, MongoDB, Full-Stack Web Application, Native Breeds.*

A Conceptual Framework for a Mental Wellness Analyzer Using Stress and Lifestyle Patterns

Ms. G. A. Harshavartini
UG Scholar,
Department of Data Science
harshavartini.24bds@kclas.ac.in

Dr.R.K.Kavitha
Assistant Professor
Department of Computer Applications
kavitha.rk.mca@kct.ac.in
Kumaraguru College of Liberal Arts and Science, Coimbatore.

Abstract

The escalating importance of mental health on a global scale necessitates a comprehensive understanding of its determinants, particularly the influence of stress and lifestyle factors on overall well-being. Current assessment methodologies often fall short, primarily focusing on clinical diagnoses or isolated lifestyle metrics. This limitation underscores the imperative for a holistic, accessible, and adaptive framework capable of proactively evaluating mental wellness within the context of daily life. This document outlines a conceptual framework for a Mental Wellness Analyzer. This system integrates stress indicators with lifestyle patterns to provide personalized insights. The analyzer considers parameters including sleep quality, work-life balance, dietary habits, exercise routines, screen time, and social interactions, in conjunction with stress-related inputs such as workload intensity and mood variations. Unlike existing methodologies, this framework incorporates a dynamic feedback loop, enabling continuous self-assessment to adapt scoring and recommendations over time. This adaptive design ensures that users receive guidance that evolves in response to their changing lifestyle and stress levels. The framework's output comprises a personalized wellness score, risk level categorization (healthy, moderate risk, high risk), and individualized preventive recommendations. Its innovative aspect is its multi-factor, adaptive methodology, prioritizing proactive care and early detection over reactive diagnosis. While this study concentrates on conceptual design rather than empirical validation, it establishes a basis for the future development of mobile and web-based applications. By providing accessible, evolving, and personalized insights, the proposed Mental Wellness Analyzer seeks to facilitate access to mental health support.

Keyword: *Mental Wellness, Stress Analysis, Lifestyle Patterns, Wellness Monitoring, Survey-Based Analysis*

Quick Cart: E-Commerce Platform Using MERN Stack

Mr. S. Deepak,
Irshadh Moulana M.M,
UG Students,

Ms. K.Parvathavarthini
Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING,
VISTAS.

Abstract

E-commerce has become one of the most important platforms for trade and business in today's fast-paced digital world. The rapid growth of online shopping has changed consumer behavior and forced companies to adopt new, user-friendly online solutions. Traditional retail models have issues like limited customer reach, high operating costs, and geographic limits. E-commerce platforms, however, overcome these challenges by offering a convenient, affordable, and accessible marketplace 24/7. Despite this growth, many small and medium-sized businesses struggle to set up strong e-commerce platforms due to high development costs and complex technical needs. This project aims to tackle these issues by using the MERN stack to create a full stack e-commerce website. MongoDB, Express.js, React.js, and Node.js are known for their flexibility, scalability, and cost-effectiveness.

Keywords: React.js, Node.js, Express.js, MongoDB, Web Application, Online Shopping, User Authentication, Scalable Architecture.

Jarvis AI Assistant: A Voice-Controlled Personal AI System

Mr. M.Murugan
Mr. Mohammad Ashik
Student

Ms. K. Parvathavarthini
Assistant Professor

Department of Computer Science and Engineering, VISTAS.

Abstract

Modern human–computer interaction increasingly relies on voice as a natural modality for user input. However, many commercially available voice assistants prioritize cloud-based reasoning and conversational capabilities while providing limited local control over system-level tasks. This paper presents JARVIS, a desktop voice assistant that integrates Google Gemini as the reasoning engine with local automation agents that directly control the host OS. JARVIS performs speech recognition, natural language understanding, TTS (text-to-speech), web and application control, automated typing and scrolling, media control, and persistent memory storage using MySQL. The assistant is designed with a hybrid approach: rule-based local handlers for deterministic system commands and Gemini-powered reasoning for open-ended queries. We describe architecture, implementation details, evaluation, privacy considerations, and future enhancements. Results from functional testing show reliable local automation and clear, concise AI answers; user trials indicate improved productivity for common desktop tasks. JARVIS demonstrates how combining cloud AI with local agents yields a powerful, personalized desktop assistant.

Keywords: Voice assistant, Gemini AI, speech recognition, local agents, PyQt5, Flask, MySQL, human–computer interaction

A Theoretical Framework of Impact of Blockchain Technology in Human Resource Management

Saran Raj S

Assistant Professor,
Department of Information Systems Management
Ramakrishna Mission Vivekananda College

Abstract

This paper proposes a theoretical framework to explore the impact of block chain technology on Human Resources Management (HRM). Blockchain technology has the potential to transform HRM by enhancing transparency, security, and efficiency in various HR processes. The framework outlines the key applications of block chain in HRM, including verification of credentials, secure storage of employee data, streamlined hiring processes, and automated payroll and benefits. This study examines the potential benefits and challenges of adopting block chain technology in HRM, providing a foundation for future research on the impact of block chain on HRM and its potential to revolutionize the field.

Keywords: Blockchain Technology, Human Resources Management, Transparency, Security, Efficiency, Theoretical Framework.

A Spatio-Temporal Generative Framework with Structural Learning for Advanced Prediction of Chronic Kidney Disease Progression

Ms. M. S. Jayaprabha
Research Scholar

Dr.V.Vishwa Priya
Assistant Professor
Department of Computer Science and Information Technology
Vels Institute of Science, Technology and Advanced Studies, Chennai.

Abstract

Chronic Kidney Disease (CKD) is a progressive global health crisis demanding early and accurate prediction. While machine learning models have shown promise, their evolution from standard classifiers to sophisticated deep learning architectures capable of handling the temporal and spatial intricacies of clinical data remains underexplored. This study presents a comparative analysis of two advanced deep learning frameworks for CKD prediction. The first model integrates an Automated Stochastic Deep Radial Basis network (SDRB) for feature extraction, a Residual Neural Network (ResNet) for classification, and a Bidirectional LSTM (BiLSTM) for temporal sequencing. The second, more advanced framework employs a Planar Flow-based Variation Auto encoder (PF-VAE) for generative feature extraction, a Structural-RNN to model spatial dependencies among clinical features, and a Convolutional LSTM (ConvLSTM) with spatial autocorrelation for spatio-temporal progression modeling. Both models were rigorously evaluated on the standard UCI CKD dataset under identical experimental conditions. The SDRB-ResNet-BiLSTM model achieved a strong accuracy of 92.75%, with precision, recall, and F1-scores of 93.10%, 91.82%, and 92.45% respectively. The PF-VAE-Structural RNN-ConvLSTM model demonstrated superior performance, achieving a state-of-the-art accuracy of 97.40%, with precision, recall, and F1-scores of 96.80%, 97.20%, and 97.00% respectively. The latter model showed a significant enhancement in capturing complex, non-linear feature relationships and their temporal evolution. The explicit modeling of spatial correlations among clinical variables through Structural-RNN and the generative feature learning capability of PF-VAE provide a significant performance advantage over powerful temporal-only models. This study confirms that the integration of spatio-temporal and generative architectures represents the next frontier in robust and accurate clinical prediction models. The PF-VAE-StructuralRNN-ConvLSTM framework emerges as a superior tool for early CKD diagnosis and progression forecasting.

Keywords: *Chronic Kidney Disease, Deep Learning, Predictive Modeling, Variational Autoencoder, Structural-RNN, Spatio-Temporal Analysis, Comparative Study*

HealthTech for Women: A Machine Learning Integrated Web Application for Early Diagnosis, Monitoring, and Lifestyle Management of PCOS

Ms. S. Sakthi
Ms. Udaiyar Shalini Mathiayan
Student

Ms. K. Parvathavarthini
Assistant Professor
Department of Computer Science and Engineering
Vels Institute of Science and Advanced Studies, Chennai.

Abstract

Women of reproductive age are frequently affected by PCOS, a common endocrine illness that can lead to irregular menstrual cycles, hormonal imbalances, and long-term problems include infertility and metabolic disorders. Monitoring and early diagnosis are critical to successful treatment. HealthTech for Women: A Machine Learning Integrated Web Application for Early Diagnosis, Monitoring, and Lifestyle Management of PCOS is presented in this work. It makes use of a curated dataset that includes clinical, demographic, hormonal, and lifestyle factors for prediction. Several machine learning methods are used by the system to improve prediction accuracy. These include linear models (Linear and Logistic Regression), tree-based models (Random Forest, Decision Tree, XGBoost, AdaBoost), and additional classifier (SVM, KNN, Naive Bayes, Stochastic Gradient Descent). The web app also has a period cycle tracker that lets users record their menstrual cycles and symptoms for individualized insights. Moreover, it offers lifestyle advice based on the user's health profile, including suggestions for nutrition, exercise, and stress reduction techniques. The solution supports rapid clinical choices, encourages long-term wellness, and facilitates proactive health management by fusing structured monitoring with dataset-driven predictive analytics.

Keywords: PCOS, Machine Learning, Dataset, Menstrual Cycle Tracker, Random Forest, Decision Tree, XGBoost, Logistic Regression, Lifestyle Management, Early Diagnosis.

Discovering Hidden Student Potential and Guiding them Towards the Passion through Models in Education

Ms. Renuga.S
Research Scholar,

Dr.R.Padma
Assistant Professor,
Department Of Computer Science and Information Technology
Vels Institute of Science, Technology and Advanced Studies, Chennai

Abstract

Conventional education systems do not recognize and develop the potential of individual students, which results in poor career guidance and academic performance. This study proposes an algorithmic model based on Support Vector Machine (SVM) that is aimed at identifying latent student potential and directing them to their areas of passion by using the extensive data analysis and predictive modeling. In today's rapidly evolving academic landscape, identifying the hidden talents and passions of students has become an essential component of holistic education. Traditional evaluation systems often overlook individual strengths, leading to a mismatch between students' potential and their chosen career paths. Recent advancements in artificial intelligence (AI) and machine learning (ML) have paved the way for personalized educational guidance. Among these, Support Vector Machine (SVM) models have emerged as robust tools for analyzing complex educational datasets, classifying student attributes, and predicting suitable career directions. This paper examines existing research on SVM-based approaches while also exploring complementary algorithms such as k-Nearest Neighbors (kNN), Random Forests, and Deep Learning models. This paper highlights how these models can integrate academic records, behavioral indicators, and psychometric evaluations to provide accurate insights into student capabilities. A complete SVM model was created with consideration of student academic performance, behavioural patterns, and preference indicators. This model uses feature selection methods, hyper parameter optimization, and cross-validation to make the performance robust.

Keywords: Support Vector Machine, Educational Data Mining, Student Potential Discovery, Machine Learning, Predictive Modeling, Academic Performance.

Hybrid Machine Learning Models for Uterine Cancer Detection Using Medical Imaging Data

Ms K .Manochithira
Research scholar,

Dr. A.S.Arunachalam
Professor,
Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies (VISTAS)

Abstract

The article has shed light in understanding how hybrid machine learning models is used for uterine cancer detection through help of medical imaging data. Uterine cancer can be considered as the most prevalent gynaecological malignancies on a global basis and imposes significant challenges due to delayed detection of the cancerous cells growing. Challenge happened due to overlapping images and features with benign uterine conditions. In such a situation, there is application of hybrid machine learning (ML) models with deep learning-based features along with traditional machine learning classifiers so that detection rate can be improved. It has been identified that is use of convolutional neural networks (CNNs) so that the automated features can be extracted and also there is combination of different machine learning algorithms used such as Support Vector Machines (SVMs), Random Forests and Gradient Boosting so that an optimized classification can happen. Robustness of the machine learning (ML) algorithm increases through merging different features of deep learning and helps in having better medical imaging analysis. It has helped in closing in the gap in between the advanced computational intelligence with that of practical clinical utility so that uterine cancers of different kinds such as endometrial cancer and cervical cancer can happen effectively.

Keywords: Hybrid Machine Learning, Convolutional Neural Networks, Uterine Cancer, Support Vector Machines, Endometrial Cancer, Cervical Cancer, Artificial Intelligence, Deep Learning Network, Medical Imaging Analytics

An Integrated Multi-Clustering Approach for Patient Stratification and Disease Prediction Using Electronic Medical Records in Vector-Borne Diseases

Dr. S. Muthukumaran

Assistant Professor

Department of Advanced
Computing and Analytics

Vels Institute of Science and Advanced Studies, Chennai.

Ms. V. Nivathra

Assistant Professor

Department of Pharmaceutical Chemistry and Analytics,
Vels Institute of Science and Advanced Studies, Chennai.

Dr. P. Arivazhagan

Assistant Professor

Department of Advanced
Computing and Analytics

Vels Institute of Science and Advanced Studies, Chennai.

Abstract

An electronic medical record comprises both structured and unstructured pieces of patient information, including but not limited to personal identifiers, vital signs, lab test results, physician's notes for diagnosis, and prescriptions. Clustering techniques are one means to EMR data to cluster patients having similar symptoms based on their laboratory results and to identify anomalies, for example, due to faulty data entry. **Objective:** However, the very large numbers of records and the extremely high dimensionality of EMR datasets make the computational burden really substantial in developing machine learning models for predicting diseases. **Methods:** In this regard, this research anticipates the communication of an **Integrated Multi-Clustering Algorithm (IMCA)** that combines and integrates all the three clustering methods: K-Means, Agglomerative, and DBSCAN clustering algorithms. A dataset on Vector borne Diseases for the study consists of 64 manifestations in relation to 11 different types of fever. The output of an individual clustering algorithm in the IMCA was then evaluated on a personal basis to classify patients with like disease conditions or symptoms and to assist a clinician in identifying subgroups benefiting from specific effective treatments. **Results:** Calinski-Harabasz Index (CHI), Davies-Bouldin Index (DBI), and Silhouette Score are various measures for evaluating the clustering algorithms' performance. The results claimed that clustering by K-means was producing more balanced clusters with a more favourable CHI value but slightly lower compactness compared with Agglomerative and DBSCAN methods.

Keywords: *Vector borne Disease, K-Means Clustering, Agglomerative Clustering, DBSCAN Clustering.*

Federated Learning and Adaptive Edge Scheduling for Secure and Scalable Intelligent Monitoring in Electricity Business Halls

Ms. T. Vani,
Research Scholar,
vanimadhu.m@gmail.com

Dr. S. Sathya,
Associate Professor,
Department of Computer Science and Information Technology,
School of Computing Sciences
Vels Institute of Science, Technology and Advanced Studies,
ssathya.scs@vistas.ac.in

Abstract

Electricity business halls face inefficiency, long waiting times, and rising data security concerns. Traditional monitoring methods relying on manual or cloud-only systems struggle with high latency, limited adaptability, and risks of data leakage. While 5G and edge computing offer improvements, existing scheduling algorithms remain static and fail to adapt to real-time workload and network fluctuations. This research proposes a hybrid framework combining Adaptive Edge Scheduling with Artificial Intelligence (AI) and Federated Learning (FL). The reinforcement learning-based scheduler dynamically allocates tasks such as flow counting and face recognition according to resource conditions, while FL preserves privacy by keeping raw customer data at the edge and sharing only model updates. The framework enhances efficiency, reduces response time, improves scalability, and strengthens data security, offering a practical solution for digital transformation in electricity business halls with potential applications in sectors like banking, healthcare, and transportation.

Keywords: Edge Computing, 5G, Adaptive Scheduling, Artificial Intelligence, Federated Learning, Data Security, Electricity Business Hall

AI-Based Code Debugging Assistant

Ms. Kanna Deepa

Ms. C. Aarthi

UG Students,

Department of Computer Science and Engineering

Vels Institute of Science, Technology & Advanced Studies (VISTAS),

Pallavaram, Chennai, Tamil Nadu

Abstract

Debugging is one of the most crucial and time-consuming phases in the software development lifecycle. Developers often spend a significant portion of their time identifying, understanding, and correcting errors in code. Traditional debugging methods such as manual code inspection, use of compilers, and standard debugging tools require high technical expertise and can still leave scope for oversight. This project proposes the design and development of an AI-based Code Debugging Assistant that utilizes Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to automate the debugging process. The system will be trained on large datasets of code samples, error logs, and correction patterns across multiple programming languages including Python, Java, and C++. By learning from these datasets, the assistant will be capable of recognizing syntax errors, logical flaws, and runtime exceptions, as well as providing optimized solutions. Unlike traditional debugging tools, the AI assistant will not only point out the location of the error but also explain the cause in simple terms and suggest corrective measures. It will also provide real-time debugging support as the user writes code, thereby minimizing delays caused by trial-and-error methods. Furthermore, the system can be extended to suggest best practices, optimize code performance, and act as an intelligent tutor for learners.

Keywords — Artificial Intelligence, Machine Learning, Natural Language Processing, Debugging Assistant, Software Development

SA-TDGFormer: A Hybrid Spatiotemporal GCN–Transformer Architecture for Human Action Recognition

Ms. S. Jayanthi

Research Scholar,

Dr. Sabin Begum

Assistant Professor

Computer Application

B.S Abdur Rahman crescent

Institute of science And Technology, Chennai

Abstract

Accurate skeleton-based action recognition remains challenging, as most existing methods employ sequential combinations of Graph Convolutional Networks (GCNs) and attention mechanisms that treat the skeleton as a complete structure, overlooking correlations among indirectly connected joints. To address this limitation, we propose SA-TDGFormer, a parallel framework that unifies GCNs and Transformers to jointly capture local topological features and global inter-joint dependencies. The architecture employs a dual-stream design— one spatiotemporal GCN stream and one spatiotemporal Transformer stream—whose complementary outputs are integrated through a late fusion strategy, enabling richer motion representations. Extensive evaluations on NTU RGB+D 60, NTU RGB+D 120, and Kinetics- Skeleton datasets demonstrate that SA-TDGFormer consistently improves recognition accuracy, achieving up to 5% gains on NTU RGB+D 60 over state-of-the-art methods.

Keywords : Action Recognition, Graph Convolutional Networks, Transformer

Enhancing E-Commerce Product Recommendations Engines with Generative AI

Dr. Prita Davidson
Assistant Professor

Ms. G. V. Aruna Shivani
Ms. K. B. Aishwarya
Student
Valliammal college for Women, Chennai.

Abstract

This paper presents a novel approach to enhancing e-commerce product recommendation systems by integrating a Generative AI framework. While traditional collaborative filtering models are effective at identifying relevant products, they often fail to deliver a personalized and engaging customer experience. This research addresses this limitation by developing a hybrid system that leverages a classic recommendation model to predict user preferences and then utilizes the Gemini to generate dynamic, personalized textual content. The methodology involves a two-stage pipeline. A collaborative filtering model is first trained on a transactional dataset to suggest products. These recommendations are then passed to the Gemini, which crafts compelling, context-aware descriptions tailored to the user's inferred interests. The system's effectiveness is evaluated through a human-in-the loop study, comparing its performance against a rule-based baseline to assess metrics like perceived personalization and engagement.

Keywords: *Generative AI, E-commerce, Recommender Systems, Personalization, Gemini, Collaborative Filtering.*

Algorithms for Addressing Data Security Challenges in Cloud Computing Environments

Mr. R. Suresh
Research Scholar

Dr T.R. Nisha dayana
Assistant Professor
Department of Computer Science and Information Technology
Vels Institute of Science, Technology & Advanced Studies, Chennai

Abstract

The study aims to identify the issues present in current data security algorithms and explore the development of a hybrid algorithm designed to enhance resilience in the cloud computing. The increasing rate of data breaches has exposed the vulnerabilities of cloud computing as a platform for online data storage and processing. Studies in cloud and quantum computing indicate that traditional security techniques face notable limitations, with AES—though currently effective—projected to falter against quantum computing power. These concerns highlight the necessity of continuously innovating and refining security mechanisms, with hybrid algorithms offering a promising path toward safeguarding sensitive data in future cloud environments.

Keywords: Data security, Hybrid algorithms, AES, Quantum Computing, Threats, Limitations, Improved Security Outcomes.

Avian Species Identification using Audio Signal Processing and Neural Network

Dr. G. Parthiban
Professor

Ms. A.Ezhil Arasi
Student

Department of computer Application & Technology
SRM Arts And Science Collage , Kattankalathur

Abstract

It suggests combining multiple picture and audio classification methods in conjunction with a neural network model and machine learning classifiers to identify the bird call. By instructing the model to pay particular attention to the spectrogram pictures produced from the audio clippings of birds, it is possible to identify the bird call in the audio file. The first step was creating an ideal dataset that included all of the bird species' sound recordings. The sound snippets were then put through a variety of sound preprocessing procedures, including reconstruction, framing, silence removal, and pre-emphasis. For each reconstructed sound clip, spectrograms were produced. In the subsequent stage, a neural network was set up and given the spectrograms as input. Neural Networks (ANN), which are a subset of supervised machine learning, are based on the input features, and our dataset will contain both the input and the matching output. Finding a technique to transfer this input to the appropriate output is our main goal. Regression and classification issues can both be resolved with Artificial Neural Network and LSTM. On unheard audio samples, the trained neural network is then utilized to identify the species of birds. The network predicts the most likely bird species using the features it learnt during training, after pre processing the audio data to extract those features. To evaluate the effectiveness of the suggested system, experiments and evaluations are carried out. The performance of categorization is gauged using metrics including accuracy, precision, recall, and F1 score. Examined is the system's resistance to changes in audio quality, background noise, and various bird vocalizations. The outcomes show that the suggested technique for identifying bird species achieves high accuracy and can successfully categorize bird species based on acoustic inputs. A promising method for automated bird species identification using neural networks and acoustic signal processing techniques has been developed. This method can help with ecological study, biodiversity monitoring, and conservation activities. For the system mentioned above, a real-time implementation model was also created and put into practice.

Keywords: *Audio signal processing, Deep learning, Convolutional Neural Network, Recurrent Neural Network (RNN), Audio feature extraction.*

An Ensemble-Based Hybrid Machine Learning Model for Autism Spectrum Disorder Classification

Ms. G. Divya

Research Scholar,

gdivya19.mca@gmail.com

Dr. V. Mani Raj

Research Supervisor,

Department Of Computer Science

A.V.V.M Sri Pushpam College

Poondi, Thanjavur

manirajv61@gmail.com

Abstract

Autism Spectrum Disorder (ASD) poses significant challenges to the Affected individuals daily lives, making early intervention crucial for mitigating its severity. In this paper, we present a comprehensive framework aimed at evaluating various Machine Learning (ML) techniques for the early detection of ASD) with a focus on real-time prediction through a user-friendly graphical user interface (GUI). These strategies are applied to feature –scaled datasets, which are then subjected to classification using seven different and popular ML algorithms: Ada Boost (AB), Random Forest (RF), k-Nearest Neighbors (KNN), Gaussian Naive Bayes (GNB), Logistic Regression (LR), Support Vector Machine (SVM), and Linear Discriminant Analysis (LDA). We conducted experiments on four standard ASD datasets representing different age groups (Toddlers, Adolescents, Children and Adults). To evaluate the classification outcomes, we employ various statistical measures such as Accuracy, Receiver Operating Curve (ROC) curve, F1 score, Specificity, Sensitivity etc. Through this analysis, the best-performing classification methods for each ASD dataset have been identified. Overall, our purposed framework Demonstrates promising results compared to existing approaches for the early detection of ASD.

Keywords – Autism Spectrum Disorder (ASD), Early Detection, Machine Learning, Classification, Feature Scaling, Statistical Measures.

Traffic Flow Prediction

Dr. P. Loganathan

Assistant Professor,

loganathancat@srmasc.ac.in

Ms. T.Harshini

Student

Department of Computer Applications and Technology

harshinitmca@srmasc.ac.in

SRM Arts And Science Collage , Kattankalathur

Abstract

Efficient traffic management is vital in modern urban settings with rising vehicular density and dynamic conditions. This project presents a real-time traffic flow prediction system using advanced machine learning techniques, including Long Short-Term Memory (LSTM) networks and Graph Neural Networks (GNNs). By integrating historical and real-time traffic along with external factors such as weather, infrastructure, and events, the system captures complex spatial-temporal dependencies. Key components involve data preprocessing, feature engineering, and scalable model training. The system is designed to handle missing data, adapt to unforeseen scenarios, and support proactive traffic strategies. Evaluation will focus on accuracy, latency, and robustness, contributing to safer, more efficient, and sustainable urban transportation. The system's scalability ensures its applicability to large cities with diverse traffic conditions, offering a robust tool for long-term urban planning. It also provides insights into the impact of external factors like weather and events on traffic flow. The proposed solution is positioned to aid traffic authorities in optimizing resource allocation and reducing overall congestion.

Keywords: *Traffic flow prediction, LSTM, Graph Neural Networks, urban mobility, real-time analytics, smart transportation, spatial-temporal modeling, traffic management.*

Hybrid Transformer for Intelligent Traffic Flow Forecasting

Mr. M. Subashini
Research Scholar

Dr. V. Poornima
Assistant Professor

Department of Computer Science and Information Technology

Vels Institute of Science, Technology & Advanced Studies, Chennai

Abstract

Traffic flow forecasting is needed in intelligent transportation systems and urban traffic management. This paper suggests new hybrid transformer architecture to combine spatial- temporal attention modules and graph neural networks to improve the accuracy of traffic flow prediction. The model suggested takes advantage of the local traffic behavior and global dependencies to model the complex traffic behaviours. It is a hybrid model which integrates convolutional neural networks, long short-term memory networks, and transformer attention mechanisms to extract spatial features, model time, and learn long-range dependencies, respectively. We show in our experiments on real-world traffic datasets that our approach is more accurate than the state-of-the-art approaches up to 15.3 percent and 12.7 percent improvements in mean absolute error and root mean square error, respectively. The model is especially strong in the capability to manage the non-periodic traffic patterns and respond to the dynamic cities.

Keywords: Traffic Flow Forecasting, Transformer Networks, Spatial-Temporal Modeling, Graph Neural Networks, Intelligent Transportation Systems

AI-Driven Automated Exploitation: A Threat Analysis of HexStrike-AI and Emerging Offensive Frameworks

Dr.V.Shoba,
Assistant Professor
shobacs@srmasc.ac.in.

Ms.A. Monica Antony
MCA Student
Department of Computer Applications and Technology,
monicaantony02@gmail.com
SRM Arts And Science Collage , Kattankalathur

Abstract

The emergence of artificial intelligence (AI)-powered offensive tools has redefined the cybersecurity landscape, enabling adversaries to automate vulnerability discovery, exploitation, and lateral movement at unprecedented speed. HexStrike-AI, a recently developed exploitation framework, integrates large language models (LLMs) with more than 150 cyber security tools to autonomously identify and exploit vulnerabilities such as zero-day and critical CVEs. This research presents an in-depth threat analysis of HexStrike-AI, examining its architecture, attack capabilities, and potential implications for critical infrastructures. Through controlled simulations, we evaluate the efficiency of AI-driven exploitation compared to conventional methods, highlighting reductions in attack timelines and increased success rates. Finally, we propose defensive countermeasures, including AI-enhanced intrusion detection, real-time patch deployment, and automated adversarial testing frameworks, to mitigate the risks posed by such tools. By providing a holistic evaluation of AI-driven automated exploitation, this study contributes to strengthening cyber resilience against the next generation of intelligent threats.

Keywords: *Artificial Intelligence (AI) - HexStrike-AI - Automated Exploitation – Cyber Security Vulnerability Discovery - Zero-Day Exploits - Critical CVEs - Large Language Models (LLMs) - Threat Analysis - Defensive Countermeasures - Intrusion Detection - Cyber Resilience.*

Predicting Soil Nutrient Deficiencies Using Advanced Machine Learning Techniques for Sustainable Crop Yield Enhancement

Ms. S. Deepa

Research Scholar

ds6963@srmist.edu.in

Dr. R. Renuga Devi

Associate professor

drrenugadevi86@gmail.com

Department of Computer Science and Applications

Faculty of Science and Humanities,

SRMIST, Ramapuram Campus

Abstract

Machine Learning (ML) is a potent method for forecasting a number of agricultural parameters, including soil fertility. Predicting soil nutrient deficiency is crucial for optimizing fertilizer use, increasing crop yields, reducing costs, and protecting the environment by preventing over-fertilization and runoff. It allows for timely intervention before severe deficiencies cause visible symptoms like stunted growth, yellowing leaves, and fruit deformation, leading to healthier plants and a more reliable food supply. Soil fertility is influenced by complex interactions between multiple physical, chemical, and biological factors in the soil ecosystem, all of which play a critical role in determining agricultural output. An accurate assessment of soil fertility is essential to maximize farming practices, enhance crop productivity, and ensure sustainable land management over the long term. Traditional methods for measuring soil fertility often rely on labor-intensive and time-consuming laboratory analyses. These approaches are particularly difficult to scale for large-scale agricultural operations and may fail to capture the temporal and spatial variability of soil properties with sufficient accuracy. In contrast, machine learning offers a viable alternative by leveraging large datasets of soil samples, environmental parameters, and agronomic practices. By applying advanced machine learning algorithms, predictive models can be developed that accurately and efficiently estimate soil fertility indicators such as nutrient levels (Nitrogen, Phosphorus, Potassium), pH, organic matter content, and others. The primary goal of this study is to investigate the ability of machine learning algorithms to reliably forecast soil fertility based on a wide range of soil characteristics and environmental factors. Soil nutrient deficiencies significantly impact crop yield and overall agricultural productivity, posing a major challenge to sustainable farming practices worldwide. This research focuses on developing an advanced machine learning-based framework to accurately predict soil nutrient deficiencies, specifically Nitrogen (N), Phosphorus (P), and Potassium (K) levels, using tabular datasets containing key soil properties such as pH, Electrical Conductivity (EC), Organic Carbon (OC), and existing nutrient measurements. Compared to traditional rule-based methods, the machine learning approach provides superior generalization and adaptability across different soil types and regions. The research also identifies gaps in existing studies, including the limited availability of large-scale, open-access soil datasets and the lack of integration with real-time IoT-based soil monitoring systems. Further, existing models often overlook the influence of environmental factors such as rainfall and temperature on nutrient dynamics, which presents a direction for future work. This study demonstrates the potential of machine learning as a powerful tool in precision agriculture by enabling data-driven fertilizer recommendations, minimizing chemical overuse, promoting environmental sustainability, and enhancing crop yield. The study offers a scalable solution to empower farmers with timely and actionable insights into soil health management. Overall, intelligent approaches to soil nutrient detection promise a more productive, sustainable, and economically viable future in farming.

Keywords: *Soil Fertility Prediction , Agriculture, Nutrient Deficiency Detection Automation, Fertilizer Predictive Modeling*

Machine-Learning-Based Agriculture Crop Recommendation System

Mr. P. Kuppan
Research Scholar

Dr. V. Vishwa Priya
Assistant Professor
Department of Computer Science & Information Technology
VISTAS.

Abstract

In machine learning, a broad range of techniques are used to create models that can infer prediction rules from available data and forecast yet-to-be-observed future data. One of the newest study areas is the application of machine learning techniques to agriculture. Reducing agricultural costs in order to gain profit is the primary goal of smart agriculture. Smart farming is made possible by machine learning algorithms' predictive nature. Astute farmers employ this strategy in their farms to boost productivity. Among the many crucial elements that must be taken into account is the soil. Many farmers are unaware that the pH, texture, and nutritional makeup of the soil can have significant effects on crop growth. Choosing the appropriate crop for the farmland's soil composition is one of the most crucial factors in agriculture. This poll is mostly concerned with crop recommendations and ideas. This paper provides a brief overview of machine learning approaches applied in agricultural recommendation systems, considering soil and crop yield prediction systems.

Keywords: Machine learning, prediction, Classification, crop yield Prediction, Random Forest algorithm, K Nearest Neighbour Classifier.

Visionary Automation in Smart Building designs using AI and ML

Dr. A. Jayasudha

Associate Professor

Mr. J. David

Student

Ms. A. Anitha

Student

Computer Applications and Technology
SRM Arts and Science College, Kattankulathur

Abstract

The smart buildings evolution is increasingly accelerated through the seamless incorporation of Artificial Intelligence (AI) and Machine Learning (ML). Traditional building architectural framework replace with smart and efficient systems that continuously adapt to optimize performance. **Findings:** Apart from automation, AI and ML are becoming main foundational technology of smart building innovation. This study explores how they serve as catalysts for intelligent operation, environmental efficiency, and occupant focused design. A modular AI driven framework for self-learning smart buildings that adapt in real-time to environmental changes and occupant behaviour using reinforcement learning, federated learning, and generative design enabling energy efficient, autonomous, and privacy preserving intelligent infrastructure. **Implementation:** Predictive Maintenance that helps in detecting failures before they happen to reduce downtime using Random Forest algorithm. Occupancy Detection and Space Utilization are predicting room usage to optimize layout and energy using Convolutional Neural Networks (CNN) and K-Means Clustering. Adaptive Comfort Control which can personalize environmental conditions (temperature, lighting) per occupant using Reinforcement Learning. **Usage:** Digital twins, generative design, and federated learning also open new possibilities in virtual simulation, ethical AI deployment, and privacy preserving smart environments. Such advancements signal a transformative toward buildings that are not only smart but self-learning and context aware, paving the way for the future of intelligent urban infrastructure.

Keywords: *Artificial Intelligence (AI), Machine Learning (ML), Reinforcement Learning, Federated Learning, Generative Design, Predictive Maintenance, Random Forest Algorithm, Occupancy Detection, Space Utilization, Convolutional Neural Networks (CNN), K-Means Clustering.*

Intelligent Aquaculture: Leveraging Machine Learning and IoT for Sustainable Water Quality Management in fish ponds

Mrs. V Sundari

Associate Professor,

Mrs. M R Nithya

Associate Professor,

Ms. Neelaveni M A

Department of Computer Science Engineering,
Meenakshi Sundararajan Engineering College, Chennai, India

Abstract

Aquaculture, a vital contributor to global food security, faces numerous challenges, including maintaining optimal water quality. To address this issue, researchers have turned to Machine Learning (ML) and Internet of Things (IoT) technologies to enhance water quality monitoring and prediction. By integrating IoT sensors, ML models, and cloud-based systems, farmers can receive real-time alerts and take proactive measures to prevent water quality degradation, ensuring a healthy environment for fish and aquatic organisms. ML algorithms, such as Random Forest, Support Vector Machine (SVM), Decision Trees, K-Nearest Neighbours and Artificial Neural Networks, have been successfully applied to predict water quality parameters, including dissolved oxygen, pH, pond temperature, foul smell and turbidity. These models can learn from historical data and identify patterns, enabling farmers to make informed decisions about water quality management. Moreover, ML can also be used to detect fish diseases and identify the underlying causes, allowing for timely interventions and reducing the risk of disease outbreaks. The integration of ML and IoT in aquaculture has the potential to revolutionize the industry, enabling farmers to optimize production, reduce costs, and minimize environmental impacts. With the ability to monitor and predict water quality in real-time, farmers can respond quickly to changes in water conditions, ensuring a healthy and sustainable environment for fish and aquatic organisms. As the demand for seafood continues to grow, the adoption of ML and IoT technologies in aquaculture is crucial for meeting the world's food security needs while protecting the environment.

Keywords: *Machine Learning, Internet of Things, Artificial Intelligence*

Early Detection and Risk Stratification of Uterine Fibroid using Deep Learning-A Review

K. Bilkees

Research Scholar

Department of Computer Science

Vels Institute of Science, Technology and Advanced studies,
Chennai, Tamil Nadu, India

Dr. K Kasturi

Associate Professor

Department of Applied computing and Emerging
Technologies

Vels Institute of Science, Technology and Advanced studies,
Chennai, Tamil Nadu, India

Abstract

Uterine fibroids represent a significant global health burden, affecting a majority of women by age 50. Current diagnostic methods, primarily ultrasound and MRI, are effective for anatomical description but are subjective and lack predictive capability for individual risk stratification. This review synthesizes the emerging application of deep learning (DL) in transforming the management of uterine fibroids. We detail how convolutional neural networks (CNNs) and architectures like U-Net automate detection and segmentation with high accuracy, outperforming traditional methods. Crucially, we explore the potential of DL to integrate imaging radiomics with clinical and genetic data to predict future fibroid behavior, enabling a shift from reactive to proactive, personalized care. Despite promising results, challenges including data scarcity, algorithmic bias, and model explainability remain significant hurdles to clinical adoption. The review concludes that overcoming these barriers through interdisciplinary collaboration and robust validation is essential to realize the full potential of DL in improving early detection, risk stratification, and ultimately, patient outcomes for uterine fibroids.

Keywords: *Uterine Fibroids, Deep Learning, Risk Stratification, Explainable AI, Medical Imaging.*

Real-Time Public Transport Tracker for College Bus

Ms. I. Pavitra

UG Scholar

Department of Computer Science
Bishop Heber College, Tiruchirappalli

Dr. B. Gayathri

Associate Professor

Department of Computer Science
Bishop Heber College, Tiruchirappalli

Abstract

Efficient and reliable transportation plays a critical role in the daily functioning of educational institutions. This paper presents the design and development of a Real-Time Public Transport Tracker for College Buses, aimed at improving the commuting experience for students and staff. The system integrates GPS tracking, mobile applications, and cloud-based data processing to provide live location updates of buses. Users can access estimated arrival times, route details, and notifications regarding delays or changes in schedule. The proposed solution minimizes uncertainty, reduces waiting time, and enhances safety by enabling real-time monitoring of bus movement. The methodology involves modular development including data collection, location tracking, user interface, and server-side processing. This research demonstrates how IOT and mobile technologies can be effectively combined to optimize college transport management. The project not only improves convenience for commuters but also ensures better resource utilization for institutions. Guardian, focusing on its impact on self-awareness, stress reduction, and effective time utilization. The results of this project demonstrate the potential of technology to provide preventive care and productivity enhancement, ultimately fostering healthier and more sustainable student lifestyles.

Keywords: Mental Health, Productivity, Students, Stress Detection, Sentiment Analysis, Time Management, AI Companion

Early Stage Generalized Autism Spectrum Disorder Detection Using Machine Learning

Mr. Rajashekar Reddy
Assistant professor

Ms. G. Sree Varshini

Ms. M. Sneha Reddy

Ms. P. Lakshmi Varshita

Ms.M.Pragnitha

Department of Information Technology
Bvrit Hyderabad College of Engineering for Women
Hyderabad

Abstract

Autism Spectrum Disorder (ASD) refers to a group of complex underdevelopment disorders with repetitive patterns of behavior and difficulties in social interaction, speech and nonverbal communication. Diagnosis can be further complicated because ASD has a diverse clinical presentation, but is diagnosed via standardized diagnostic scores and early intervention services may improve quality of life for those with the disorder. Early recognition can result in appropriate early referral, and hence effective preventive measures to impede disability as it progressively impairs functioning throughout developmental life. Past researchers created separate models for each age group, which can make them less generalizable. In this proposed model we aim to introduce a relatively universal framework for augmenting the earlier diagnosis of ASD with advanced Machine Learning (ML) methodologies that generalizes across all ages. Our goal is to maximize the prediction of ASD risk factors, by concentrating on multiple feature scaling approaches and classification strategies. We use a model that integrates age-related variance and allows us to create tangible tools of cohesive relevance through the lifespan. The results of this study may help all those working in the field to understand more about ASD and help in its early detection which can in turn lead to better services, thereby creating numerous possibilities for development.

Keywords: Autism Spectrum Disorder (ASD), Early Diagnosis, Machine Learning (ML), Predictive Modeling, Feature Scaling, Classification Techniques, Age-related Variance, Universal Framework, Neurodevelopmental Disorders, Early Intervention

Deep Learning-Based Intelligent Road Surface Damage and Lane Recognition System for Autonomous Vehicles

Mrs. T. Sujatha Jayakrishnan

Assistant Professor
Department of Computer Applications & Technology
SRM Arts and Science College Kattankulathur

Ms. D. Monisha

MCA Student
Department of Computer Applications & Technology
SRM Arts and Science College Kattankulathur

Abstract

Road surface conditions and clear lane detection are crucial for the safe operation of autonomous vehicles. Surface damages such as cracks, potholes, and irregularities pose significant challenges for navigation and vehicle stability, while midsection of lane markings can lead to collisions and inefficient routing. This research proposes a deep learning-based intelligent system for simultaneous detection of road surface damages and lane markings using advanced convolution neural networks (CNN s) and semantic segmentation models. The system integrates image per-processing, feature extraction, and classification techniques to achieve accurate detection under varying lighting and weather conditions. Experimental results demonstrate that the proposed framework enhances road safety and transportation efficiency by providing reliable real-time analysis, achieving higher accuracy and robustness compared to traditional image processing methods.

Keywords: Autonomous Vehicles-Road Surface Damage Detection- Lane Detection- Convolution Neural Networks (CNNs)- Semantic Segmentation- Deep Learning- Computer Vision-Real-Time Processing- Road Safety- Intelligent Transportation Systems.

Multi-Task Deep Learning for Real-Time Traffic Sign Detection and Recognition

Mrs. S. Jayasree

Assistant professor
Department of Computer Science
SRM Arts and Science College, Kattankulathur
jayasreecat@srmasc.ac.in

M. Rashethunnisa

M.Sc Computer Science Student
Department of Computer Science
SRM Arts and Science College, Kattankulathur
rashethun@gmail.com

Abstract

Traffic sign detection and recognition are crucial components of advanced drive assistance systems and autonomous vehicles. Accurate interpretation of traffic signs under various real-world conditions, such as occlusion, illumination changes, and motion blur, remains a significant challenge. This paper proposes a multi-task deep learning framework combining detection and classification within a single architecture, leveraging transfer learning on DenseNet201 and Inception ResNetV2 backbones. The model achieves high accuracy on benchmark datasets such as GTSRB and BTSD and operates in real time on edge devices. Extensive experiments demonstrate superior performance compared to conventional single-task models, highlighting the benefits of joint learning and robust data augmentation.

Keywords: Traffic Sign Detection- Traffic Sign Recognition-Deep Learning- Multi-task Learning- Transfer Learning- DenseNet201- InceptionResNetV2- GTSRB-BTSD-Autonomous Vehicles- Edge Computing- Data Augmentation.

Recommended System for Student's Co-Curricular Activities Using Data Mining Techniques

Mr. M. Kannan

Head i/c
Department of Computer Science with Data Science
SRM Arts and Science College Kattankulathur

Ms. J. Harini

Student
Department of Computer Science with Data Science
SRM Arts and Science College Kattankulathur

Abstract

Recommended system for student's Co-Curricular activities using data mining techniques is based on recommending system for student's co-curricular activities. In this paper is specially designed for the students of SRM Arts and Science College. The main motto of the paper is when the student searches some events means; it will display details about the events. Here we used tree search algorithms that view the elements as vertices of a tree, and traverse that tree in some special order. Examples of the latter include the exhaustive methods such as depth-first search and breadth-first search, as well as various heuristic-based search tree pruning methods such as backtracking and branch and bound. Unlike general met a heuristics, which at best work only in a probabilistic sense, many of these tree-search methods are guaranteed to find the exact or optimal solution, if given enough time. Burst Search Algorithm that has less time complexity compared with other searching algorithm of data mining. It is helpful to search and display about the events. If the student want to participate the event means, register the event through online.

Keywords: Recommendation System, Co-curricular Activities, Student Profiling

Deaf Assistance through Smart Alerts for Non Hearing Users

Ms. S Shanmathy

UG Scholar
Department of Computer Science
Bishop Heber College, Tiruchirappalli

Dr. B. Gayathri

Associate Professor
Department of Computer Science
Bishop Heber College, Tiruchirappalli

Abstract

Deaf and hard-of-hearing individuals often face challenges in perceiving essential auditory signals such as phone rings, doorbells, alarms, or emergency sounds. The proposed system, “Deaf Assistance through Smart Alerts,” provides real-time smart notifications to non-hearing users. It integrates visual (flashlight), tactile (vibration), and automated text message responses to bridge the communication gap. By leveraging smartphone sensors, IoT devices, and automation, the system ensures that deaf users remain informed, safe, and connected in various situations.

Keywords: Deaf Assistance, Smart Alerts, Accessibility, Vibration Notification, Flashlight Alerts, Auto Message System, Inclusive Technology

AI-Driven Approaches to Fertilizer Optimization in Sustainable Agriculture

Mrs. J. Porkodi

Head and Assistant Professor
Department of Computer Applications (UG)
V.V.Vanniaperumal College for Women
Virudhunagar, Tamilnadu
porkodi@vvvcollege.org

Mr. A. Nagavaratharajan

Associate Professor
nagavaratharajan@vhnsnc.edu.in

Mr. M. Sivasankar

Assistant Professor
Department of Commerce with Computer Application
V.H.N.Senthikumara Nadar College
Virudhunagar, Tamilnadu
sivasankar@vhnsnc.edu.in

Abstract

Agriculture remains a cornerstone of global food security and sustainable development, making the optimization of farming practices increasingly vital. Fertilizer application, when tailored to the specific needs of soil and environmental conditions, significantly influences crop productivity and quality. This study proposes a predictive framework that employs machine learning techniques to generate efficient and eco-friendly fertilizer strategies. By utilizing comprehensive datasets incorporating soil properties such as nitrogen (N), phosphorus (P), potassium (K), and pH, along with environmental factors including temperature, humidity, and rainfall, we preprocess and integrate the data to develop robust model inputs. Several supervised machine learning algorithms, including K-Nearest Neighbors, Decision Trees, Random Forests, and Gradient Boosting, are evaluated for their effectiveness in optimizing fertilizer recommendations. The integration of real-time data and sensor-driven insights enhances the adaptability and accuracy of these models, enabling precise nutrient management. The proposed approach not only minimizes fertilizer waste and reduces environmental impacts but also improves crop yield and quality. Ultimately, this work advances the vision of sustainable agriculture by supporting precision farming practices and contributing to the global effort of meeting rising food demands with reduced resource consumption and ecological footprint.

Keywords: *Sustainable Agriculture, Precision Farming, Machine Learning, Fertilizer Recommendation, Soil Nutrient Analysis, Environmental Parameters, Predictive Modeling, Crop Yield Optimization*

AI-Driven Deep Learning Model For Sustainable Plant Disease Detection And Classification

Ms. S. Meenakshi

UG Scholar

Department of Computer Science
Bishop Heber College, Tiruchirappalli

Dr. B. Gayathri

Associate Professor

Department of Computer Science
Bishop Heber College, Tiruchirappalli

Abstract

Artificial Intelligence (AI) has become a transformative technology in solving complex classification problems across multiple domains such as healthcare, agriculture, and security. This project presents an AI-based Image Classification System that leverages deep learning techniques to accurately identify and categorize images. The system employs a pipeline of pre-processing, feature extraction, and classification using Convolutional Neural Networks (CNNs). Pre-processing steps ensure consistency in image size, clarity, and quality, while CNNs automatically extract high-level visual features for robust classification. The model is trained on a large dataset and optimized through hyper parameter tuning and regularization strategies to enhance accuracy and generalization. Performance evaluation is conducted using standard metrics including accuracy, precision, recall, and F1-score. Results demonstrate that the proposed system achieves high accuracy while maintaining scalability for real-world applications. This research highlights the potential of AI-driven image classification in developing intelligent systems that support decision-making, improve efficiency, and contribute to sustainable technological advancements.

Keywords: Artificial Intelligence, Deep Learning, Image Classification, Convolutional Neural Networks (Cnn), Feature Extraction, Preprocessing, Model Evaluation

A Unified Framework Approach for Automated Vulnerability Assessment

Dr. B. Kamatchy

Assistant Professor

Department of Advanced Computing and Analytics

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

kamatchi6282@gmail.com

Dr. N. Kalaichelvi

Assistant Professor

Department of Advanced Computing and Analytics

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

kalaichelvi.scs@vistas.ac.in

Abstract

The rapid growth of cyber security threats has made vulnerability assessment crucial for organizational defense strategies. Conventional methods rely on running separate scanners individually and manually linking their results. This approach is becoming less efficient, uses more resources, and is more likely to involve human error. These issues not only slow down the assessment process but also increase the chance of overlooking important vulnerabilities. To address these challenges, this paper introduces a unified framework for automated vulnerability assessment. The framework combines several widely used security scanners into one automated system that handles tool execution, brings together results, and smartly connects findings. This unified process cuts down on redundancy, improves detection accuracy, reduces false positives, and ensures that even subtle or less obvious vulnerabilities are found and prioritized. By automating the entire workflow, the framework significantly reduces assessment time. This allows cyber security professionals to focus on remediation, incident response, and strategic defense. In the end, this approach improves efficiency, consistency, and resilience in vulnerability management, giving organizations a stronger defense against changing cyber threats.

Keywords: *Automated Vulnerability Assessment, Unified Framework, Cyber Security, Threat Detection, Risk Management, Security Automation.*

Review on Emotions Recognition using Machine Learning

Mr. M. Ganeshbabu

Student

Department of Computer Applications and Technology
SRM Arts and Science College, Kattankulathur

Dr. R. Rathidevi

Assistant Professor

Department of Computer Applications and Technology
SRM Arts and Science College, Kattankulathur

Abstract

Numerous disciplines, including computer science, psychology, and philosophy, study emotion. Our decisions, plans, thoughts, and responses to situations are all influenced by our emotions. Automated Human Emotion Recognition (AHER) is a key topic in computer science, with applications in marketing, human-robot interaction, video games, e-learning, and more. It is helpful in any circumstance when knowing someone's emotional condition is crucial. AHER detects emotions using a variety of data formats, including facial expressions, written language, speech, and physiological signs such as EEG (brainwaves), blood volume pulse, ECG (heart impulses), and others. These signals can be utilized separately (uni-modal) or together (multi-modal). The majority of extant investigations are conducted in laboratories, using models adapted to specific individuals. Recent research has shifted its focus toward real-world testing and the development of broad models. This work provides a complete assessment and evaluation of the most recent AHER approaches based on machine learning from a computer science perspective, as well as suggestions for future research.

Keywords: Emotion Recognition, Machine Learning, Facial Expression Analysis, Speech Emotion Recognition, Affective Computing

Identity Sentinel: Sophisticated Social Media Doppelganger Detection and Fake ID Prevention

Dr. N. Kalaichelvi

Assistant Professor

kalaichelvi.scs@vistas.ac.in

Dr. B. Kamatchy

Assistant Professor

Department of Advanced Computing and Analytics

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

kamatchi6282@gmail.com

Abstract

Identity Sentinel: Advanced Fake ID Prevention and Doppelganger Detection for Social Media offers a robust and multi-faceted solution to the escalating challenges of identity fraud, duplicate accounts, and profile impersonation on social media platforms. This system integrates cutting-edge image similarity detection, comprehensive database verification, and secure authentication protocols to ensure genuine user registrations. By proactively identifying and blocking fake identities, detecting subtle variations in user details intended for duplicate accounts, and flagging similar profile pictures used for impersonation, Identity Sentinel significantly enhances platform integrity and user trust. Its real-time verification capabilities and scalable architecture provide an efficient means to combat fraudulent activities, fostering a safer and more authentic online environment for all users. The proliferation of social media has unfortunately been accompanied by a surge in malicious activities, including the creation of fake accounts, identity theft, and the proliferation of doppelganger profiles, which undermine platform integrity and user security. Existing authentication methods often fail to detect minor alterations in user information used for creating deceptive accounts. To address these critical vulnerabilities, Identity Sentinel meticulously establishes a secure and rigorously verified user registration process. By leveraging advanced image analysis, thorough database validation, and fortified authentication, it effectively prevents fake identities, duplicate accounts, and profile impersonation. This integrated approach substantially curtails fraudulent activities and cultivates a more trustworthy and authentic user experience, ultimately aiming to restore confidence in online interactions by ensuring reliable identity verification and protection against malicious impersonation.

Keywords: Identity Verification, Doppelganger Detection, Fake ID Prevention, Social Media Security

From Classical Algorithms to Deep Learning: Advances in Biomedical Image Analysis

Ms. A. Kamatchi

Research Scholar
Department of Computer Science
V.V.M.Sri Pushpam College(Autonomous) Poondi,Thanjavur(Dt)

Dr. V. Maniraj

Associate Professor,
Research Supervisor,
Head of the Department,
Department of Computer Science,
V.V.M.Sri Pushpam College (Autonomous) Poondi,Thanjavur(Dt)

Abstract

In this study, we explored multiple techniques for image segmentation and processing, specifically applied to MRI scans of human medical data and to bioinformatics tasks involving brain cell segmentation. Our objective was to introduce and analyze a range of mathematical and computational methods for enhancing medical and biological image interpretation. We developed new software tools and methodologies aimed at improving the accuracy and efficiency of segmentation in both domains. This approach opens new possibilities for enhancing diagnostic precision in medical imaging and cellular-level analysis, such as in iron content assessment. Among the implemented methods, we utilized the GrabCut algorithm, developed in C++ for robust segmentation, and introduced a fuzzy logic-based approach for tissue classification, implemented in MATLAB. Additionally, we employed machine learning techniques, notably a U-Net-based deep learning architecture, to perform precise segmentation and extract key parameters of brain cells. These methods enabled us to process data that traditional techniques could not handle effectively. Overall, our contributions led to more accurate segmentation results with clearer object boundaries and improved edge definition. The proposed approaches also have the potential to be extended to a wide range of other biomedical and biological imaging applications.

Keywords: Image Partitioning, Digital Image Analysis, Fuzzy Logic-Based Image Analysis, Graph-Based Segmentation, Convolutional Neural Networks (CNNs)

Investigating The Impact of Medical Record Search Engines on Clinical Workflows

Ms. Narmatha.S

Research Scholar
Department of Computer Science
AVVM Sri Pushpam College (Affiliated to Bharathidasan University)
Tiruchirappalli, Poondi, Thanjavur Tamilnadu, India

Dr. V. Maniraj

Research Advisor, PG & Research
Department of Computer Science
AVVM Sri Pushpam College (Affiliated to Bharathidasan University)
Tiruchirappalli, Poondi, Thanjavur Tamilnadu, India

Abstract

This study explores the challenges of utilizing domain-specific ontologies to enhance the classification of online content, with a specific focus on categorizing medical documents through the use of the Medical Subject Headings (MeSH) thesaurus. The research aims to develop a more effective content representation model grounded in semantic knowledge. To assess the proposed approach, it was benchmarked against traditional stemming methods using two prominent data mining algorithms: C4.5 and K-Nearest Neighbors (KNN). The findings reveal that the ontology-based method consistently outperforms standard techniques. By incorporating MeSH-derived concepts and hierarchical structures such as hypernyms, the resulting feature vectors were significantly more informative, leading to improved classification outcomes. When applied to the Ohsumed biomedical dataset, the proposed method demonstrated a substantial 30% improvement in performance compared to stem-based models. These results highlight the benefits of embedding domain-specific ontologies into medical text classification workflows.

Keywords: *Medical Record Search Engines, Clinical Workflows, Electronic Health Records (EHR), Health Informatics, Information Retrieval*

Blockchain Consortium Algorithms: A Comprehensive Study

Ms. B. Gayathri

Research scholar
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. V. Vishwapriya

Assistant Professor
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

Consortium block chains represent a hybrid model that bridges the gap between public and private distributed ledgers. Unlike fully decentralized public block chains, consortium models restrict participation to a predefined group of trusted organizations, thereby enhancing efficiency, scalability, and regulatory compliance. Consensus algorithms play a pivotal role in ensuring data integrity and network resilience within these systems. This paper explores major consortium block chain algorithms including Practical Byzantine Fault Tolerance (PBFT), Raft, and Proof of Authority (PoA). It also provides a comparative analysis of their efficiency, fault tolerance, scalability, and security, highlighting their applications across domains such as finance, supply chain, and healthcare. Finally, the paper outlines the challenges and future directions for the evolution of consortium block chain algorithms.

Keywords: *Blockchain, Consortium Blockchain, Consensus Algorithms, PBFT, Proof of Authority, Raft, Distributed Ledger*

Using AI and Real-Time Data for Predictive Maintenance in the Automotive Sector

Ms. P. Vennila

Research Scholar (FT)
Department of Computer Science
AVVM Sri Pushpam College (Affiliated to Bharathidasan University)
Tiruchirappalli, Poondi, Thanjavur Tamilnadu, India

Dr. V. Maniraj

Research Advisor, PG & Research
Department of Computer Science
AVVM Sri Pushpam College (Affiliated to Bharathidasan University)
Tiruchirappalli, Poondi, Thanjavur Tamilnadu, India

Abstract

The use of cutting-edge technologies like sensors, telematics, and onboard diagnostics (OBD) is rapidly changing the automotive sector. The majority of cars still rely on reactive or scheduled maintenance methods in spite of these advancements, which are unable to predict unexpected malfunctions and lead to more expenses and downtime. Through the early detection of anomalies and component degradation, predictive maintenance (PdM), which is fueled by real-time data analytics and machine learning (ML), provides a proactive solution. This study suggests a sophisticated predictive maintenance framework for smart cars that combines real-time diagnostic decision-making, machine learning-based anomaly detection, and IoT-enabled data collection. To tackle the issues of precision, computational effectiveness, and implementation in resource constrained automotive settings, we assess a variety of machine learning models, such as Random Forest (RF), Long Short-Term Memory (LSTM) networks, and Extreme Gradient Boosting (XGBoost). According to experimental results, XGBoost offers a compromise between accuracy and processing complexity, whereas LSTM-based temporal models perform better than conventional algorithms at collecting sequential sensor data patterns. In next-generation smart cars, the suggested framework shows promise for lowering unscheduled maintenance, increasing vehicle uptime, and improving safety.

Keywords: *Predictive Maintenance, Automotive Industry, Telematics, Machine Learning, Onboard Diagnostics, Anomaly Detection.*

Beyond Functionality Enhancing User Experience Through Human – Computer Interaction

Ms. V. Prasanna Shreeya
Student

Ms.R. Lakshana
Student

Dr. L. Mary Louis
Professor
Kumaraguru college of Liberal Arts and Science
Coimbatore

Abstract

Human–Computer Interaction (HCI) explores how people connect with technology, with the goal of making digital systems more intuitive, inclusive, and meaningful to use. This study set out to understand how thoughtful design can improve the way individuals interacts with computers and smart devices in everyday life. To achieve this, we reviewed established design principles and then tested prototype interfaces with real users through task-based experiments. Alongside performance measures such as speed and accuracy, participants shared their experiences and challenges through direct feedback. The findings suggest that when interfaces are designed with simplicity, accessibility, and adaptive feedback in mind, users complete tasks more efficiently and report greater satisfaction. Features that personalize the experience like context-aware prompts or subtle emotional cues—further enhance engagement by making the technology feel more responsive and human-centered. In today’s world, where digital systems influence learning, healthcare, business, and even social interaction, these insights are especially relevant. The research highlights the importance of designing technology that adapts to people, rather than forcing people to adapt to machines. By focusing on usability and inclusivity, HCI has the potential to shape a future where technology is not just functional but genuinely supportive of human needs.

Keywords: Usability, User Experience (UX), Interface Design, Accessibility, Adaptive Feedback, Personalization, Task-based Evaluation, Inclusive Technology, Human-Centered Design.

Hybrid ML Approach for Ransomware & Malware Forensics in IoT

Ms. F. Mashiya Afroze

Research scholar
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. V. Poornima

Associate Professor
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

The rapid growth of the Internet of Things (IoT) has revolutionized automation and connectivity, but it has also expanded the attack surface for ransomware and malware, creating critical challenges for digital forensics and incident response. Traditional signature-based detection and isolated machine learning (ML) approaches often fail to detect polymorphic and zero-day attacks, resulting in incomplete forensic investigations. This paper proposes a **Hybrid Machine Learning (HML) framework** that combines supervised and unsupervised models for robust ransomware and malware forensics in IoT environments. The proposed system integrates feature extraction from network traffic, system logs, and memory artifacts, followed by a two-stage detection pipeline: an ensemble-based supervised classifier (Random Forest + XGBoost) for known threats and an unsupervised anomaly detector (Autoencoder) for novel attack patterns. To optimize detection accuracy and reduce false positives, a meta-heuristic algorithm is employed for feature selection. Experimental evaluation using benchmark IoT datasets (IoT-23, TON_IoT) demonstrates that the hybrid approach achieves **98.6% detection accuracy**, significantly outperforming standalone ML models, while maintaining computational efficiency suitable for edge devices. Furthermore, the framework generates forensic-ready evidence by correlating malicious events with timestamps and affected nodes, improving attribution and incident reconstruction. The results suggest that the proposed hybrid ML approach enhances both proactive threat detection and post-attack forensic analysis, offering a scalable and explainable solution for securing IoT ecosystems against ransomware and malware threats.

Keywords: *Hybrid Machine Learning (HML), IoT Forensics, Ransomware Detection, Signature-based Detection, Ensemble Learning.*

Credit Card Fraud Detection in Banking Transactions Using Advanced Machine Learning Techniques

Mr. R Barath Ramesh

Assistant Professor

Department of Computer Science with Artificial Intelligence,

SRM Arts and Science College

Affiliated to the University of Madras

Kattankulathur, Chengalpattu- 603203,

barathrameshcat@srmasc.ac.in

Abstract

A crucial role in banking transactions is fraud detection, which identifies unusual or unauthorized activity to protect clients and financial institutions from financial loss and damage to their reputations. An advanced machine learning-based framework is presented in this paper with the goal of improving the effectiveness and dependability of systems for identifying fraud. The major goal is to increase detection accuracy and decrease false positives while preserving real-time transaction processing. The methodology is a hybrid approach, combining supervised classification techniques like Gradient Boosting and Random Forest with unsupervised anomaly detection to identify unusual events. Model performance is optimized through the incorporation of class imbalance handling, data standardization, and comprehensive feature engineering. Experimental evaluation of large-scale transactional datasets yields better outcomes than standard rule-based and standalone models. The suggested framework obtains superior accuracy, recall, and AUC values, showing that it has the potential to be a scalable and effective solution for successful fraud detection in contemporary financial contexts.

Keywords: Anomaly detection, Cyber security, Machine learning, Fraud detection, Banking transactions, Financial Fraud

Assessment of Lung Cancer by Pathologists using CT Scans with Deep Learning Methods: A Review

Ms. F. Leema Raina

Research scholar

Department of Computer Science and Information Technology

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. C. Anbarasi,

Assistant Professor

Department of Computer Science and Information Technology

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

Lung cancer remains a leading cause of cancer-related mortality worldwide, with early and accurate diagnosis being critical for improving survival outcomes. Histopathological examination of biopsy tissue continues to be the gold standard for definitive diagnosis, offering insights into tumor type and molecular markers. However, with the increasing reliance on imaging— particularly computed tomography (CT) scans—for lung cancer screening, staging, and monitoring, there is a growing need for automated, robust, and scalable diagnostic support tools. Deep learning, particularly convolutional neural networks (CNNs), has emerged as a powerful technique for the automated analysis of CT images, enabling accurate detection, segmentation, and classification of lung nodules. This review consolidates current advancements in the application of deep learning models to CT imaging, covering key methodologies, segmentation architectures (e.g., U-Net, V-Net), classification frameworks (e.g., 3D CNNs, ResNet), and radiogenomic approaches that bridge imaging features with molecular data. Key public datasets such as LIDC-IDRI and NSCLC-Radiomics have been instrumental in training and validating these models. The integration of AI into clinical workflows has demonstrated the potential to augment pathologists' capabilities by providing rapid, consistent, and quantitative assessments. Nonetheless, challenges persist regarding model interpretability, data variability, and regulatory approval for clinical deployment. This review emphasizes the synergistic role of CT-based deep learning and histopathological evaluation in achieving precise and personalized lung cancer diagnosis.

Keywords: *Lung Cancer Diagnosis, Computed Tomography (CT), Deep Learning, Convolutional Neural Networks (CNNs), Histopathology, Radiogenomics.*

From Noise Reduction to Diagnosis: A Review of Machine Learning Techniques in Lung Image Enhancement

Ms. V. Nandhini

Research scholar
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. R. Parameswari

Professor & Head
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

The early and accurate diagnosis of lung diseases, including lung cancer and tuberculosis, is critical for effective treatment and improved patient outcomes. However, the diagnostic process is often hindered by limitations in conventional imaging techniques, such as Low-Dose Computed Tomography (LDCT) and chest X-rays, which can result in images with low contrast, noise, and artifacts. The emergence of machine learning (ML) and deep learning (DL) algorithms has revolutionized the field, offering promising solutions to enhance lung image quality and diagnostic accuracy. This comprehensive review explores the advancements in ML/DL methodologies for improving lung image quality. It delves into challenges associated with conventional imaging techniques, data scarcity, and annotation limitations, alongside discussing noise reduction, contrast enhancement, and segmentation techniques. The paper further evaluates ML/DL-based algorithms, including convolutional neural networks (CNNs), generative adversarial networks (GANs), and hybrid approaches, highlighting their impact on image enhancement and disease detection. By analyzing performance metrics and comparative studies, the review underscores the potential of these technologies to transform clinical workflows. The findings emphasize the need for robust datasets, enhanced generalize ability, and explain ability to facilitate widespread clinical adoption and improve patient outcomes.

Keywords: *Low-Dose CT (LDCT), Chest X-Ray Analysis, Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs)*

Review of Machine Learning and Artificial Intelligence Techniques in Cyber Security

Dr. V. S. Padmavathy

Assistant Professor
Department of Computer Applications and Technology,
SRM Arts and Science College Kattankulathur

Mr. S. Praveen Aadithya

Student
Department of Computer Applications and Technology
SRM Arts and Science College
Kattankulathur

ABSTRACT

Cyber security threats continue to evolve in scale, scope, and complexity. Classical signature-based defences are increasingly inadequate for detecting zero-day attacks, polymorphic malware, advanced persistent threats (APTs), insider threats, and other emerging vectors. In response, Machine Learning (ML) and Artificial Intelligence (AI) approaches have gained prominence for their ability to learn from data, detect anomalous patterns, adapt to changing threat landscapes, and automate defensive tasks. With the exponential rise of cyber threats, traditional defense mechanisms have proven insufficient in combating sophisticated attacks. Artificial Intelligence (AI) and Machine Learning (ML) are transforming cybersecurity by enabling real-time threat detection, automated incident response and predictive defense. The Internet itself has become critical for governments, companies, financial institutions, and millions of everyday users. Networks of computers support a multitude of activities whose loss would all but cripple these organizations. As a consequence, cyber security issues have become national security issues. This paper explores how AI and ML techniques are applied in cyber security, their benefits, challenges, and the future outlook for creating more resilient digital infrastructures.

Keywords: Machine Learning ML, Artificial Intelligence AI, Cyber Security

Cloud-Agnostic Architectures to Mitigate Vendor Lock-In

Ms. SUJITHA S

Department of Computer Applications,
Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India.

Ms. SWETHA S

Department of Computer Applications,
Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India.

Ms. DHEEKSHA K

Department of Computer Applications,
Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India.

Dr. S. RAJA RAJESWARI SETHURAMAN

Assistant Professor (SG),
Department of Computer Applications,
Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India.

Abstract

Cloud computing has become the backbone of digital transformation, offering scalability, flexibility, and cost efficiency. However, one of the major challenges organizations face is vendor lock-in, where heavy reliance on a single cloud provider creates barriers to migration, increases long-term costs, and reduces strategic flexibility. Proprietary tools, unique APIs, and non-standardized services make it difficult for businesses to adopt a multi-cloud approach or switch providers when needed. This paper addresses the problem of cloud vendor lock-in and proposes solutions through cloud-agnostic architectures, containerization technologies such as Docker and Kubernetes, and hybrid or multi-cloud deployment models. By adopting these strategies, organizations can reduce dependency on a single provider, enhance resilience, optimize costs, and leverage the best services across multiple platforms.

Keywords: Cloud Computing, Docker, Storage, Migration, Services.

Block chain - Driven Cyber Security Framework for Next-Generation Networks

Ms. S. Harshini
Student

Mr. V.S. Nishok
Student

Ms. R. Viji
Student

Ms. R. Nikhasini
Student

Department of Computer Applications
Dr. Mahalingam College of Engineering and Technology
Pollachi, Tamil Nadu, India.
rnikashini@gmail.com

Abstract

The emergence of next-generation networks such as 5G, 6G, and large-scale IoT ecosystems has transformed the way digital services are delivered. While these systems promise ultra-low latency and massive connectivity, they are also exposed to serious security risks including unauthorized access, data tampering, and network intrusions. This study introduces a block chain-driven framework that addresses these challenges by providing decentralized, transparent, and tamper-resistant security mechanisms. The main objective is to design a model that strengthens trust, safeguards data integrity, and enables secure communication in distributed environments. The proposed methodology combines block chain with advanced cryptographic protocols to support identity management, secure data exchange, and intrusion resistance. Experimental evaluation and simulated network scenarios are used to analyze the framework's scalability, latency, and overall security performance. Preliminary results show that the model can minimize single-point vulnerabilities, improve authentication efficiency, and enhance resilience against coordinated cyber attacks. Compared with traditional centralized security systems, the block chain-based approach demonstrates better fault tolerance and trust management, especially in large-scale, heterogeneous networks. The significance of this work lies in its applicability to critical domains such as healthcare, finance, transportation, and smart cities, where secure and reliable communication is essential. By integrating block chain with network security, the research contributes toward building trustworthy, scalable, and sustainable infrastructures for future communication technologies.

Keywords: Blockchain, Cyber security, Next-Generation Networks, IoT, 5G/6G, Secure Communication

Generative Adversarial Networks (GANs) for Image Super- Resolution

Ms. P. Brindha

Student

Ms. S. Harshini

Student

Ms. G. Preethi

Student

Ms. M. Subhashini

Student

Dr. S. Raja Rajeswari Sethuraman

Assistant Professor,

Department of Computer Applications,

Dr. Mahalingam College of Engineering and Technology,

Pollachi, Tamil Nadu, India

Abstract

Super-resolution (SR) is the process of reconstructing a high-resolution (HR) image from a low-resolution (LR) input, a crucial task in modern computer vision. Traditional interpolation and CNN-based methods often produce blurred or unrealistic outputs, especially at high upscaling factors. Generative Adversarial Networks (GANs), with their adversarial learning paradigm, have emerged as an effective solution for generating photo-realistic images with fine details. This paper explores state-of-the-art GAN-based SR methods—focusing on architectures like SRGAN and ESRGAN—demonstrating their advantages for practical domains like medical imaging, satellite observation, and video restoration.

Keywords: Deep Learning, Generative Adversarial Networks, Image Super-Resolution, SRGAN , Computer Vision , high Resolution Reconstruction .

A Survey on Prediction of Human Diseases Using Machine Learning with Big Data Analytics

Mr. M. Nagarajan

Research scholar
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. A.S. Arunachalam

Professor
Department of Computer Science and Information Technology
School of Computing Sciences
Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

The rapid growth of healthcare data, which includes electronic health records (EHRs), medical imaging, wearable devices, and genetic sequencing, has created new opportunities and challenges for disease prediction. Traditional data processing methods are unable to handle the volume, speed, and variety of medical big data. Machine Learning (ML), when combined with Big Data Analytics (BDA), has emerged as a powerful paradigm for disease prediction, diagnosis, and tailored treatment. This study presents a thorough examination of strategies for combining machine learning and big data analytics in healthcare. It investigates advanced models, data sources, analytical frameworks, and disease prediction applications. This paper also examines challenges such as data heterogeneity, scalability, security, interpretability, and ethical problems, as well as potential future research directions.

Keywords: *Electronic Health Records (EHRs), Human Diseases, Machine Learning (ML), and Big Data Analytics (BDA).*

A Survey on Multi-Agent-Based Framework for Resource Allocation in Cloud Computing

Mr. S. Pugazhendi

Research scholar

Department of Computer Science and Information Technology

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Dr. A.S. Arunachalam

Professor

Department of Computer Science and Information Technology

School of Computing Sciences

Vels Institute of Science Technology and Advanced Studies (VISTAS) Pallavaram

Abstract

Cloud computing has become the cornerstone of modern information technology because it offers scalable, affordable, and on-demand access to computational resources. Efficient resource allocation remains a significant challenge in cloud systems due to varying workloads, varied resources, and service-level agreement (SLA) requirements. As a promising paradigm for describing, coordinating, and negotiating distributed environments, Multi-Agent Systems (MAS) have attracted a lot of attention. This review looks at the latest research on multi-agent-based frameworks for allocating cloud computing resources. An overview of coordination systems, performance metrics, allocation strategies, and agent-based models is provided. It also discusses significant challenges, open issues, and future research directions for creating multi-agent-driven cloud resource management.

Keywords: Cloud Computing, Multi Agent System, Resource Allocation, Service-Level Agreement (SLA)

Advancing Oncology Diagnostics: AI-Enabled Early Detection of Lung Cancer through Hybrid Histological Image Analysis

Dr V. Malathi

Assistant Professor
Department of Computer Applications and Technology
SRM Arts and Science College, Kattankulathur

Ms. D. Nivetha

Student
Department of Computer Applications and Technology
SRM Arts and Science College
Kattankulathur

Abstract

Lung Cancer (LC) remains one of the leading causes of cancer-related mortality worldwide, with early detection being crucial to improving patient outcomes. Traditional diagnostic methods often struggle with issues of accuracy, speed, and accessibility, particularly in early stages. The proposed project addresses these challenges by leveraging Artificial Intelligence (AI) to enhance LC detection through advanced histological image analysis. The study evaluates Machine Learning (ML) algorithms, including Light Gradient Boosting Machine (LightGBM) and Extreme Gradient Boosting (XGBoost), selected for their strong performance with large datasets. A dual-platform approach using MATLAB and Python enables a comprehensive comparison of these algorithms across key metrics such as accuracy, sensitivity, specificity, and computational efficiency. The findings offer valuable insights into each algorithm's strengths and limitations, contributing to the refinement of AI-based techniques for early LC detection. By bridging theoretical advancements with practical applications, this research highlights the transformative potential of AI in medical imaging and aims to guide future clinical implementations to improve diagnostic accuracy and patient outcomes.

Keywords: *Artificial Intelligence (AI), Lung Cancer Detection, Oncology Diagnostics, Histological Image Analysis*

Blockchain based Privacy-Preserving Authentication and Key Verification using PBDS Framework

Mr. M.P. Kumar

Research Scholar,

Department of Computer Science,

Vels Institute of Science, Technology & Advanced Studies (VISTAS),

kumarm.p.mca@gmail.com

Dr. A. Akila

Associate Professor,

Department of Computer Science and Information Technology,

Vels Institute of Science, Technology & Advanced Studies (VISTAS),

akila.scs@vistas.ac.in

Abstract

The block chain based Industrial Internet of Things (IIoT) system is confirmed a secure data sharing, which maintained user privacy by decentralization and cryptographic methods. It confirms privacy through protecting a data during a transmission, permit only for authorized entities to communicate with IIoT networks. However, IIoT secure and privacy preserving is difficult, a traditional method addressed only privacy and authentication, while static block chain consensus due to incurring overhead and lack of adaptive access control for dynamic environments. To overcome these challenges, the proposed Password Based Key Derivation Function2 (PBKDF2) integrated with Blowfish Encryption (BE) and Hash Message A along with Digital Signature named as PBDS which method is associated with block chain, Zero Knowledge Proof (ZKP) are used to prevent a data sharing and handling an unauthorized access. The PBDS is generating an adaptive and fine-grained access control, confirming a various untrusted entity is flexible and securely manage resources relies on contexts and policies. The experimental results show that proposed PBDS attains a less Encryption Time (ET) and Decryption Time (DT) of 6.4513 and 5.9836 respectively.

Keywords: Authentication, Blockchain, Blowfish Encryption, Digital Signature, Hash Message Authentication Code, Zero Knowledge proof.

Impact of Preprocessing Methods on Skin Lesion Analysis: A Comparative Study

Ms. G. Evanjalini Indira

Research Scholar

Department of Computer Science and Applications
The Gandhigram Rural Institute (Deemed to be University)
Gandhigram, India.
evanjalini83@gmail.com

Dr. M. Mary Shanthi Rani

Professor,

Department of Computer Science and Applications
The Gandhigram Rural Institute (Deemed to be University)
Gandhigram, India.
drmmsrani@gmail.com

Abstract

Skin cancer is one of the worst diseases in the world, and delayed or erroneous diagnostics contribute to high fatality rates. Early detection is difficult for improving survival rates, but clinical diagnosis remains tough. Deep learning-based frameworks have shown great promise in confront these difficulties by allowing for robust images pre-processing, automatic feature extraction, and accurate categorization. This research finds the effects of pre-processing approaches on the presentation of skin cancer detection systems. Initially, Contrast Limited Adaptive Histogram Equalization (CLAHE) and Adaptive Histogram Equalization When compared to previous research, AHEs performed badly. To solve this problem, Gaussian blur was pre-owned with CLAHE and AHE to lessen noise while improving image quality. The better pre-processing pipeline was trial on the HAM10000 dataset, which consists of couple images of benign and malignant skin lesions. Pair testing revealed that combining Gaussian blur with CLAHE or AHE enhanced detection performance while outperforming unique approaches. The findings highlight the importance of effective pre-processing procedures in improving image quality and diagnostic accuracy, clarifying for the advancement of deep learning-based skin cancer detection frameworks.

Keywords: *Skin Cancer Detection, Deep Learning, Image Detection, Feature Extraction, Image Segmentation, Pre-processing, Contrast Enhancement, HAM10000 Dataset, Benign Vs Malignant Lesions.*

AI-Powered Cardiac Risk Classification: Hyper Capsule LSTM–GRNN with Optimal Swarm Intelligence for Arrhythmia Prediction

Mr. G. S. Manish

Research scholar

Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai-600117, Tamilnadu, India.
manishgurajala.s@gmail.com

Dr.S. Perumal

Professor

Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai-600117, Tamilnadu, India.
perumal.scs@velsuniv.ac.in

Abstract

The rising prevalence of cardiovascular diseases, particularly arrhythmias, highlights the urgent need for advanced predictive models that can enhance early diagnosis and treatment. This study proposes the Cardiotech Next AI Model, an intelligent healthcare framework that integrates a hybrid predictive architecture combining Hyper Capsule Long Short-Term Memory (LSTM) and Gated Recurrent Neural Networks (GRNN), supported by an Optimal Swarm Intelligence Technique (OSIT) for superior accuracy in arrhythmia and cardiovascular disease prediction. The methodology begins with C-score normalization, which standardizes cardiac data to ensure optimal model performance. Next, the Cardiac Disease Impact Behavioural Scaling Rate is applied to evaluate the influence of behavioural factors on heart health, offering a deeper understanding of risk contributors. Feature selection is then performed using OSIT, which mimics swarm intelligence to identify the most significant predictors of cardiovascular risk, resulting in a robust and generalizable model. In the final stage, the Hyper Capsule LSTM-GRNN architecture is employed to classify patients into multiple disease risk categories by leveraging the temporal dependencies present in cardiac signals. This approach not only enhances prediction accuracy but also improves interpretability, providing meaningful insights into the dynamics of arrhythmias and related cardiovascular disorders. Experimental results demonstrate substantial improvement over conventional methods, achieving precision = 0.97, accuracy = 0.82, recall = 0.84, and F1-score = 0.91. These outcomes confirm the effectiveness of combining advanced AI architectures with intelligent feature selection for proactive cardiovascular health management and risk classification using multiclass labels.

Keywords: *Cardio tech Next AI Model, Hyper Capsule LSTM, Gated Recurrent Neural Networks, Optimal Swarm Intelligence Technique, C-score Normalization, Cardiovascular Risk Prediction.*

Cataloging of Soil Using Res Net Pre-Trained Model in Deep Learning

Dr G Lalitha

Department of Computer Science and Applications (MCA)
SRMIST Ramapuram, Chennai - 600089
lalithaunom@gmail.com

Dr K Hazeena

Department of Computer Application
BS Abdur Rahman Crescent Institute Of Science and Technology.
Chennai -600048
haseenakader67612@gmail.com

Abstract

For appropriately managing fields and crops, it is essential to understand soil properties. There are drawbacks to the conventional methods currently used for collecting a large amount of data from agricultural lands. A convolutional neural network is a deep learning algorithm that specializes in image classification, and developing soil property prediction techniques using this algorithm will be extremely beneficial to soil management. We present convolution neural network models for estimating water content and dry density using soil surface images. Soil surface images were taken with a conventional digital camera. The range of water content and dry density were determined considering general upland soil conditions. Each image was divided into segmented images and used for model training and validation. The developed model confirmed that the model can learn soil features through appropriate image argumentation of a few original soil surface images. Additionally, it was possible to predict the soil water content in a situation where various soil dry density conditions were considered.

Keywords: VGG, VGGNet, Convolutional Neural Network, Ensemble model.

A Comparative Study of AHE and CLAHE Image Enhancement Techniques for Chest X-Ray Pneumonia Diagnosis

Ms. M. Shanmuga Priya

Research Scholar

Department of Computer Science and Applications
The Gandhigram Rural Institute (Deemed to be University)
Gandhigram, India.

Dr.M.Mary Shanthi Rani

Professor

Department of Computer Science and Applications
The Gandhigram Rural Institute (Deemed to be University)
Gandhigram, India.

Abstract

Pneumonia is a significant cause of death worldwide, especially for susceptible populations including children under five and adults over 65. A large collection of microorganisms, which differ widely throughout the world, can cause it, including viruses, fungi, and bacteria. One or both each of the human lungs are impacted by this illness. One-third of deaths in India happen due to pneumonia, according to the World Health Organization, underscoring the urgent demand for better diagnosis techniques. While traditional methods may be laborious and subjective, new developments in deep learning and image processing present encouraging solutions. However, pre-processing techniques are crucial since the performance of such models is highly dependent on image quality. In order to improve diagnostic accuracy, techniques like noise reduction, histogram equalization, contrast enhancement, and normalizing are used to input images. In the present study, 5,863 JPEG images from the Kermany dataset are used to compare two enhancement approaches, CLAHE and AHE, applied to chest X-ray images. To determine the statistical significance of the differences between the two approaches, a paired t-test is utilized. A variety of evaluation criteria, such as Mean Squared Error (MSE), Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR), are used to assess the performance of the enhanced images. The objective is to identify the method that enhances image quality the most for more precise and reliable pneumonia identification. To further statistically assess the enhancing techniques and ascertain the significance of their variations, a paired t-test is applied.

Keywords: Pneumonia, CXR Images, Kermany dataset, Image Pre-processing, Image Enhancement Techniques, Histogram Equalization, CLAHE, AHE, Deep Learning.

A Review on Cloud Computing in Cyber Forensics: Opportunities, Challenges and Future Directions

Dr. S.Sweetlin Susilabai

Assistant Professor

Department of Computer Science w/s in Cyber Security,
Faculty of Science and Humanities
SRM Institute of Science and Technology,
Ramapuram, Chennai

Mr. B. Prem Chander

Student

Department of Computer Science w/s in Cyber Security
SRM Institute of Science and Technology,
Ramapuram, Chennai

Mr. R.Sharlin

Student,

Department of Computer Science w/s in Cyber Security,
SRM Institute of Science and Technology,
Ramapuram, Chennai

Abstract

Cloud computing has transformed the digital ecosystem by providing scalable, on- demand resources for data storage, computation, and communication. However, it has also created new challenges for the field of cyber forensics, which is concerned with the investigation and analysis of digital crimes. As distributed, virtualized, and multi- tenant cloud environments replace traditional physical infrastructures, investigator face difficulties in locating, obtaining, preserving, and analyzing evidence. Forensic procedures are made more difficult by jurisdictional concerns, data volatility, reliance on cloud service providers, and encryption. Notwithstanding these limitations, cloud computing has a lot to offer forensic investigations, including scalable storage for large datasets, distant evidence accessibility, and collaborative analysis via cloud- based forensic platforms. New ideas that aim to improve evidence processing and speed up investigations include Forensics-as-a-Service (FaaS), block chain-enabled chain of custody, and AI-driven cloud forensic tools. In order to handle crimes in cloud environments, this article examines the role of cloud computing in cyber forensics, stressing its uses, difficulties, and possible solutions. It also highlights the necessity of international legal collaboration and cutting-edge forensic techniques.

Keywords: Cloud Computing, Cyber Forensics, Digital Evidence, Forensics-as-a- Services (FaaS) , Cloud Services Provides (CSPs), Digital Investigation .

Advance Multi Modal Transformer Based Model Short and Long Term Stock Price Movement Forecasting

Ms. R. Vindhiya

Student

Department of Computer Science and Engineering
Vels Institute of Science and Advanced Studies, Chennai.

Ms. V. Kirthika

Student

Department of Computer Science and Engineering
Vels Institute of Science and Advanced Studies, Chennai.

Dr. A. Saritha

Assistant Professor

Department of Computer Science and Engineering
Vels Institute of Science and Advanced Studies, Chennai.

Abstract

Stock market forecasting remains a challenging task due to the highly volatile, dynamic, and non-linear nature of financial data. Traditional statistical and machine learning approaches often fail to capture complex dependencies across multiple modalities such as historical price data, financial news, social sentiment, and macroeconomic indicators. In this work, we propose an advanced multi-modal transformer-based model that integrates heterogeneous data sources to forecast both short-term and long-term stock price movements. The model leverages transformer architectures to capture temporal dependencies, attention mechanisms for feature prioritization, and cross-modal fusion for combining structured (numerical time-series) and unstructured (textual or sentiment) information. Experimental results on benchmark financial datasets demonstrate improved forecasting accuracy, reduced error rates, and enhanced robustness compared to conventional deep learning baselines. The proposed framework offers significant potential for investors, traders, and financial analysts in making data-driven decisions under uncertainty.

Keywords: Multi-Modal Learning, Transformer Model, Stock Price Forecasting, Time Series Analysis

Analysis of Predicting Residential Property Prices with Multi-Feature Machine Learning Approaches

Mr. S. Vetri velan

Research Scholar,
School of Computer Application,
Karpagam College of Engineering,
Coimbatore.

Dr. P. Shanthi

Professor,
School of Computer Application,
Karpagam College of Engineering,
Coimbatore

Abstract

Accurate prediction of residential property prices is crucial for informed decision-making in the real estate sector. This study presents a comparative analysis of three prominent machine learning models—Linear Regression (LR), Random Forest (RF), and XGBoost (Extreme Gradient Boosting)—for house price prediction using multi-feature datasets. The dataset incorporates a wide range of property attributes, including structural details (e.g., area, number of rooms), locational factors (e.g., proximity to schools, transport), and categorical variables (e.g., property type). While Linear Regression offers simplicity and interpretability, it falls short in modeling nonlinear relationships. Random Forest and XGBoost, as ensemble-based methods, demonstrate superior performance by capturing complex feature interactions and reducing overfitting. The models are evaluated using RMSE, MAE, and R2 score, and the results indicate that XGBoost consistently outperforms other models in prediction accuracy, followed closely by Random Forest. The study highlights the importance of feature-rich datasets and ensemble techniques in building robust and scalable real estate valuation systems. The outcomes of this research serve as a foundation for developing intelligent real estate analytics tools, enabling better market valuation and investment strategies.

Keywords: House Price Prediction, Machine Learning, Random Forest, XGBoost, Linear Regression, Multi-Feature Dataset, Real Estate Analytics, Ensemble Learning, Structured Data, Property Valuation, Predictive Modeling, Regression Algorithms

Explainable Quantum-Enhanced Machine Learning for Diabetes Prediction

Mr. N. Lokesh

Student

School of Computer Applications
Karpagam College of Engineering,
Coimbatore.

Dr. P. Shanthi

Professor

School of Computer Applications
Karpagam College of Engineering,
Coimbatore

Abstract

Diabetes is a chronic metabolic disorder characterized by high blood sugar levels over a long time. It may be due to the insufficient production of insulin in the body or the ineffective use of the latter. When early diagnosis is not performed, or the condition is left untreated, harm begins to develop, such as in cardiovascular diseases, neuropathy, visual disturbance, and nephropathy. In recent years, diabetes has become an uncontrollable epidemic worldwide, especially in regions of poor medical infrastructure and healthcare access. In these communities, the lack of timely diagnostic tools and treatment highlights the immediate need for intelligent, flexible, and accessible solutions for early detection. This study attempts to propose a hybrid model of diagnosis aimed at solving this wide concern by combining traditional machine-learning techniques with quantum computing capabilities and an interpretable framework for model decisions. The model isolates important clinical features from patient data using an XGBoost algorithm, and the quantum classification task using quantum-enhanced computation is executed using quantum support vector machines (QSVM). The interpretative technique SHAP (SHapley Additive exPlanations) is then applied to show how changes in the value of each feature affect the output of the model for more transparency among clinicians. This overall approach gives a reliable prediction work and, hence, one that the doctors will trust and understand its recommendations. The amalgamation of these cutting-edge technologies thus offers a robust, scalable, and pragmatic tool for diabetes diagnosis in real-life healthcare settings.

Keywords: *Diabetes Prediction, Quantum Machine Learning, Explainable AI, SHAP, XGBoost, QSVM, Medical Diagnosis, Hybrid AI Systems, Clinical Decision Support, Transparent Healthcare AI.*

Knowledge Hub Sharing and Gaining Knowledge using MERN Staack

Ms. D. Harini

Student, Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies, Chennai, Tamil Nadu, India

Mr. V. Divakar

Student, Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies, Chennai, Tamil Nadu, India

Mr. A. Vengadanathan

Student, Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies, Chennai, Tamil Nadu, India

Dr. A. Saritha

Assistant Professor, Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies, Chennai, Tamil Nadu, India

Abstract

In the rapidly evolving landscape of technology, knowledge sharing and acquisition have become pivotal for both individual and organizational growth. This paper explores the concept of knowledge hubs and the role of the MERN stack (MongoDB, Express.js, React.js, Node.js) in facilitating knowledge sharing and gaining knowledge. The research highlights the significance of knowledge hubs in educational and professional settings, emphasizing how the MERN stack can be leveraged to create efficient, scalable, and user-friendly applications that promote collaborative learning and knowledge exchange. Through a comprehensive literature review, the paper identifies existing methodologies and technologies that support knowledge sharing, followed by a detailed analysis of the MERN stack's components and their contributions to developing knowledge-sharing platforms. The findings suggest that the MERN stack not only enhances the functionality of knowledge hubs but also fosters a culture of continuous learning and collaboration. The paper concludes with implications for future research and practical applications in various domains.

Keywords: *Web Application Development, MongoDB, Express.js, React.js, Node.js*

Real-Time Cyber Threat Detection Using Adaptive AI Security Monitoring: A Machine Learning Approach

Krishna Kadel

Student

Department of Computer Science and Engineering

VELS Institute of Science, Technology & Advanced Studies (VISTAS) Chennai, Tamil Nadu, India

kadelk598@gmail.com

Vivek Kumar.R

Student

Department of Computer Science and Engineering

VELS Institute of Science, Technology & Advanced Studies (VISTAS) Chennai, Tamil Nadu, India

Fahad Sheriff

Student

Department of Computer Science and Engineering

VELS Institute of Science, Technology & Advanced Studies (VISTAS) Chennai, Tamil Nadu, India

P. V. Hemavathi

Assistant Professor

Department of Computer Science and Engineering

VELS Institute of Science, Technology & Advanced Studies (VISTAS) Chennai, Tamil Nadu, India

pvhemavathi.se@vistas.ac.in

Abstract

Traditional signature-based cyber security systems are inadequate against sophisticated modern threats including phishing, malware, and unauthorized access. To address this, an AI powered threat detection framework is proposed using machine learning algorithms for proactive defense. Through supervised learning, anomaly detection, CNNs, RNNs, and UEBA behavioral analytics, the system monitors network traffic and user patterns. When threats are detected, automated alerts and containment protocols activate instantly. NLP components analyze communications for social engineering. Real-time threat intelligence enables adaptive learning. Model validation prevents adversarial attacks. Continuous retraining maintains accuracy. As a result, organizational security is enhanced through intelligent AI-driven cyber security defense.

Keywords: *AI-artificial Intelligence, CNN-Convolutional Neural Network, RNN-Recurrent Neural Network, UEBA-User Entity Behaviour, Analytics, NLP-Natural Language Processing*

CONNECTEASE

Nanthini K

Pre-final Year,

B.Tech Computer Science and Engineering – Specialization in Artificial Intelligence and Data Science
Vels Institute of Science, Technology and Advanced Studies, Chennai, India
nanthini19052005@gmail.com

Rakshana R. P

Pre-final Year,

B.Tech Computer Science and Engineering – Specialization in Artificial Intelligence and Data Science
Vels Institute of Science, Technology and Advance Studies, Chennai, India
rakshanarprakshu@gmail.com

Abstract

SMS-based control system has been developed to enable the reliable operation of digital and IoT devices in regions with limited or no internet connectivity. In many rural or resource-constrained environments, the absence of stable internet access renders conventional IoT solutions impractical. The proposed system, ConnectEase, addresses this limitation by utilizing the wide availability of GSM networks, allowing users to send and receive operational commands through standard SMS communication. This approach ensures accessibility, reduces dependency on smartphones or data networks, and extends the reach of smart technologies to underserved areas. Keywords-Remote access, smartphone control, mobile security, SMS-based communication, data protection, real-time alerts, device management, encrypted messaging, Android implementation, offline access.

Keywords: Communication Platform, Collaboration Tool, Social Connectivity, Real-Time Chat,

Online Retail Recommendation System

Ms. PT Sreeyakrishna

Student

Department of CSE AI & DS,
Vels Institute of Science, Technology and Advance Studies

Ms. Nikitha Menon

Student,

Department of CSE AI & DS,
Vels Institute of Science, Technology and Advance Studies

Ms. V Bharathi

Assistant Professor

Department of CSE AI & DS,
Vels Institute of Science, Technology and Advance Studies

Abstract

In today's digital marketplace, personalization has become essential for enhancing customer satisfaction and driving business growth. With the rapid increase in online shopping, consumers are often overwhelmed by the vast number of available products, making it difficult to identify items of real interest. To address this challenge, we propose an artificial intelligence (AI) based Online Retail Recommendation System capable of generating personalized product suggestions in real time. The system leverages collaborative filtering techniques, which analyze historical purchase data and behavioral patterns to identify similarities between users and predict likely preferences. Data preprocessing ensures that raw transaction records are cleaned and structured, after which a customer-item interaction matrix is constructed. Key features such as purchase frequency, co-occurrence of items, and user similarity measures are extracted and analyzed to generate recommendations. The system is trained on diverse datasets of user activity, enabling it to refine accuracy while minimizing irrelevant suggestions. Potential enhancements using hybrid approaches combine collaborative filtering with content-based and demographic methods to further improve reliability. Once recommendations are generated, they can be dynamically presented to customers through e-commerce platforms, personalized dashboards, or targeted campaigns. For businesses, the system provides actionable insights into consumer trends, enabling improved decision-making in inventory management, marketing, and customer engagement strategies. Its adaptability allows deployment across various online retail domains, ranging from fashion and electronics to groceries and lifestyle products. By translating raw behavioural data into meaningful recommendations, this work demonstrates how data-driven approaches enhance customer experiences, strengthen brand loyalty, and create measurable impacts on revenue generation.

Keywords: Online Retail, Recommendation System, E-commerce, Personalized Suggestions, Data Mining

Crime Prevention through Scream Detection

Sudharsan G

Student

Department of CSE AI & DS,

Vels Institute of Science, Technology and Advanced Studies , Chennai, India

Kishore V J

Student

Department of CSE AI & DS,

Vels Institute of Science, Technology and Advanced Studies , Chennai, India

Raghul Sekar

Student

Department of CSE AI & DS,

Vels Institute of Science, Technology and Advanced Studies , Chennai, India

Ms.Kalpana

Assistant Professor

Department of CSE AI & DS,

Vels Institute of Science, Technology and Advanced Studies , Chennai, India

Abstract

The increasing rate of crimes, especially in urban areas, calls for the development of advanced technologies to ensure public safety and timely intervention. Crime Prevention through Scream Detection, presents an AI-powered system designed to detect human screams in real-time using audio analysis techniques. The primary objective is to identify distress situations, such as assaults or emergencies, where a scream is likely to occur, and automatically trigger alerts to nearby authorities or security systems. The system employs machine learning models trained on a dataset containing both positive (scream) and negative (non-scream) audio samples. By extracting relevant features such as pitch, frequency, and amplitude, the model effectively differentiates between normal environmental sounds and actual screams. Once a scream is detected, the system can be configured to activate alarms, notify law enforcement, or record the event for further investigation. The system can be integrated with smart surveillance cameras, home security systems, and mobile applications, making it versatile for deployment in residential areas, public spaces, transportation networks, and institutional environments. Real-time detection enables faster response and timely intervention, thereby preventing crimes or minimizing their severity.

Keywords: *Crime Prevention, Scream Detection, Audio Analysis, Machine Learning, Real-time Detection, Public Safety.*

Predictive Modelling for Customer Purchase Behaviour: A Logistic Regression Approach Based on Age and Estimated Salary

Selvakumar S

UG Student

Department of CSE AI&DS

VISTAS

Yogeshwaramoorthi K

UG Student

Department of CSE AI&DS

VISTAS

M.G.Jegathambal

Assistant Professor

Department of CSE AI&DS

VISTAS

Abstract

Customer purchase prediction has become a critical requirement in the insurance industry, where businesses strive to maximize customer acquisition while minimizing marketing costs. Accurate forecasting of whether a potential customer will purchase an insurance policy allows companies to focus on high potential leads and optimize their strategies. In this study, we propose a predictive modelling approach using logistic regression to classify customers based on two key demographic features: Age and Estimated Salary. A dataset of over 1,000 customer records was pre-processed, visualized, and divided into training and testing subsets using an 80:20 ratio. The logistic regression model was trained to identify significant patterns influencing purchase decisions and to estimate the probability of policy adoption. To enhance usability, the trained model was deployed in a Streamlit based web application that includes secure user authentication, interactive input fields, decision boundary visualization, and a leaderboard to track predictive outcomes. Experimental results demonstrate that the logistic regression model achieves an accuracy of approximately 90%, with strong interpretability through coefficient analysis and decision boundary visualization. This work highlights the potential of combining machine learning models with lightweight, interactive applications to support business analysts and decision-makers. Future work will focus on incorporating additional demographic and behavioral features, applying advanced ensemble models, and integrating large-scale real world datasets to further enhance prediction performance.

Keywords: Logistic Regression, Customer Purchase Prediction, Insurance Analytics, Streamlit Application, Decision Boundary Visualization, Predictive Modeling

360 Degree Business Analysis of Online Delivery Apps Data Analytics with Power BI

Ms. P.S. Haripriya

Student

Department of Information Technology and Computer Science
Vels Institute of Science, Technology and Advanced studies, Chennai

Ms. V. Bhuvaneshwari

Student

Department of Information Technology And Computer Science
Vels Institute of Science, Technology and Advanced studies, Chennai

Abstract

In the digital world, Online food ordering system is mainly designed primarily function for use in the food delivery industry. This system will allow hotels and restaurants to increase online food ordering such type of business. The customers can be selected food menu items just few minutes. In the modern food industries allows to quickly and easily delivery on customer place. Restaurant employee then use these orders through an easy to delivery on customer place easy find out navigate graphical interface for efficient processing.

Keywords: Power BI, Food Delivery apps, Data Analytics, Real-Time Order Tracking, Predictive Analysis.

Bridging the Language Barrier: AI for Tribal and Regional Languages in India

R Ramyadevi , Assistant Professor
ramyadeviresearch@gmail.com

Naveena T , Assistant Professor
naveena.220597@gmail.com

S. Abinaya , Student
sa0110@srmist.edu.in

Punitha G, Student
pg6394@srmist.edu.in

Department of Computer Science & Applications, SRMIST, Chennai, Ramapuram

Abstract

India's linguistic landscape is one of the richest in the world: hundreds of regional and tribal languages coexist alongside major scheduled languages. Yet many of these languages are severely under-represented in digital datasets and AI systems, creating barriers to access, education, governance, and cultural preservation. This study examines the role of artificial intelligence in bridging the language barrier for tribal and regional languages in India, combining dataset creation, algorithmic adaptation, and community-centric deployment strategies. We present a structured framework for (1) data collection and ethical curation in low-resource settings; (2) model adaptation using transfer learning, self-training and multilingual pretraining; and (3) participatory deployment that centers native speakers in annotation and evaluation. The literature review surveys recent works across speech corpora, ASR techniques, neural machine translation, TTS, and policy/ethics perspectives to identify practical methods and sustainable interventions. Methodology combines mixed methods: quantitative benchmarking of adapted ASR/MT models against baseline systems and qualitative interviews with community stakeholders to measure adoption readiness and cultural acceptability. Illustrative (simulated) results show substantial WER gains after targeted adaptation for under-represented languages and highlight resource thresholds where incremental data collection yields diminishing returns. We discuss governance, intellectual property, and capacity-building recommendations - emphasizing community ownership, open datasets, and modular tooling to allow low-cost local deployments. The paper concludes with a prioritized research roadmap for policymakers, technologists, and tribal organizations aimed at preserving linguistic diversity while delivering tangible socioeconomic value. The approach balances technical feasibility with social responsibility and can be adapted for India's diverse linguistic contexts.

Keywords: *Automatic Speech Recognition (ASR), Neural Machine Translation (NMT) , Text-to-Speech (TTS)*

Improving Skin Lesion Image Quality with Adaptive Histogram Equalization

Naveena T, Assistant Professor
naveena.220597@gmail.com

R Ramyadevi, Assistant Professor,
ramyadeviresearch@gmail.com

A Zaara Farhath, Student
za3434@srmist.edu.in

Prisila Sheryl, Student,
ps6220@srmist.edu.in

Department of Computer Science & Applications
SRMIST, Chennai, Ramapuram

Abstract

Skin acts as a natural indicator of the health. If the health is good it will be visibly indicated in our skin in the same way if our health is not in a good condition and we are suffering from any kind of disease it is also clearly indicated in our skin. Even if the particular disease can not be identified it is for sure it can be identified that the person is suffering from some kind of disease. If the person has skin disease they may also undergo certain psychological problems such as depression, fear of facing the society, they may be anxious when the disease will get cured. The skin disease affected image mostly will not be clear in order to get the clear cut picture of the affected image the image has to be pre processed. In pre processing the image will be enhanced so that the affected area will be more clear and visible for diagnosing the disease. The unwanted details in the image will be ignored so it is clearer to view the image. For the process of pre processing the histogram equalization is implemented. The histogram equalization is the process in the image contrast can be improved by adjusting the intensity values. The accurate diagnosis of dermatological conditions, particularly skin cancers, is critically dependent on the visual quality of skin lesion images. However, dermatoscopic and clinical photographs often suffer from poor contrast, uneven illumination, and shadows, which can obscure subtle features such as lesion boundaries, color variations, and textural patterns. These visual artifacts pose a significant challenge for both human clinicians and automated AI-based diagnostic systems, potentially leading to misdiagnosis. In this paper how the histogram equalization is used in pre processing can be seen clearly mainly for the skin disease images.

Keywords : *histogram equalization, pre processing, image processing, skin disease images*

Deepfake Image Detection Using Color Channel Co-occurrence Metrics and Convolutional Neural Networks

Mr. U. Velmurugan

Student,

School of Computer Applications,
Karpagam College of Engineering, Coimbatore

Dr. P. Shanthi

Professor,

School of Computer Applications,
Karpagam College of Engineering, Coimbatore

Abstract

Deepfake technology has advanced rapidly, enabling the creation of highly realistic yet manipulated visual content, which poses significant threats to digital trust, privacy, and security. This paper presents a robust deepfake image detection method that leverages color channel co-occurrence matrices in combination with Convolutional Neural Networks (CNNs). The approach begins by extracting co-occurrence matrices from each RGB channel of the input image, capturing statistical dependencies between neighboring pixel intensities. These matrices encode subtle texture and noise patterns introduced by generative adversarial networks (GANs), which are often imperceptible to the human eye. The extracted co-occurrence features are fed into a CNN architecture to automatically learn discriminative representations for classification into real or fake categories. Experimental evaluations on benchmark datasets such as FaceForensics++ and StyleGAN-generated images demonstrate detection accuracies between 88% and 93%, proving the method's effectiveness in identifying GAN-specific artifacts. The proposed approach achieves high detection rates while maintaining computational efficiency, making it suitable for real-time and resource-constrained applications. This work contributes to the growing need for reliable deepfake detection solutions and highlights the potential of combining statistical texture analysis with deep learning frameworks.

Keywords: *Deepfake detection, Co-occurrence matrix, Convolutional Neural Network, GAN artifact analysis.*

A Comprehensive Analysis of Machine Learning Methods for Predicting Attention Deficit Hyperactivity Disorder (ADHD)

Mr. Kalasha

Research scholar

Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai-600117

Dr.S. Perumal

Professor

Department of Computer Science & Information Technology
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai-600117

Abstract

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most common neurodevelopmental diseases, characterized by inattention, impulsivity, and hyperactivity that impair academic, vocational, and social performance. Early and precise ADHD prediction is essential for timely interventions and personalised treatment plans. Machine learning (ML) has evolved in recent years as an effective technique for detecting complicated patterns in clinical, behavioral, neuroimaging, and genetic data that traditional diagnostic approaches may ignore. This paper provides a comprehensive analysis of various machine learning methods used to predict ADHD, including supervised learning approaches like support vector machines, decision trees, random forests, and deep learning models, as well as unsupervised techniques for data clustering and feature extraction. The study highlights the benefits and drawbacks of each approach, concentrating on issues such as feature selection, data imbalance, interpretability, and generalisability. The study also examines how multimodal data integration and big data analytics may improve prediction accuracy. The findings indicate that when combined with effective preprocessing techniques, hybrid and ensemble models hold a great deal of promise to enhance ADHD diagnosis and prognosis. This comprehensive review aims to provide clinicians and scholars with crucial information on the most recent advancements, challenges, and possible directions in the use of machine learning to detect ADHD.

Keywords: *Attention Deficit Hyperactivity Disorder (ADHD), convolutional neural networks, deep learning, deep neural networks, disease diagnosis, heart disease, machine learning.*

VOCALMORPH – Voice Identity Conversion in Music Videos using Deep Learning

Ms. Shifaali B R,

UG STUDENT, Department of CSE AI&DS, VELS University, Chennai, India

Ms. Roshitaahari

UG STUDENT, Department of CSE AI&DS, VELS University, Chennai, India

roshu250704@gmail.com

Mr. Varadharajan S

Asst.Prof, Department of CSE, VISTAS, Chennai, India
Department of Data Science and Information Technology

Abstract

The growth of artificial intelligence and deep learning has transformed the way multimedia is created, processed, and consumed. This project presents VOCALMORPH, an AI based system designed to convert the singer's voice in a music video into the vocal identity of a user while preserving pitch, rhythm, and musical background. Unlike traditional speech voice conversion systems, Vocal Morph focuses on singing voices, which are far more complex due to expressive elements such as tonal variation, emotion, and tempo. The system leverages HuBERT for self-supervised feature extraction, HiFi-GAN for neural vocoding, and speaker embedding's for personalized voice identity synthesis. Only 5–8 minutes of recorded speech are required from the user to generate a realistic singing voice. The objective of this work is to showcase deep learning applications in personalized multimedia, specifically in music videos, with practical use cases for musicians, dubbing artists, and multimedia developers.

Keywords: Voice Conversion, Deep Learning, HuBERT, HiFi-GAN, Music Personalization, Neural Vcoders

Smart Mining: Survey on Rock Type Classification using Physical Parameters and AI

Mr. S.Yogesh

Student, II MCA,
School of Computer Applications,
Karpagam College of Engineering,
Coimbatore

Dr. P. Shanthi

Professor,
School of Computer Applications,
Karpagam College of Engineering,
Coimbatore

Abstract

The Rock classification is a crucial task in mining and geotechnical engineering, directly impacting the selection of excavation techniques, safety planning, and equipment usage[3],[9]. Traditional classification methods often rely on expert knowledge, field surveys, and laboratory tests, which are resource-intensive and prone to subjective variations[4],[6]. Recent developments in artificial intelligence (AI) and machine learning (ML) have introduced efficient, data-driven alternatives for rock type prediction using physical and mechanical properties such as uniaxial compressive strength (UCS), porosity, density, and hardness[5]. This survey explores and compares existing research efforts that utilize AI-based models—including Support Vector Machines (SVM), Random Forests (RF), k-Nearest Neighbors (kNN), and Deep Learning approaches—for rock classification[1]. Studies such as Bayram (2025) have demonstrated that SVM can achieve a classification accuracy of over 95%, underscoring the effectiveness of data mining implementations[1]. The paper discusses datasets, preprocessing methods, algorithm performance, and software tools such as WEKA and Python[2],[. Common challenges, including overfitting, missing data, and limited dataset generalization, are critically analyzed[1]. The survey concludes by outlining future directions, such as the integration of real-time sensor data, hybrid deep learning frameworks, and mobile-based field deployment to enhance the scalability and practical adoption of AI in rock classification[12].

Keywords: *Rock classification, machine learning, support vector machine (SVM), random forest, data mining, physical properties, mechanical properties, artificial intelligence, WEKA, geotechnical engineering, mining automation.*

Child Malnutrition Prediction Using Machine Learning Algorithm

A.Amuthavalli

Research Scholar

Department of Computer Science

Vels Institute of Science, Technology & Advanced Studies (VISTAS)

Dr.K.Nandhini

Assistant Professor

Department of Computer Applications (PG)

Vels Institute of Science, Technology & Advanced Studies (VISTAS)

Abstract

Malnutrition remains one of the most pressing public health challenges, particularly in developing nations, where children under five years of age are highly vulnerable. Early prediction and intervention are essential to reduce morbidity, mortality, and long-term developmental impairments associated with undernutrition. With the rapid advancement of data-driven technologies, Machine Learning (ML) offers promising avenues for predicting malnutrition by analyzing complex health, demographic, and socio-economic data. This study focuses on malnutrition prediction through optimized machine learning algorithms. The proposed approach leverages diverse datasets that include anthropometric measurements, dietary intake patterns, household income, sanitation, and access to healthcare services. After extensive preprocessing and feature selection, multiple machine learning models such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, and Gradient Boosting were trained and evaluated. An optimization strategy was applied to enhance predictive performance and reduce model complexity. Furthermore, the study highlights the interpretability of feature importance, showing that maternal education, sanitation access, and dietary diversity are significant predictors. By enabling early detection, such predictive systems can guide targeted nutritional interventions, improve resource allocation, and contribute towards achieving Sustainable Development Goal 2: Zero Hunger.

Keywords— *Machine Learning, Malnutrition, Prediction, Public Health*

A Comparative Analysis of Classical and Deep Learning-Based Image Segmentation: K- Means vs. U-Net

Mr.K. Manishankar

Scholar, Department of Information Systems Management RKMVC, Chennai,Tamilnadu India. Email: kmanishankar174@gmail.com

Mr. K, R. Vishnu Aravindh

Scholar, Department of Information Systems Management RKMVC, Chennai,Tamilnadu India. Email: vishnuaravindhr@gmail.com

Dr. Ch. Sasi Dhar Rao

Assistant Professor,Department of Computer Science, RKMVC , Chennai, Tamilnadu India. Email: sdrteachings@gmail.com

Abstract

Image segmentation, the process of partitioning a digital image into meaningful regions, is a cornerstone of modern computer vision. This paper presents a comparative analysis of two paradigmatic approaches to this task: the classical, unsupervised K-means clustering algorithm and the supervised, deep learning-based U-Net architecture. The objective is to elucidate the fundamental differences in their methodologies, evaluate their performance trade-offs, and assess their suitability for various applications. The methodology involves a detailed examination of each algorithm's operational mechanics—K-means' iterative, centroid-based pixel clustering versus U-Net's hierarchical feature learning within an encoder-decoder framework. Key findings from a review of empirical studies demonstrate that deep learning architectures like U-Net achieve significantly superior performance, with accuracy and Dice scores often exceeding 98%, compared to classical methods which struggle to surpass 80% accuracy on complex datasets. This performance gap is attributed to U-Net's ability to learn intricate spatial and semantic features directly from data, a capability classical methods lack. This analysis is relevant for researchers and practitioners in fields such as medical imaging, autonomous systems, and remote sensing, providing a clear framework for selecting appropriate segmentation tools by highlighting the trade-off between the simplicity of classical algorithms and the high-fidelity results of deep learning models.

Keywords: *computer vision , K-means clustering, U-Net architecture ,unsupervised learning supervised learning, encoder-decoder framework ,centroid-based clustering, hierarchical feature learning, performance trade-offs.*

A Study and Research on Smart Ecosystem

Mr. R. Sri Hari Gajan

Department of Information Systems Management
Ramakrishna Mission Vivekananda College, Chennai
sriharigajan637@gmail.com

Mr. R. Anbu Selvan

Department of Information Systems Management
Ramakrishna Mission Vivekananda College, Chennai

Mr. V. Yuvaraj

Department of Information Systems Management
Ramakrishna Mission Vivekananda College, Chennai

Abstract

The concept of smart ecosystems has emerged as a transformative paradigm in the digital era, enabling the convergence of advanced technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), cloud computing, and big data analytics. A smart ecosystem facilitates seamless interaction among heterogeneous devices, platforms, and stakeholders, thereby fostering intelligent decision-making and resource optimization. Its applications extend across multiple domains, including smart cities, healthcare, energy management, transportation, and industrial automation, contributing to enhanced efficiency, sustainability, and quality of life. Furthermore, smart ecosystems emphasize interoperability, scalability, and resilience while addressing critical challenges such as data privacy, cyber security, and ethical considerations. As global economies transition toward digital-first infrastructures, the evolution of smart ecosystems represents a pivotal step in achieving sustainable development goals and creating adaptive socio-technical systems for the future.

Keywords: Internet of Things (IOT) Sensors, Communication Technologies, IOT Gateway, Data Storage, User Interface

Role of Ethical Hacking in Securing Cyberspace

D. Tharun

Scholar, Department of Information Systems Management RKMVC,
Chennai, Tamilnadu India.
tha.run082077@gmail.com

D. Karthi

Scholar, Department of Information Systems Management RKMVC,
Chennai, Tamilnadu India.
tnsam39@gmail.com

S. Ashwin

Scholar, Department of Information Systems Management RKMVC,
Chennai, Tamilnadu India.
ashwinshankar2210@gmail.com

Abstract

Cyber-risk is a danger or threat associated with the use of interconnected technological systems. This risk occurs when one or more attributes of information namely got impacted[1]. with cyber risk. Cyber risk is an operational risk happening in cyberspace. Cybersecurity mechanisms are always considered costly to implement and create a financial burden to organization[2]. As a result, many organizations have opted not to implement cybersecurity policies and procedures in the prevention of cyber threats that lead to increase the cyber risk level[2]. These risks can lead the organizations to incur financial losses with sensitive information leaks that impacts the business in various ways. This work examines the importance of cybersecurity and the use of ethical techniques for user data protection for organization to prevent cyber threats that assures user data protection.

Keywords — Cybersecurity; Ethical Hacking; Cyber Risk; Data Protection.

Customer Feedback Classifier: Leveraging NLP and Machine Learning for Business Insights

Ms. P. Deeksha,

Department of Data Science

Kumaraguru College of Liberal Arts and Science, Coimbatore

Dr. Soonu Aravindan

Department of Data Science

Kumaraguru College of Liberal Arts and Science, Coimbatore

Abstract

Customer reviews provide valuable insights for businesses but analyzing them manually takes a lot of time and has its limitations. This paper introduces a Customer Feedback Classifier that uses transformer-based zero-shot classification, sentiment analysis, and a specialized lexicon to categorize customer reviews efficiently. Reviews are sorted into actionable categories such as quality issues, delivery problems, feature suggestions, pricing concerns, performance praise, and customer support experiences. Visualizations, including bar charts and donut charts, summarize sentiment distribution and category frequency. The findings show that the system detects sentiment, can handle multi-category and mixed sentiment reviews, and gives businesses a more expansive set of insights. This illustrates the potential of Natural Language Processing (NLP) and machine learning in making better-informed decisions on how to manage customer experiences.

Keywords: Artificial Intelligence, Machine Learning, Predictive Analytics, Decision Intelligence, Data-driven Systems

Building Scalable and Secure Web Applications: Best Practices and Challenges

Dr. G. Savitha

Department of Computer Science w/s in Cyber Security,
Faculty of Science and Humanities,
SRM Institute of Science and Technology, Ramapuram

Mr. M. Murali,

Student, Department of Computer Science w/s in Cyber Security,
SRM Institute of Science and Technology, Ramapuram

Mr. G. Bharathwaj

Student, Department of Computer Science w/s in Cyber Security,
SRM Institute of Science and Technology, Ramapuram

Abstract

The demand for scalable and secure web applications has grown significantly with the rise of digital platforms, cloud computing, and global user bases. Scalability ensures that applications can handle increasing workloads, while security protects data, users, and services from malicious attacks. This paper explores the major challenges faced in scalability and security, highlights best practices for addressing them, presents a comparative analysis, and concludes with recommendations for sustainable development. Suppose only large-scale web applications are to be developed. In that case, the architecture of the back end has to be scalable to support heavier volumes of users, data requests, and performance expected on the web.

Keywords: Scalability; Cloud Computing, Web Security, SQL Injection, DDoS, Authentication Bypass, Zero Trust, Mitigation Strategies

SafeBite: Food Label Ingredient Risk Analyzer

Shivani S,

UG STUDENT, Department of CSE AI&DS,
Vels Institute of Science, Technologies and Advanced Studies, Chennai, India

S Madhumitha

UG STUDENT, Department of CSE AI&DS,
Vels Institute of Science, Technologies and Advanced Studies, Chennai, India

Soundaraya D

UG STUDENT, Department of CSE AI&DS, Vels Institute of Science, Technologies and Advanced
Studies, Chennai, India

S Rajan

Asst.Prof, Department of CSE, Vels Institute of Science, Technologies and Advanced Studies,
Chennai, India

Abstract

In today's health-conscious world, understanding the nutritional and chemical content of packaged food is essential. However, food labels often include technical or unfamiliar terms, making interpretation difficult—especially for individuals with allergies, dietary restrictions, or conditions like diabetes and hypertension. To address this, the proposed study introduces a data-driven system that analyzes food label ingredients using Optical Character Recognition (OCR), regular expressions (regex), and machine learning. The system extracts text from food package images via OCR, and then uses regex to detect additives, E-numbers, sugars, and preservatives. These ingredients are categorized as “Healthy,” “Neutral,” or “Harmful” using a Random Forest classifier trained on a curated dataset. To enhance accuracy and reliability, classifications are validated against guidelines from regulatory bodies such as FSSAI, FDA, and EFSA. The system also provides personalized allergen alerts and a visual health score to support user decisions. Achieving 91% accuracy, this tool offers a practical solution for real-time ingredient risk analysis, promoting healthier eating choices and improving transparency in food labeling.

Keywords: *Optical Character Recognition (OCR), Food Label Analysis, Ingredient Classification, Regular Expressions (Regex), Random Forest Classifier, Additive Detection, Nutritional Analysis*

Car Price Prediction Using Machine Learning and Streamlit Web Application

Kishore.S, Prajwal Ram.R, Sabarish.M.L, Mrs. S.Gayathri Devi

Department of CSE AI&DS,
Vels Institute of Science, Technology and Advanced Studies, Chennai, India

Abstract

The automobile industry is undergoing rapid expansion, with the used-car market emerging as a critical component of global automotive sales. Accurate valuation of pre-owned vehicles is essential for informed decision-making by buyers, sellers, dealerships, financial institutions, and insurance providers. However, traditional appraisal methods—such as depreciation tables, manual inspections, and blue book references—often lack the precision and adaptability required to reflect real-time market dynamics. This paper presents a machine learning (ML)-based car price prediction system that leverages structured data, advanced preprocessing techniques, and regression modeling to estimate used vehicle prices with high accuracy. The system is deployed through a locally hosted Streamlit web application at <http://localhost:8501/>, offering an interactive user interface for inputting vehicle attributes such as brand, year of manufacture, engine size, mileage, and fuel type. The application supports secure user authentication, session-based history, a dynamic leaderboard for manual estimation comparison, and instantaneous price prediction through on-the-fly model inference. Among the models evaluated, Linear Regression was selected for its balance between interpretability and performance, achieving an R^2 score of 0.85, along with low Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The proposed system demonstrates the practical integration of machine learning in real-world scenarios and offers a transparent, scalable, and user-centric solution for data-driven used car valuation.

Keywords: *Used car valuation, Machine Learning (ML), Linear Regression, Price prediction, Streamlit, Residual analysis, Web application, Model evaluation, MAE, RMSE, R^2 score, Data preprocessing, Feature engineering, Real-time inference.*

Enhancing Recruitment Efficiency: An Intelligent Document Processing Framework

Nithya Lakshmi N, Neela Adarsh, Rachakonda Dinesh, Korivi Sai

Department of Information Technology

Maturi Venkata Subba Rao Engineering College

Hyderabad, Telungana

Abstract

The Recruitment process of any Government / Private Organizations involves manual verification of candidate documents - such as educational qualifications, GATE scorecards, and category certificates during the recruitment process. This approach is labour-intensive, time consuming, and susceptible to human error, particularly when handling large volumes of applications in varying formats and regional languages. To overcome these limitations, this paper proposes the adoption of Intelligent Document Processing (IDP) powered by Machine Learning (ML), Deep Learning (DL), Artificial Intelligence (AI), and Natural Language Processing (NLP). The proposed solution enables automated extraction and validation of key information at the point of document submission with real-time alerts for any inconsistencies. Furthermore, advanced screening techniques are employed to ensure accurate and reliable data verification. This intelligent automation framework is designed to enhance the efficiency, accuracy, and scalability of the recruitment workflow.

Keywords: *Optical Character Recognition (OCR), Document Verification, Educational Certificate Validation, Identity Document Processing, Document Information Extraction, Intelligent Document Processing (IDP).*

A Study on Student Posture Recognition Techniques in Smart Classrooms

Mr.R.Sridhar

II MCA,

School of Computer Applications,
Karpagam College of Engineering, Coimbatore.

Dr.P.Shanthi

Professor

School of Computer Applications,
Karpagam College of Engineering, Coimbatore

Abstract

This study focuses on recognizing student postures in smart classrooms using a deep learning model known as YOLOv8. In today technology-driven educational environments, monitoring student behavior is essential for enhancing learning outcomes and classroom engagement. Posture serves as a strong indicator of a student's attentiveness and participation. The main objective of this work is to develop an intelligent system that can automatically identify various postures such as attentive, slouching, sleeping, or using a mobile phone during class. YOLOv8 has been chosen for this task due to its advanced object detection capabilities and real-time processing efficiency. A labeled dataset of classroom images was prepared, where each image is annotated to indicate specific posture categories. Using this dataset, the YOLOv8 model is trained to detect and classify postures accurately. Once trained, the model can process live classroom video streams to recognize student postures in real time. This enables teachers to easily identify students who may be disengaged or distracted without interrupting the teaching process. The system functions quietly in the background, providing insights into student behavior and helping educators deliver personalized support. Posture outputs are displayed with clear bounding boxes and class labels on the video feed. This AI-powered method offers a more scalable and efficient alternative to manual observation. It enhances smart classroom functionality by making behaviour monitoring more reliable and data-driven.

Keywords: Student posture recognition, Smart classrooms, YOLOV8, Object detection, Deep Learning

Facial Expression and Landmark-based Real-Time Fatigue Monitoring for workers using LSTM

Sandhya R, Gopinath M, Dr.R.Balakrishna
Department of CSE AI&DS, VELS University, Chennai, India

Abstract

Fatigue is a critical factor contributing to workplace accidents, particularly in industrial and monitoring environments that require sustained attention. Conventional self-reporting or manual observation techniques are prone to human error and delay. This paper proposes a vision-based fatigue detection system leveraging facial expression analysis and temporal modeling to ensure real-time monitoring and timely intervention. The system employs a Convolutional Neural Network (CNN) for feature extraction—detecting eye closure, yawning, and facial tension—followed by Long Short-Term Memory (LSTM) to model temporal changes such as blinking frequency and head movement. The final classification is performed using Light Gradient Boosting Machine (LightGBM), assigning fatigue levels as Alert, Mild, or Severe. Experimental evaluations on public datasets (NTHU-DDD, YawDD) and custom workplace recordings demonstrate the effectiveness of the proposed approach in accurately identifying fatigue patterns under varying lighting conditions and orientations. The system contributes to enhanced workplace safety, productivity, and well-being through non-invasive, real-time, and intelligent fatigue detection.

Keywords:*Fatigue Detection, CNN, LSTM, LightGBM, Facial Expression, Workplace Safety, Computer Vision*

Spam Detection with XLM-RoBERTa and Sentiment Analysis for Multilingual Emails

Sudhan J, Monish Rajan R, Sriram K

UG STUDENT, Department of CSE AI&DS, VISTAS, Chennai, India
sudhan24082000@gmail.com

Mrs. Sangeetha M.E,

Asst.Prof, Department of CSE, VISTAS, Chennai, India

Abstract

In today's digital communication ecosystem, spam emails have evolved beyond simple keyword-based deception, posing risks such as phishing, fraud, and malware. These threats are exacerbated in multilingual environments where traditional filters fail to capture contextual and semantic nuances. This paper presents a hybrid spam detection framework combining semantic analysis using XLM-RoBERTa and sentiment analysis using BERT/VADER. By integrating contextual embeddings, emotional tone, and metadata features such as sender IP and email structure, the system achieves high multilingual accuracy and real-time inference (<50ms). Experimental results demonstrate superior performance compared to traditional models, providing a scalable, language-independent, and interpretable solution for robust spam detection.

Keywords: *Spam Detection, XLM-RoBERTa, Sentiment Analysis, Multilingual NLP, Deep Learning, Email Security*



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